EXP 801 DS Wheat Heat Drought

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Porometer DATA - ANALYSIS OF LIGHT ON PORTION

- Analysis of gs increase, when light switched on

Load the libraries

```
library(knitr)
#knitr::spin("Exp801.porometer.anal_i.v1.r")
library(ggplot2)
opts_chunk$set("tidy" = FALSE, "dev" = "png", "dpi" = 300,
               "fig.width" = 7.5, "fig.height" = 10, "comment"=NA, fig.path = "figures/increase/")
library(dplyr)
##library(asreml)
library(brms) ##package used to fit bayesian regression model using Stan
```

Set up constants and files

```
options(width=90)
load("../data/porometer.v1.Rdata")
saveFile <- "../data/porometer.anal_i.v1.Rdata"
```

Fit a model to an individual Date to begin with, just for the light on portion

```
# li600.day1.dat <- subset(li600.dat, Date_Sys == min(Date_Sys) & Light == "On")
#head(li600.day1.dat)
# unique(li600.day1.dat$CEBLane)
# unique(li600.day1.dat$CEBPosition)
# unique(li600.day1.dat$CEBSide)
# unique(li600.day1.dat$CEBLocation)
# unique(li600.day1.dat$CEBBlock)
# unique(li600.day1.dat$CEBSubUnit)
# unique(li600.day1.dat$CEBCell)
# ggplot(li600.day1.dat, aes(x=CEBLane, y = CEBPosition)) +
#   geom_tile(fill = "white") +
#   geom_text(aes(label = CEBSubUnit))

##Fit a variance component model to the time course data
# li600.day1.varcomp <- asreml(gsw ~ (Device_Watering*Line) +
#                                time.elapsed +
#                                Device_Watering:time.elapsed +
#                                Line:time.elapsed,
#                              random = ~ spl(time.elapsed),# +
#                                #Device_Watering:spl(time.elapsed),# +
#                                #Line:spl(time.elapsed),# +
#                                #Device_Watering:Line:spl(time.elapsed),
#                              #random = ~ CEBSubUnit + CEBSubUnit:CEBCell,
#                              residual = ~ units,
#                              data = li600.day1.dat)
#
# li600.day1.varcomp <- update(li600.day1.varcomp)
```

```
# summary(li600.day1.varcomp)$varcomp
# wald(li600.day1.varcomp, denDF = "algebraic", ssType = "conditional")
#
# plot(li600.day1.varcomp)
#
# temp.asr <- asreml(gsw ~ (Device_Watering*Line) +
#                      time.elapsed +
#                      Device_Watering:time.elapsed +
#                      Line:time.elapsed,
#                    #random = ~ spl(time.elapsed) +
#                    #  Device_Watering:spl(time.elapsed),# +
#                      #Line:spl(time.elapsed),# +
#                      #Device_Watering:Line:spl(time.elapsed),
#                    #random = ~ CEBSubUnit + CEBSubUnit:CEBCell,
#                    residual = ~ units,
#                    data = li600.day1.dat)
# temp.asr <- update(temp.asr)
# lrt(li600.day1.varcomp, temp.asr)

# li600.day1.varcomp.pred <- predict(li600.day1.varcomp, "Device_Watering:Line:time.elapsed",
#                                    pworkspace = "5gb", workspace = "5gb")#,
#                                    #levels = list(Line = "PBI16C009-0C-0N-0N-010N-14N"))
# head(li600.day1.varcomp.pred$pvals, 30)
#
# ggplot(li600.day1.varcomp.pred$pvals, aes(x=time.elapsed, y = predicted.value)) +
#   geom_point(colour = "red") +
#   facet_grid(Device_Watering ~ Line) +
#   geom_point(data = li600.day1.dat, aes(x=time.elapsed, y = gsw))
#
# li600.day1.varcomp.spl <- predict(li600.day1.varcomp, "Device_Watering:Line:time.elapsed",
#                                    only = "spl(time.elapsed)")
# ggplot(li600.day1.varcomp.spl$pvals, aes(x=time.elapsed, y = predicted.value)) +
#   geom_point(colour = "red") +
#   facet_grid(Device_Watering ~ Line)
# li600.day1.varcomp.linear <- predict(li600.day1.varcomp, "Device_Watering:Line:time.elapsed",
#                                    ignore = "spl(time.elapsed)")
# ggplot(li600.day1.varcomp.linear$pvals, aes(x=time.elapsed, y = predicted.value)) +
#   geom_point(colour = "red") +
#   facet_grid(Device_Watering ~ Line)
```

Stomatal conductance model - McAusland from McAusland et al 2016 Now let's try the Baesian approach that was fitted to the li-6800 data

```
# nlform <- bf(gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - time.elapsed)/k)+1)),
#                g0 ~ 1 + Line*Device_Watering,
#                g1 ~ 1 + Line*Device_Watering,
#                k ~ 1 + Line*Device_Watering,
#                lambda ~ 1 + Line*Device_Watering,
#                nl = TRUE)
#
# priors <- c(prior(normal(0.025, 0.02), nlpar = "g0"),
#             prior(normal(0.3, 0.2), nlpar = "g1"),
#             # prior(normal(0, 0.01), class = "sd", nlpar = "g0"),
```

```
#                 # prior(normal(0, 0.05), class = "sd", nlpar = "g1"),
#                 prior(normal(3, 2), nlpar = "k", lb = 0),
#                 prior(normal(4, 2), nlpar = "lambda", lb = 0),
#                 prior(student_t(3, 0, 0.01), class = "sigma"))
```

Initialise the model Looks like to initialise the model, the dataframe is subset for the first plant in the first rep and for PAR == 1, which is the period where the light is switched on

```
# pick a small but valid dataset

# Configure and compile the model
# fit_init <- brm(
#   formula = nlform,
#   data = li600.day1.dat,
#   prior = priors,
#   chains = 4,
#   cores = 4,
#   iter = 3000,
#   warmup = 2000,
#   control = list(
#     adapt_delta = 0.99,
#     max_treedepth = 12,
#     metric = "diag_e"
#   )
# )
#
# print(summary(fit_init))
# plot(fit_init)
# pairs(fit_init)
# mcmc_plot(fit_init, type = "trace")
# rstan::check_hmc_diagnostics(fit_init$fit)
# loo(fit_init)
#
# pp_check(fit_init)  # compares density of the observed data with data generated from posterior predic
# pp_check(fit_init, type = "error_hist", ndraws = 12)
# pp_check(fit_init, type = "scatter_avg", ndraws = 1000)
# pp_check(fit_init, type = "stat_2d")
# pp_check(fit_init, type = "loo_ribbon")
#
# plot(gsw~time.elapsed, li600.day1.dat, pch=16)
# points(fitted(fit_init)[,1]~time.elapsed, li600.day1.dat, pch=16, col="red")
#
# ##make the above plot prettier
# li600.day1.plot <- li600.day1.dat
# li600.day1.plot$fitted <- fitted(fit_init)[,1]
#
# ggplot(li600.day1.plot, aes(x= time.elapsed, y = gsw)) +
#   geom_point() +
#   facet_grid(Device_Watering ~ Line) +
#   geom_point(aes(y=fitted), colour = "red") +
#   geom_line(aes(y=fitted), colour = "red")
#
# ##look at the effects from the model
```

```
# fixef(fit_init)
# ranef(fit_init)
# coef(fit_init)

# fit_init <- add_criterion(fit_init, "loo")
# fit_init2 <- add_criterion(fit_init2, "loo")
#
# loo_compare(fit_init, fit_init2)
```

Fit a model to an individual chamber

```
# li600.ambient.dat <- subset(li600.dat, Temperature == "Ambient" & Light == "On")
# #head(li600.ambient.dat)
# unique(li600.ambient.dat$CEBLane)
# unique(li600.ambient.dat$CEBPosition)
# unique(li600.ambient.dat$CEBSide)
# unique(li600.ambient.dat$CEBLocation)
# unique(li600.ambient.dat$CEBBlock)
# unique(li600.ambient.dat$CEBSubUnit)
# unique(li600.ambient.dat$CEBCell)
# ggplot(li600.ambient.dat, aes(x=CEBLane, y = CEBPosition)) +
#   geom_tile(fill = "white") +
#   geom_text(aes(label = CEBSubUnit))
#
# ##Fit a variance component model to the time course data
# li600.ambient.varcomp <- asreml(gsw ~ (Device_Watering*Line) +
#                                   time.elapsed +
#                                   Device_Watering:time.elapsed +
#                                   Line:time.elapsed +
#                                   Device_Watering:Line:time.elapsed,
#                                 random = ~ spl(time.elapsed) +
#                                 Device_Watering:spl(time.elapsed) +
#                                 Line:spl(time.elapsed) +
#                                 Device_Watering:Line:spl(time.elapsed) +
#                                 CEBSide + CEBSide:CEBLocation +
#                                   CEBSide:CEBLocation:CEBSubUnit +
#                                   CEBSide:CEBLocation:CEBSubUnit:CEBCell,
#                                 residual = ~ units,
#                                 data = li600.ambient.dat)
#
# li600.ambient.varcomp <- update(li600.ambient.varcomp)
# summary(li600.ambient.varcomp)$varcomp
# wald(li600.ambient.varcomp, denDF = "algebraic", ssType = "conditional")
#
# plot(li600.ambient.varcomp)

# temp.asr <- asreml(gsw ~ (Device_Watering*Line) +
#                                   time.elapsed +
#                                   Device_Watering:time.elapsed +
#                                   Line:time.elapsed +
#                                   Device_Watering:Line:time.elapsed,
#                                 random = ~ spl(time.elapsed) +
#                                   Device_Watering:spl(time.elapsed) +
```

```
#                                     Line:spl(time.elapsed) +
#                                     #Device_Watering:Line:spl(time.elapsed) +
#                                     CEBSide + CEBSide:CEBLocation +
#                                     CEBSide:CEBLocation:CEBSubUnit +
#                                     CEBSide:CEBLocation:CEBSubUnit:CEBCell,
#                     residual = ~ units,
#                     data = li600.ambient.dat)
# temp.asr <- update(temp.asr)
# lrt(li600.ambient.varcomp, temp.asr)

# li600.ambient.varcomp.pred_d1 <- predict(li600.ambient.varcomp,
#                                          "Device_Watering:Line:time.elapsed",
#                                        levels = list(Device_Watering = "D1_Unlimited"))
# head(li600.ambient.varcomp.pred_d1$pvals, 30)
# li600.ambient.varcomp.pred_d2 <- predict(li600.ambient.varcomp,
#                                           "Device_Watering:Line:time.elapsed",
#                                           levels = list(Device_Watering = "D2_Limited"))
# head(li600.ambient.varcomp.pred_d2$pvals, 30)
# li600.ambient.varcomp.pred <- bind_rows(li600.ambient.varcomp.pred_d1$pvals,
#                                         li600.ambient.varcomp.pred_d2$pvals)
#
# ggplot(li600.ambient.varcomp.pred, aes(x=time.elapsed, y = predicted.value)) +
#   geom_point(colour = "red") +
#   facet_grid(Device_Watering ~ Line) +
#   geom_point(data = li600.ambient.dat, aes(x=time.elapsed, y = gsw,
#                                            colour = Replicate))
```

Stomatal conductance model - McAusland from McAusland et al 2016 Now let's try the Baesian approach that was fitted to the li-6800 data

```
# nlform_amb <- bf(gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - time.elapsed)/k)+1)),
#              g0 ~ 1 + Line*Device_Watering +
#                (1 + Line*Device_Watering || Replicate),
#              g1 ~ 1 + Line*Device_Watering +
#                (1 + Line*Device_Watering || Replicate),
#              k ~ 1 + Line*Device_Watering,
#              lambda ~ 1 + Line*Device_Watering,
#              nl = TRUE)
#
# priors_amb <- c(prior(normal(0.025, 0.02), nlpar = "g0"),
#             prior(normal(0.4, 0.15), nlpar = "g1"),
#             prior(normal(3.5, 2), nlpar = "k", lb = 0),
#             prior(normal(5, 1), nlpar = "lambda", lb = 0),
#             prior(normal(0, 0.05), class = "sd", nlpar = "g0"),
#             prior(normal(0, 0.1), class = "sd", nlpar = "g1"),
#             prior(student_t(3, 0, 0.05), class = "sigma"))
```

Initialise the model Fit the model just sampling from the priors to begin with

```
##check model using priors only
# fit_prior_amb <- brm(
#   formula = nlform_amb,
#   data = li600.ambient.dat,
#   prior = priors_amb,
```

```
#   sample_prior = "only",
#   chains = 1,
#   cores = 4
# )
#
# pp_check(fit_prior_amb, ndraws = 25)
# summary(fit_prior_amb)


# Configure and compile the model
# fit_init <- brm(
#   formula = nlform,
#   data = li600.ambient.dat,
#   prior = priors,
#   chains = 4,
#   cores = 4,
#   iter = 5000,
#   warmup = 3000,
#   control = list(
#     adapt_delta = 0.99,
#     max_treedepth = 12,
#     metric = "diag_e"
#   )
# )

# print(summary(fit_init))
# plot(fit_init)
# pairs(fit_init)
# mcmc_plot(fit_init, type = "trace")
# rstan::check_hmc_diagnostics(fit_init$fit)
# loo(fit_init)
#
# pp_check(fit_init)  # compares density of the observed data with data generated from posterior predic
# pp_check(fit_init, type = "error_hist", ndraws = 12)
# pp_check(fit_init, type = "scatter_avg", ndraws = 1000)
# pp_check(fit_init, type = "stat_2d")
# pp_check(fit_init, type = "loo_ribbon")

# plot(gsw~time.elapsed, li600.ambient.dat, pch=16)
# points(fitted(fit_init)[,1]~time.elapsed, li600.ambient.dat, pch=16, col="red")

##make the above plot prettier
# li600.ambient.plot <- li600.ambient.dat
# li600.ambient.plot$fitted <- fitted(fit_init)[,1]
#
# ggplot(li600.ambient.plot, aes(x= time.elapsed, y = gsw, group = Replicate)) +
#   geom_point() +
#   facet_grid(Device_Watering ~ Line) +
#   geom_point(aes(y=fitted), colour = "red") +
#   geom_line(aes(y=fitted), colour = "red")
#
# ##look at the effects from the model
# fixef(fit_init)
```

```
# ranef(fit_init)
# coef(fit_init)

# fit_init <- add_criterion(fit_init, "loo")
# fit_init2 <- add_criterion(fit_init2, "loo")
#
# loo_compare(fit_init, fit_init2)
```

Fit a model to the entire dataset, for the light on portion

```
li600.on.dat <- subset(li600.dat, Light == "On")
##using full dataset, figure out the earliest time at which the light was turned on for each day
li600.on.dat <- li600.on.dat %>% group_by(Date) %>% mutate(min.time = min(time.elapsed))
##calculate adjusted time by subtracting the earliest time the light was on each day
li600.on.dat$time.adj <- li600.on.dat$time.elapsed - li600.on.dat$min.time

#head(li600.dat)
# unique(li600.on.dat$CEBRoom)
# unique(li600.on.dat$CEBLane)
# unique(li600.on.dat$CEBPosition)
# unique(li600.on.dat$CEBSide)
# unique(li600.on.dat$CEBLocation)
# unique(li600.on.dat$CEBBlock)
# unique(li600.on.dat$CEBSubUnit)
# unique(li600.on.dat$CEBCell)
# ggplot(li600.on.dat, aes(x=CEBLane, y = CEBPosition)) +
#   geom_tile(fill = "white") +
#   geom_text(aes(label = CEBSide)) +
#   facet_wrap(~CEBRoom)

##Fit a variance component model to the time course data
# li600.on.varcomp <- asreml(gsw ~ (Room_Temperature*Device_Watering*Line) +
#                              time.elapsed +
#                              Room_Temperature:time.elapsed +
#                              Device_Watering:time.elapsed +
#                              Line:time.elapsed +
#                              #Room_Temperature:Device_Watering:time.elapsed +
#                              Room_Temperature:Line:time.elapsed +
#                              Device_Watering:Line:time.elapsed,
#                            random = ~ spl(time.elapsed) +
#                              Room_Temperature:spl(time.elapsed) +
#                              Device_Watering:spl(time.elapsed) +
#                              Line:spl(time.elapsed) +
#                              #Room_Temperature:Device_Watering:spl(time.elapsed) +
#                              Room_Temperature:Line:spl(time.elapsed) +
#                              Device_Watering:Line:spl(time.elapsed) +
#                              CEBRoom:CEBSide +
#                              CEBRoom:CEBSide:CEBLocation +
#                              CEBRoom:CEBSide:CEBLocation:CEBSubUnit +
#                              CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell,
#                            residual = ~ units,
#                            data = li600.on.dat,
#                            workspace = "5gb")
```

```
#
# li600.on.varcomp <- update(li600.on.varcomp)
# summary(li600.on.varcomp)$varcomp
# wald(li600.on.varcomp, denDF = "algebraic", ssType = "conditional")
#
# plot(li600.on.varcomp)

# temp.asr <- asreml(gsw ~ (Room_Temperature*Device_Watering*Line) +
#                            time.elapsed +
#                            Room_Temperature:time.elapsed +
#                            Device_Watering:time.elapsed +
#                            Line:time.elapsed +
#                            #Room_Temperature:Device_Watering:time.elapsed +
#                            Room_Temperature:Line:time.elapsed +
#                            Device_Watering:Line:time.elapsed,
#                          random = ~ spl(time.elapsed) +
#                            Room_Temperature:spl(time.elapsed) +
#                            Device_Watering:spl(time.elapsed) +
#                            Line:spl(time.elapsed) +
#                            #Room_Temperature:Device_Watering:spl(time.elapsed) +
#                            Room_Temperature:Line:spl(time.elapsed) +
#                            Device_Watering:Line:spl(time.elapsed) +
#                            CEBRoom:CEBSide +
#                            CEBRoom:CEBSide:CEBLocation +
#                            CEBRoom:CEBSide:CEBLocation:CEBSubUnit +
#                            CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell,
#                          residual = ~ units,
#                          data = li600.on.dat,
#                          workspace = "5gb")
# temp.asr <- update(temp.asr)
# lrt(li600.on.varcomp, temp.asr)

# li600.on.varcomp.pred_d1c8 <- predict(li600.on.varcomp,
#                                       "Room_Temperature:Device_Watering:Line:time.elapsed",
#                                       levels = list(Device_Watering = "D1_Unlimited",
#                                                     Room_Temperature = "CER8_Ambient",
#                                                     time.elapsed = seq(38.5, 101.5, 1)),
#                                   workspace = "5gb",
#                                   pworkspace = "5gb")
# head(li600.on.varcomp.pred_d1c8$pvals, 30)
# li600.on.varcomp.pred_d1c7 <- predict(li600.on.varcomp,
#                                     "Room_Temperature:Device_Watering:Line:time.elapsed",
#                                     levels = list(Device_Watering = "D1_Unlimited",
#                                                   Room_Temperature = "CER7_Heat",
#                                                   time.elapsed = seq(38.5, 101.5, 1)),
#                                     workspace = "5gb",
#                                     pworkspace = "5gb")
# head(li600.on.varcomp.pred_d1c7$pvals, 30)
# li600.on.varcomp.pred_d2c8 <- predict(li600.on.varcomp,
#                                     "Room_Temperature:Device_Watering:Line:time.elapsed",
#                                     levels = list(Device_Watering = "D2_Limited",
#                                                   Room_Temperature = "CER8_Ambient",
#                                                   time.elapsed = seq(38.5, 101.5, 1)),
```

```
#                                     workspace = "5gb",
#                                     pworkspace = "5gb")
# head(li600.on.varcomp.pred_d2c8$pvals, 30)
# li600.on.varcomp.pred_d2c7 <- predict(li600.on.varcomp,
#                                     "Room_Temperature:Device_Watering:Line:time.elapsed",
#                                     levels = list(Device_Watering = "D2_Limited",
#                                                   Room_Temperature = "CER7_Heat",
#                                                   time.elapsed = seq(38.5, 101.5, 1)),
#                                     workspace = "5gb",
#                                     pworkspace = "5gb")
# head(li600.on.varcomp.pred_d1c8$pvals, 30)
# li600.on.varcomp.pred <- bind_rows(li600.on.varcomp.pred_d1c8$pvals,
#                                     li600.on.varcomp.pred_d1c7$pvals,
#                                     li600.on.varcomp.pred_d2c8$pvals,
#                                     li600.on.varcomp.pred_d2c7$pvals)
#
# ggplot(li600.on.varcomp.pred, aes(x=time.elapsed, y = predicted.value)) +
#   facet_grid(Room_Temperature + Device_Watering ~ Line) +
#   geom_point(data = li600.on.dat, aes(x=time.elapsed, y = gsw,
#                                       colour = Replicate)) +
#   geom_point(colour = "red")
```

Stomatal conductance model - McAusland from McAusland et al 2016 Now let's try the Baesian approach that was fitted to the li-6800 data

```
nlform <- bf(gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - time.adj)/k)+1)),
             g0 ~ 1 + Line*Device_Watering*Room_Temperature +
               (1 | CEBRoom:CEBSide:CEBLocation +
                  CEBRoom:CEBSide:CEBLocation:CEBSubUnit +
                  CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell),
             g1 ~ 1 + Line*Device_Watering*Room_Temperature +
               (1 | CEBRoom:CEBSide:CEBLocation +
                  CEBRoom:CEBSide:CEBLocation:CEBSubUnit +
                  CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell),
             k ~ 1 + Line*Device_Watering*Room_Temperature,
             lambda ~ 1 + Line*Device_Watering*Room_Temperature,
             nl = TRUE)

priors <- c(prior(normal(0, 0.02), nlpar = "g0"),
            prior(normal(0.4, 0.15), nlpar = "g1"),
            prior(normal(3.5, 2), nlpar = "k", lb = 0),
            prior(normal(5, 1), nlpar = "lambda", lb = 0),
            prior(normal(0, 0.05), class = "sd", nlpar = "g0"),
            prior(normal(0, 0.1), class = "sd", nlpar = "g1"),
            prior(student_t(3, 0, 0.05), class = "sigma"))

plot(0 + (0.4 - 0) * exp(-exp(((5 - seq(0, 60, 0.5))/3.5) + 1)) ~
       seq(0, 60, 0.5))
```

```
# Fit the model
fit_on_max <- brm(
  formula = nlform,
```

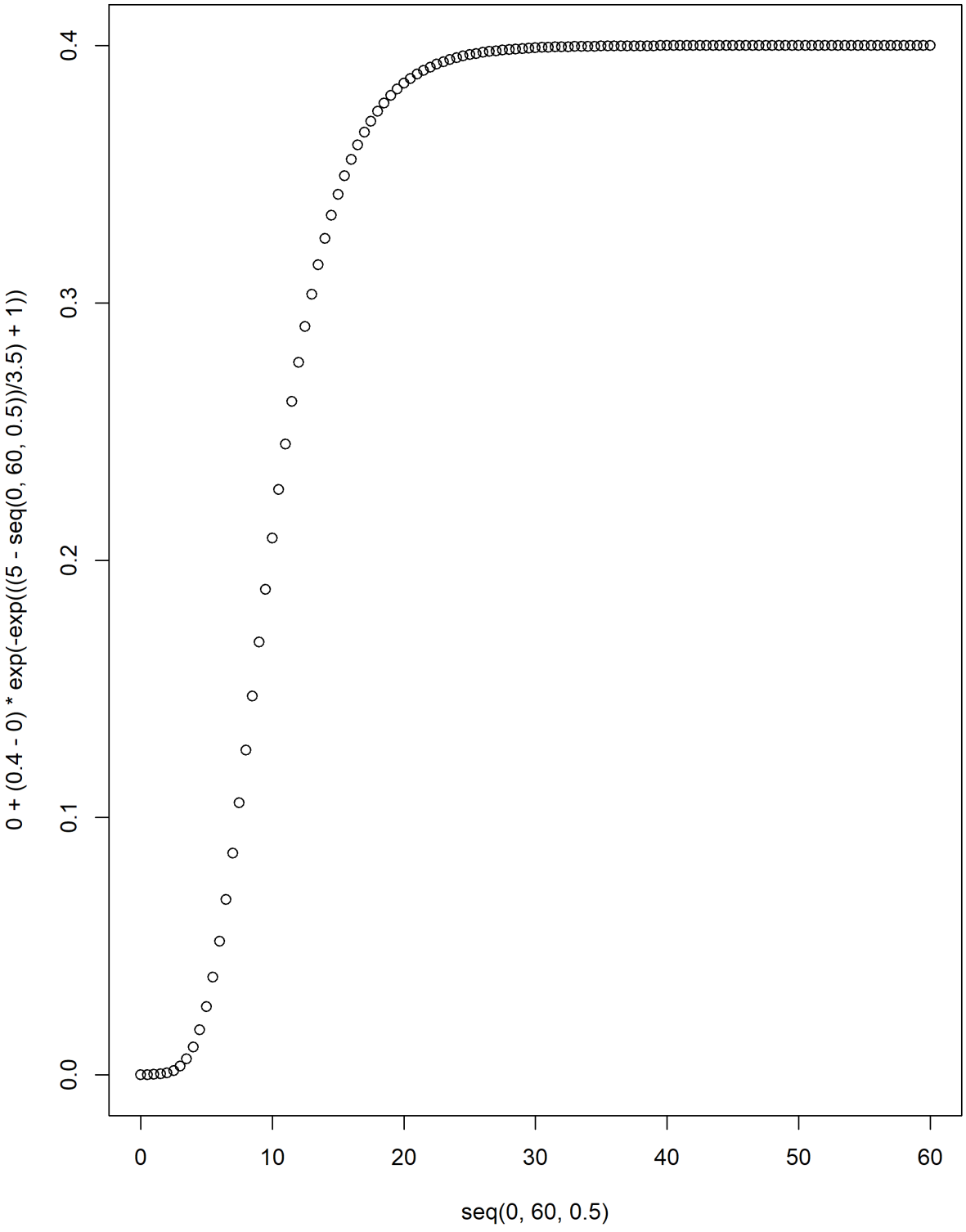


Figure 1: plot of chunk unnamed-chunk-10

```
  data = li600.on.dat,
  prior = priors,
  chains = 6,
  cores = 6,
  iter = 10000,
  warmup = 6000,
  control = list(
    adapt_delta = 0.99,
    max_treedepth = 12,
    metric = "diag_e"
  )
)
```

Initialise the model

```
Compiling Stan program...
```

```
Start sampling
```

```
print(summary(fit_on_max))
```

```
 Family: gaussian
  Links: mu = identity
Formula: gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - time.adj)/k) + 1))
         g0 ~ 1 + Line * Device_Watering * Room_Temperature + (1 | CEBRoom:CEBSide:CEBLocation + CEBRoo
         g1 ~ 1 + Line * Device_Watering * Room_Temperature + (1 | CEBRoom:CEBSide:CEBLocation + CEBRoo
         k ~ 1 + Line * Device_Watering * Room_Temperature
         lambda ~ 1 + Line * Device_Watering * Room_Temperature
   Data: li600.on.dat (Number of observations: 1682)
  Draws: 6 chains, each with iter = 10000; warmup = 6000; thin = 1;
         total post-warmup draws = 24000

Multilevel Hyperparameters:
~CEBRoom:CEBSide:CEBLocation (Number of levels: 8)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.02      0.02     0.00     0.07 1.00     7418    13208
sd(g1_Intercept)     0.05      0.04     0.00     0.14 1.00     9766    13366

~CEBRoom:CEBSide:CEBLocation:CEBSubUnit (Number of levels: 16)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.07      0.02     0.04     0.11 1.00     9538    11961
sd(g1_Intercept)     0.14      0.04     0.08     0.22 1.00     9644    14718

~CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell (Number of levels: 159)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.04      0.01     0.03     0.06 1.00     7281     9726
sd(g1_Intercept)     0.12      0.01     0.09     0.15 1.00     5957    11029

Regression Coefficients:
                                                                                Estimate
g0_Intercept                                                                        0.02
g0_LinePBI17N002M0CM0NM0NM010NM16N                                                 -0.00
g0_LinePBI17N006M0NM0NM89N                                                          0.01
g0_LinePBI17N007M0CM0NM0NM010NM8N                                                   0.01
g0_LinePBI17N013M0CM0NM0NM010NM8N                                                   0.03
g0_LinePBI17N016M0CM0NM0NM010NM27N                                                  0.00
```

g0_LinePBI19N007M0NM3N	-0.01
g0_LinePBIC15034M0CM24NM010NM1NM0N	0.01
g0_LineROCKSTAR	0.00
g0_LineSUNCHASER	0.02
g0_Device_WateringD2_Limited	0.04
g0_Room_TemperatureCER8_Ambient	0.02
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	-0.02
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	0.01
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	-0.02
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	0.03
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	0.00
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited	-0.00
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	0.02
g0_LineROCKSTAR:Device_WateringD2_Limited	-0.00
g0_LineSUNCHASER:Device_WateringD2_Limited	0.03
g0_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient	-0.00
g0_LinePBI17N006M0NM0NM89N:Room_TemperatureCER8_Ambient	0.02
g0_LinePBI17N007M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	0.00
g0_LinePBI17N013M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	0.02
g0_LinePBI17N016M0CM0NM0NM010NM27N:Room_TemperatureCER8_Ambient	0.01
g0_LinePBI19N007M0NM3N:Room_TemperatureCER8_Ambient	-0.01
g0_LinePBIC15034M0CM24NM010NM1NM0N:Room_TemperatureCER8_Ambient	-0.00
g0_LineROCKSTAR:Room_TemperatureCER8_Ambient	0.01
g0_LineSUNCHASER:Room_TemperatureCER8_Ambient	-0.02
g0_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.02
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.02
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.01
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.00
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.03
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.01
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.00
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.01
g0_LineROCKSTAR:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.00
g0_LineSUNCHASER:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.01
g1_Intercept	0.20
g1_LinePBI17N002M0CM0NM0NM010NM16N	0.03
g1_LinePBI17N006M0NM0NM89N	0.11
g1_LinePBI17N007M0CM0NM0NM010NM8N	0.08
g1_LinePBI17N013M0CM0NM0NM010NM8N	0.05
g1_LinePBI17N016M0CM0NM0NM010NM27N	-0.03
g1_LinePBI19N007M0NM3N	0.15
g1_LinePBIC15034M0CM24NM010NM1NM0N	0.16
g1_LineROCKSTAR	0.11
g1_LineSUNCHASER	0.14
g1_Device_WateringD2_Limited	-0.13
g1_Room_TemperatureCER8_Ambient	0.03
g1_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	0.15
g1_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	0.25
g1_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	-0.07
g1_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	0.09
g1_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	0.26
g1_LinePBI19N007M0NM3N:Device_WateringD2_Limited	0.20
g1_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	0.03
g1_LineROCKSTAR:Device_WateringD2_Limited	0.20

g1_LineSUNCHASER:Device_WateringD2_Limited	0.18
g1_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient	0.06
g1_LinePBI17N006M0NM0NM89N:Room_TemperatureCER8_Ambient	0.07
g1_LinePBI17N007M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	0.05
g1_LinePBI17N013M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	0.12
g1_LinePBI17N016M0CM0NM0NM010NM27N:Room_TemperatureCER8_Ambient	0.14
g1_LinePBI19N007M0NM3N:Room_TemperatureCER8_Ambient	0.11
g1_LinePBIC15034M0CM24NM010NM1NM0N:Room_TemperatureCER8_Ambient	0.10
g1_LineROCKSTAR:Room_TemperatureCER8_Ambient	0.14
g1_LineSUNCHASER:Room_TemperatureCER8_Ambient	0.09
g1_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.02
g1_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.15
g1_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.15
g1_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.16
g1_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.24
g1_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.21
g1_LinePBI19N007M0NM3N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.14
g1_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.08
g1_LineROCKSTAR:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.28
g1_LineSUNCHASER:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.21
k_Intercept	3.41
k_LinePBI17N002M0CM0NM0NM010NM16N	2.46
k_LinePBI17N006M0NM0NM89N	0.69
k_LinePBI17N007M0CM0NM0NM010NM8N	0.37
k_LinePBI17N013M0CM0NM0NM010NM8N	1.11
k_LinePBI17N016M0CM0NM0NM010NM27N	1.13
k_LinePBI19N007M0NM3N	1.67
k_LinePBIC15034M0CM24NM010NM1NM0N	7.19
k_LineROCKSTAR	2.99
k_LineSUNCHASER	3.97
k_Device_WateringD2_Limited	0.16
k_Room_TemperatureCER8_Ambient	3.46
k_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	3.16
k_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	0.55
k_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	4.18
k_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	4.32
k_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	1.31
k_LinePBI19N007M0NM3N:Device_WateringD2_Limited	0.55
k_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	3.81
k_LineROCKSTAR:Device_WateringD2_Limited	2.22
k_LineSUNCHASER:Device_WateringD2_Limited	1.96
k_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient	3.89
k_LinePBI17N006M0NM0NM89N:Room_TemperatureCER8_Ambient	2.72
k_LinePBI17N007M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	1.02
k_LinePBI17N013M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	3.45
k_LinePBI17N016M0CM0NM0NM010NM27N:Room_TemperatureCER8_Ambient	3.77
k_LinePBI19N007M0NM3N:Room_TemperatureCER8_Ambient	5.15
k_LinePBIC15034M0CM24NM010NM1NM0N:Room_TemperatureCER8_Ambient	2.40
k_LineROCKSTAR:Room_TemperatureCER8_Ambient	2.62
k_LineSUNCHASER:Room_TemperatureCER8_Ambient	4.37
k_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	8.18
k_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	4.26
k_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	4.37
k_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	4.01

```
k_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient          3.91
k_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient         3.96
k_LinePBI19N007M0NM3N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                      4.16
k_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient          4.16
k_LineROCKSTAR:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                             4.39
k_LineSUNCHASER:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient                            4.13
lambda_Intercept                                                                                 0.46
lambda_LinePBI17N002M0CM0NM0NM010NM16N                                                           3.93
lambda_LinePBI17N006M0NM0NM89N                                                                   3.28
lambda_LinePBI17N007M0CM0NM0NM010NM8N                                                            2.39
lambda_LinePBI17N013M0CM0NM0NM010NM8N                                                            3.50
lambda_LinePBI17N016M0CM0NM0NM010NM27N                                                           3.84
lambda_LinePBI19N007M0NM3N                                                                       4.53
lambda_LinePBIC15034M0CM24NM010NM1NM0N                                                           4.97
lambda_LineROCKSTAR                                                                              3.52
lambda_LineSUNCHASER                                                                             4.25
lambda_Device_WateringD2_Limited                                                                 0.45
lambda_Room_TemperatureCER8_Ambient                                                              0.77
lambda_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited                                 4.83
lambda_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited                                         3.37
lambda_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited                                  4.99
lambda_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited                                  4.99
lambda_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited                                 4.04
lambda_LinePBI19N007M0NM3N:Device_WateringD2_Limited                                             3.48
lambda_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited                                 4.70
lambda_LineROCKSTAR:Device_WateringD2_Limited                                                    4.03
lambda_LineSUNCHASER:Device_WateringD2_Limited                                                   4.23
lambda_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient                              4.60
 [ reached 'max' / getOption("max.print") -- omitted 18 rows ]

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.10      0.00     0.09     0.10 1.00    16427    16951

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
plot(fit_on_max, ask = FALSE)
```

```r
# pairs(fit_on_max)
# mcmc_plot(fit_on_max, type = "trace")
rstan::check_hmc_diagnostics(fit_on_max$fit)
```

```
Divergences:

0 of 24000 iterations ended with a divergence.

Tree depth:

0 of 24000 iterations saturated the maximum tree depth of 12.

Energy:
```

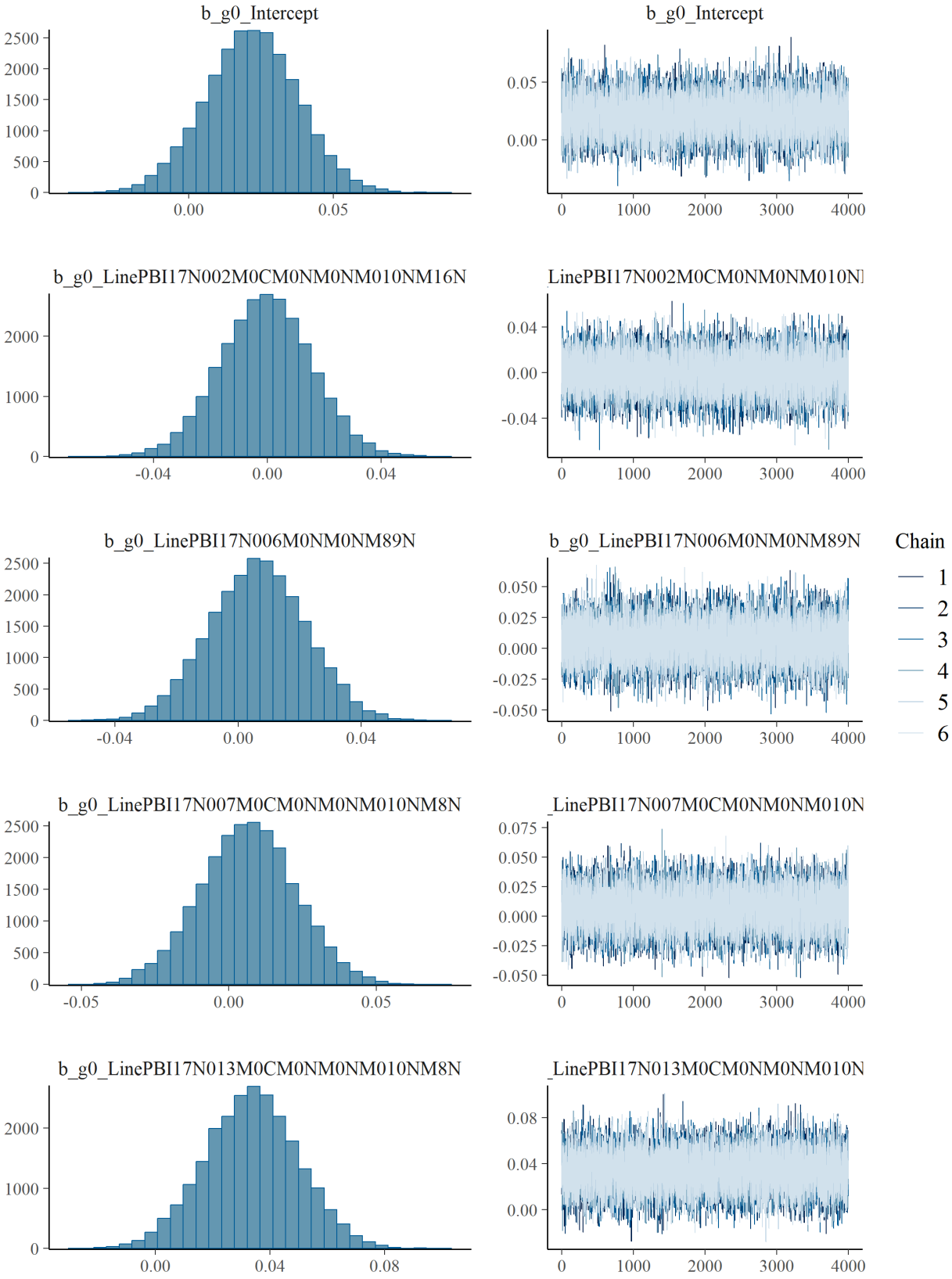


Figure 2: plot of chunk unnamed-chunk-11

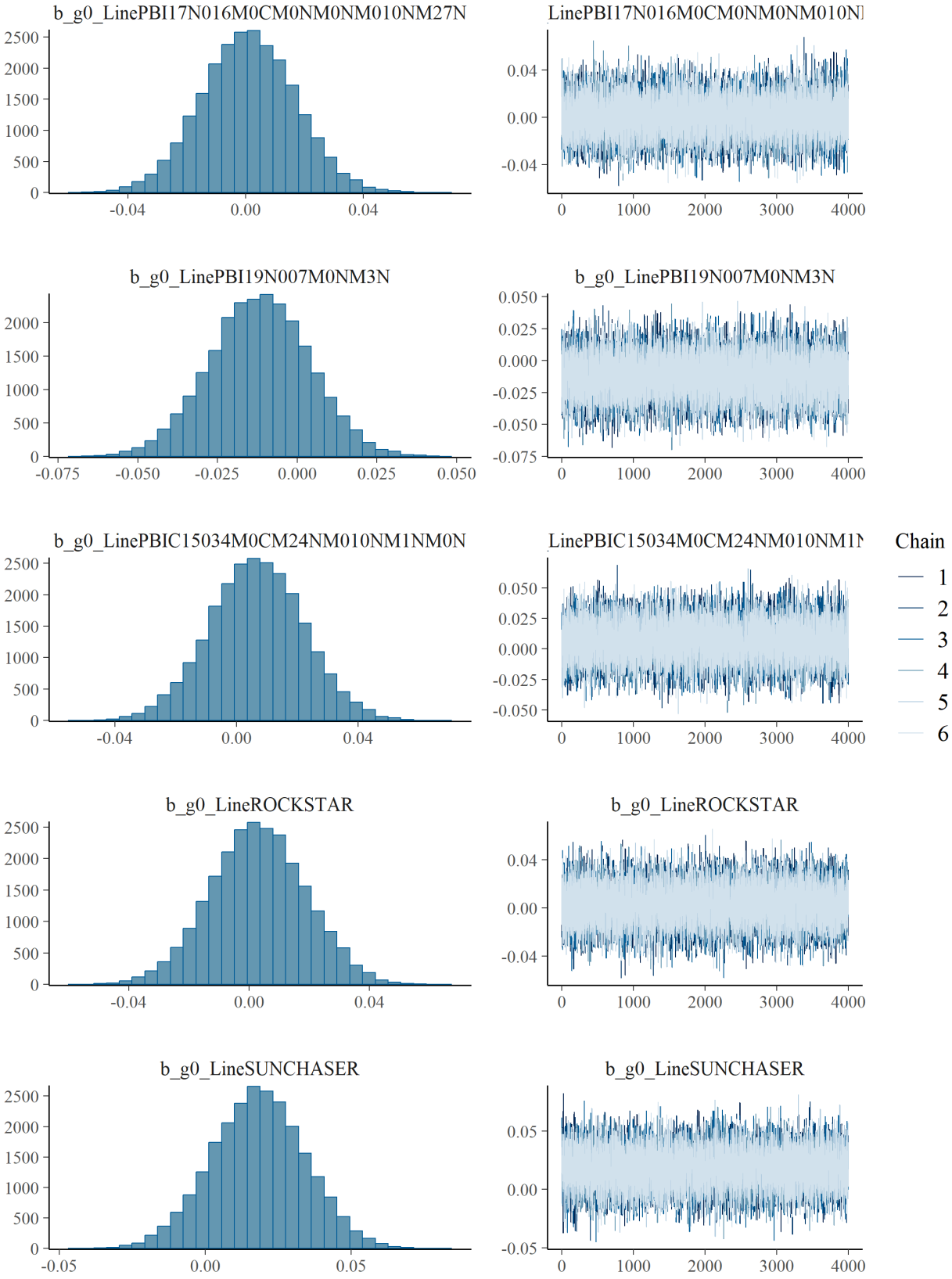


Figure 3: plot of chunk unnamed-chunk-11

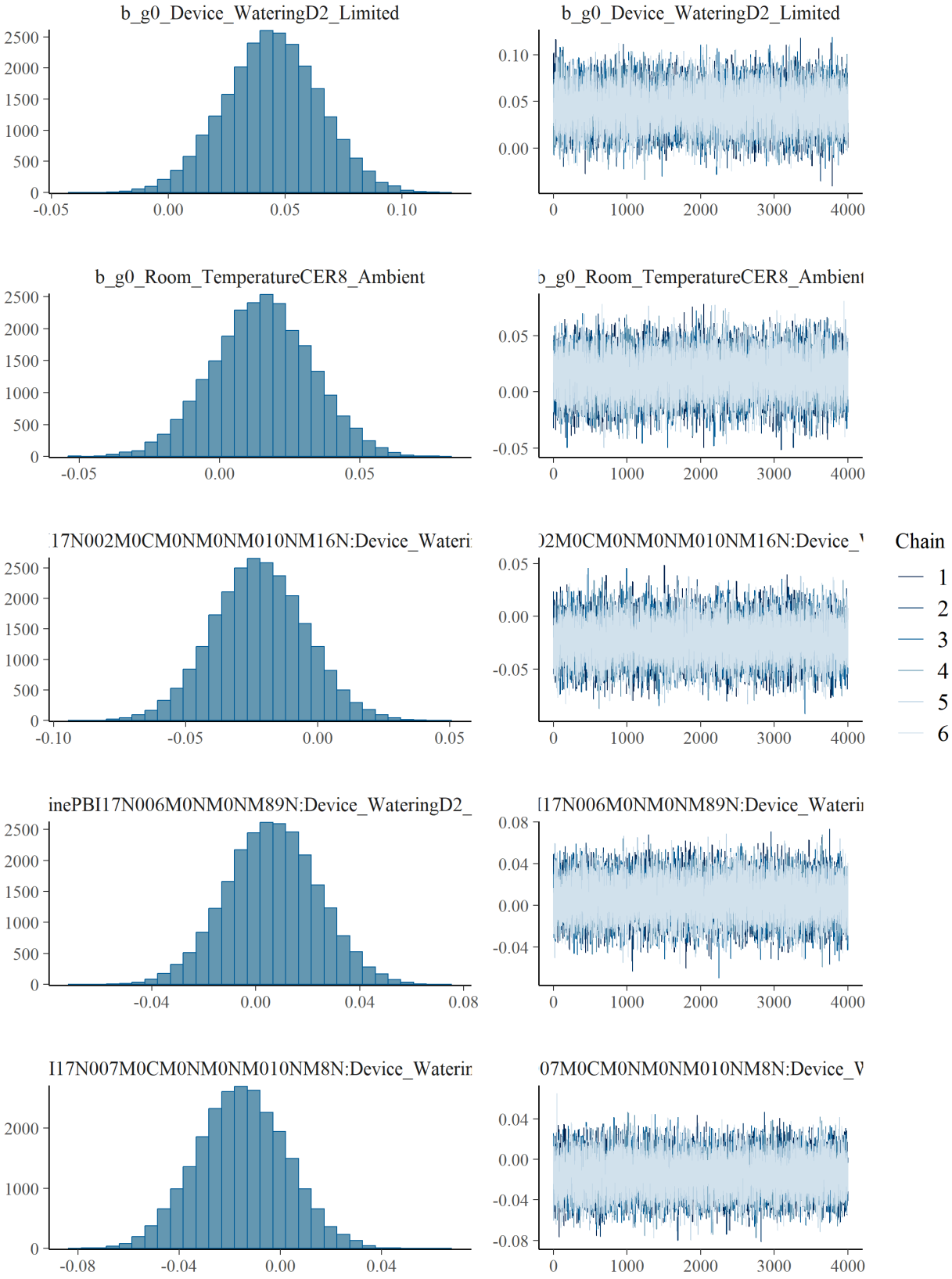


Figure 4: plot of chunk unnamed-chunk-11

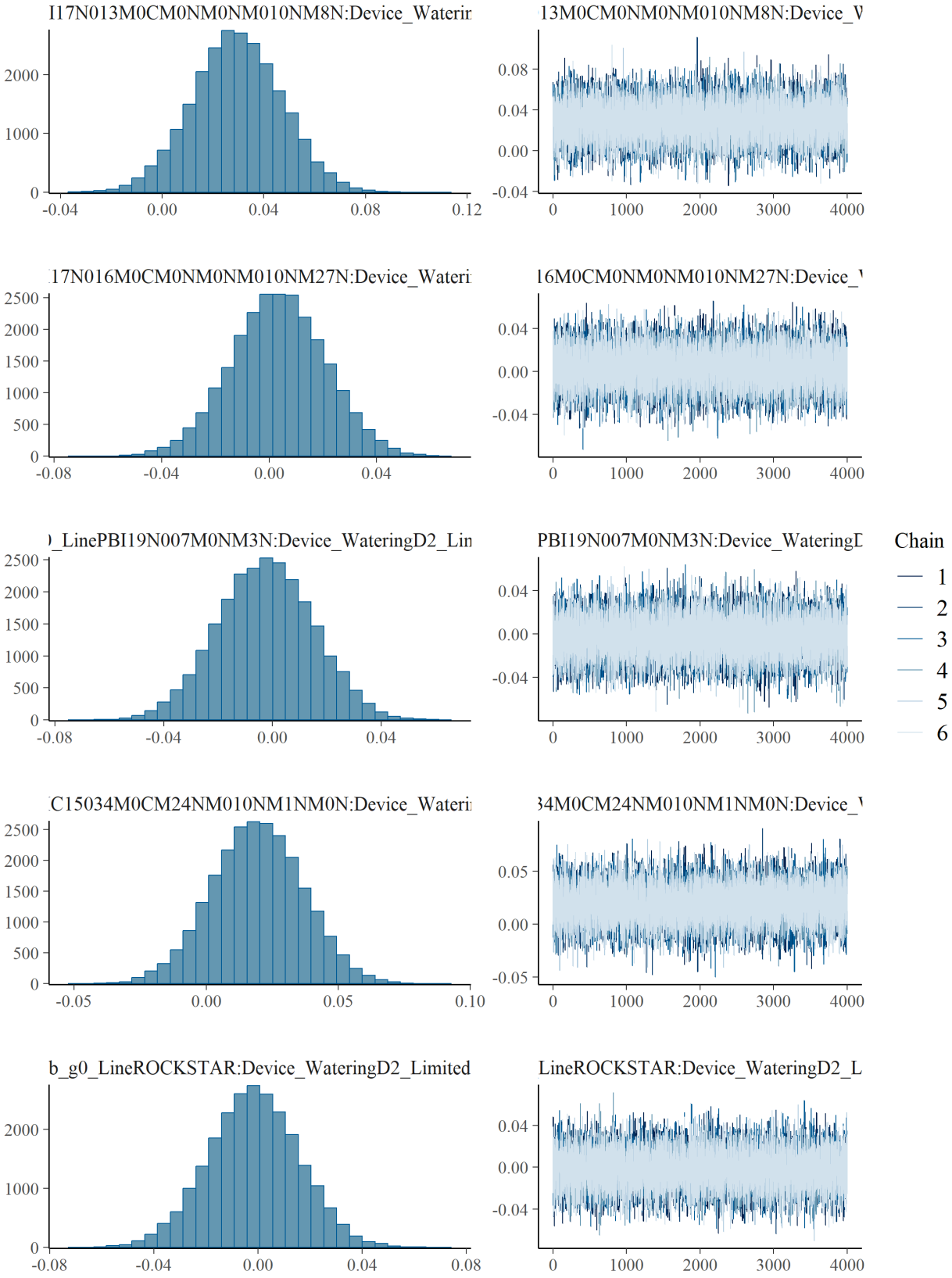


Figure 5: plot of chunk unnamed-chunk-11

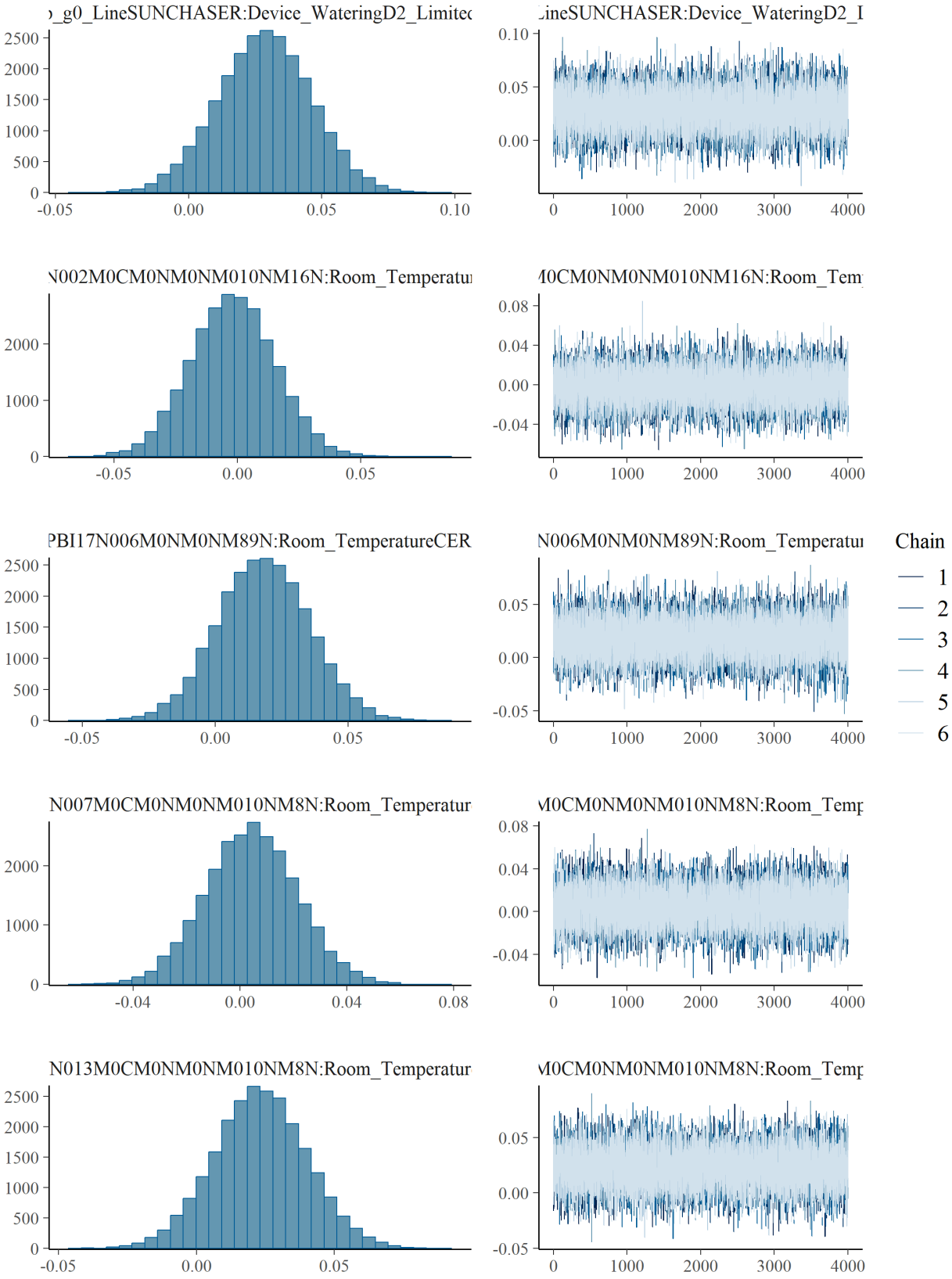


Figure 6: plot of chunk unnamed-chunk-11

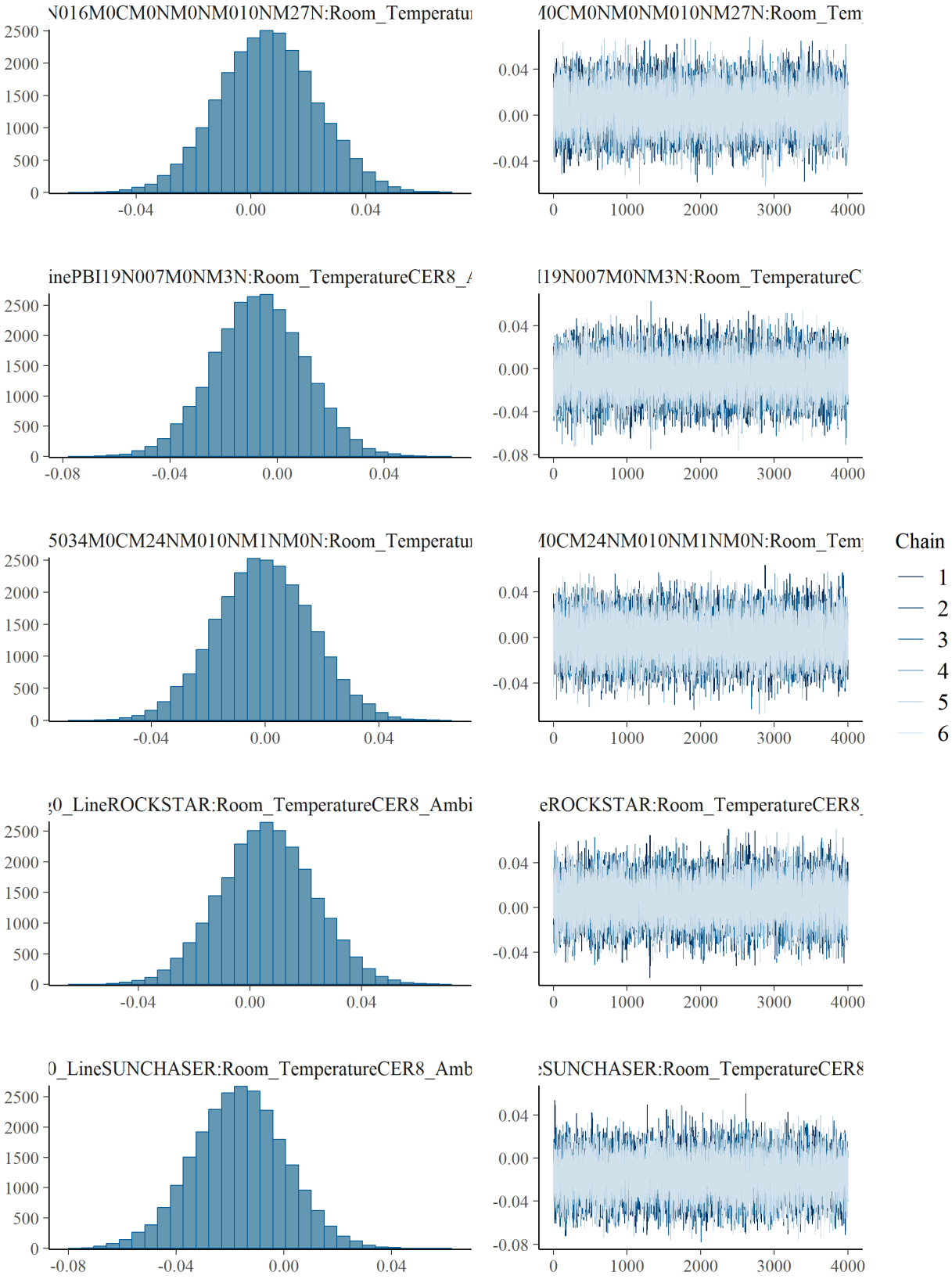


Figure 7: plot of chunk unnamed-chunk-11

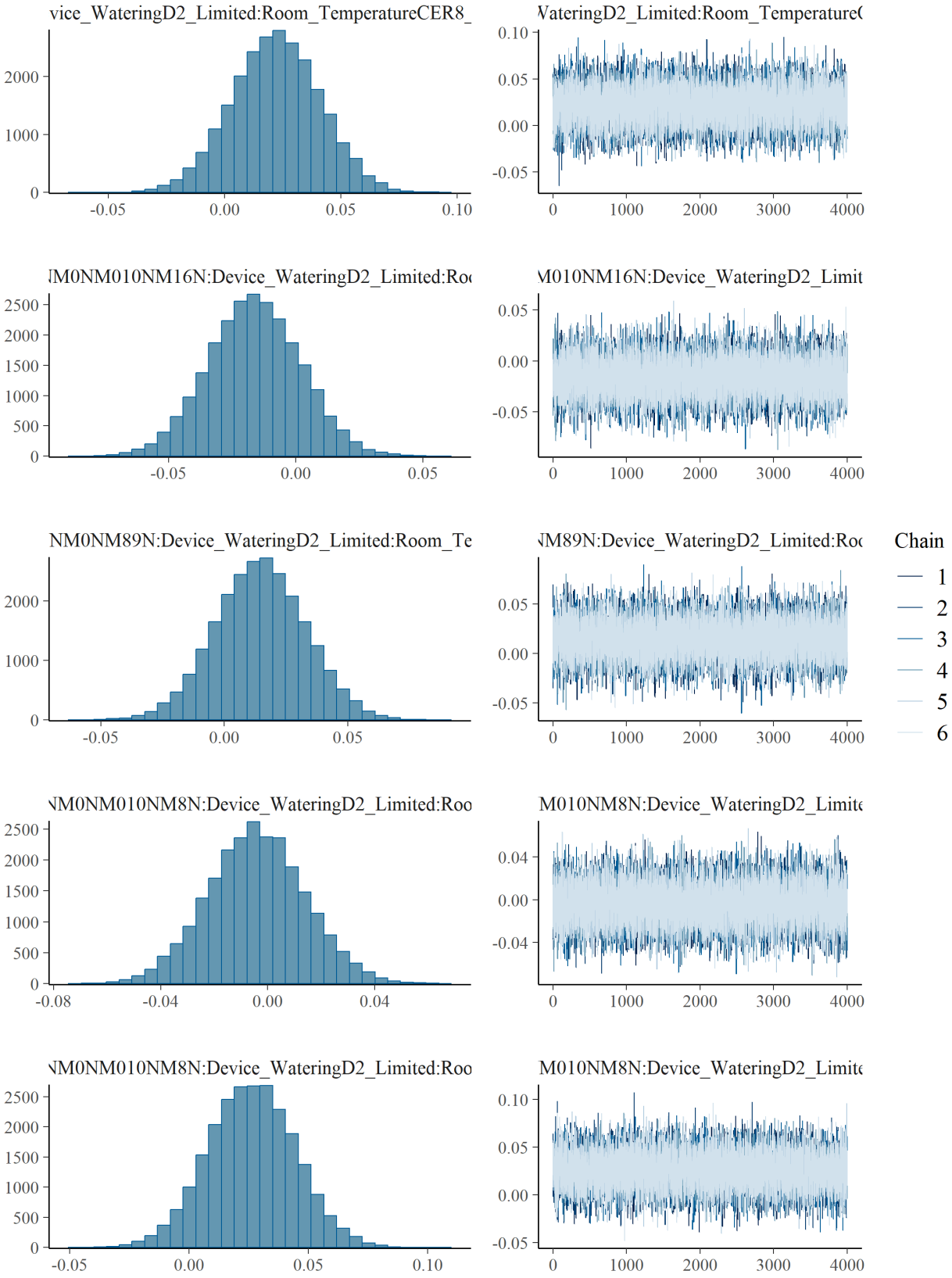


Figure 8: plot of chunk unnamed-chunk-11

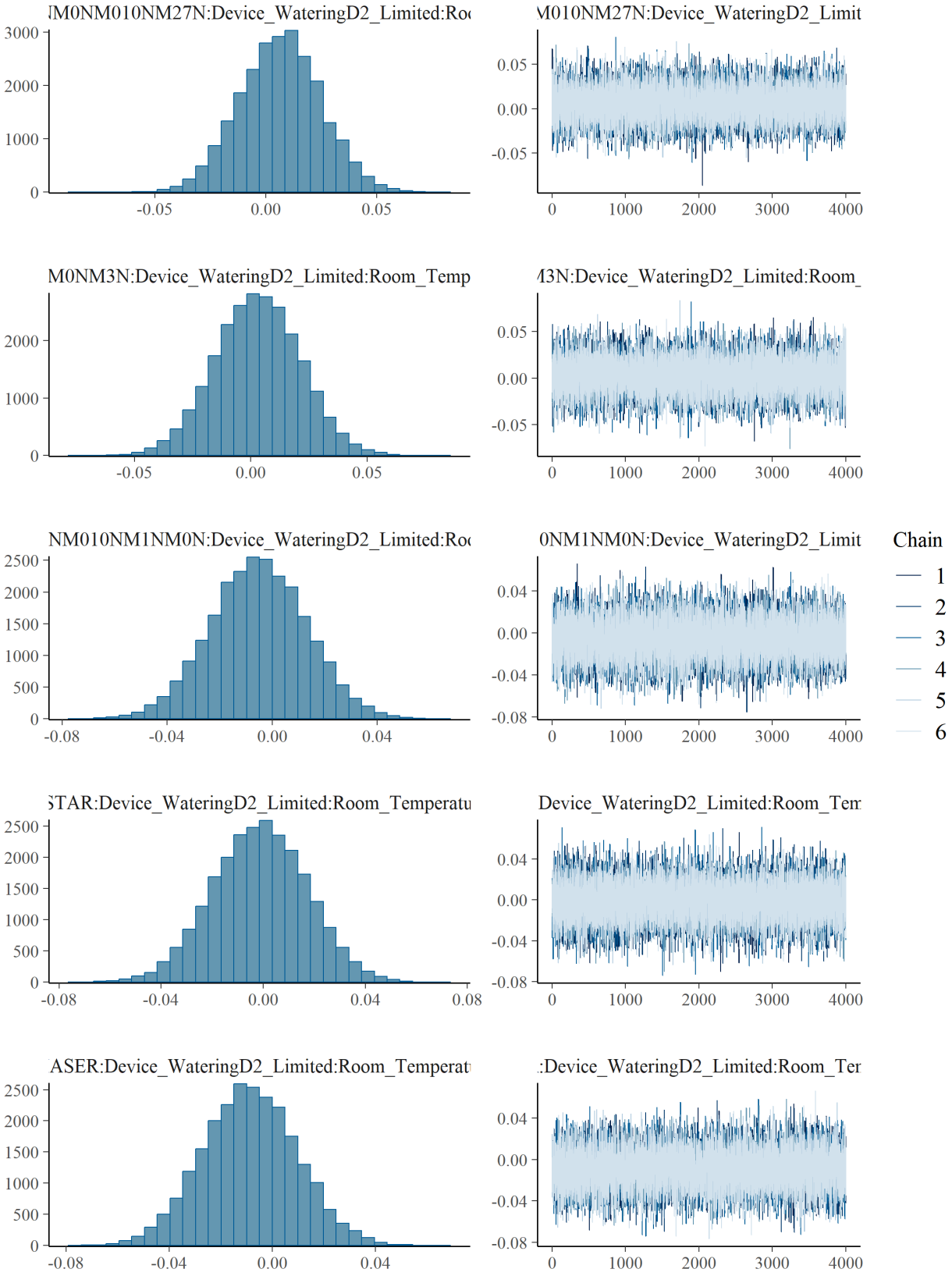


Figure 9: plot of chunk unnamed-chunk-11

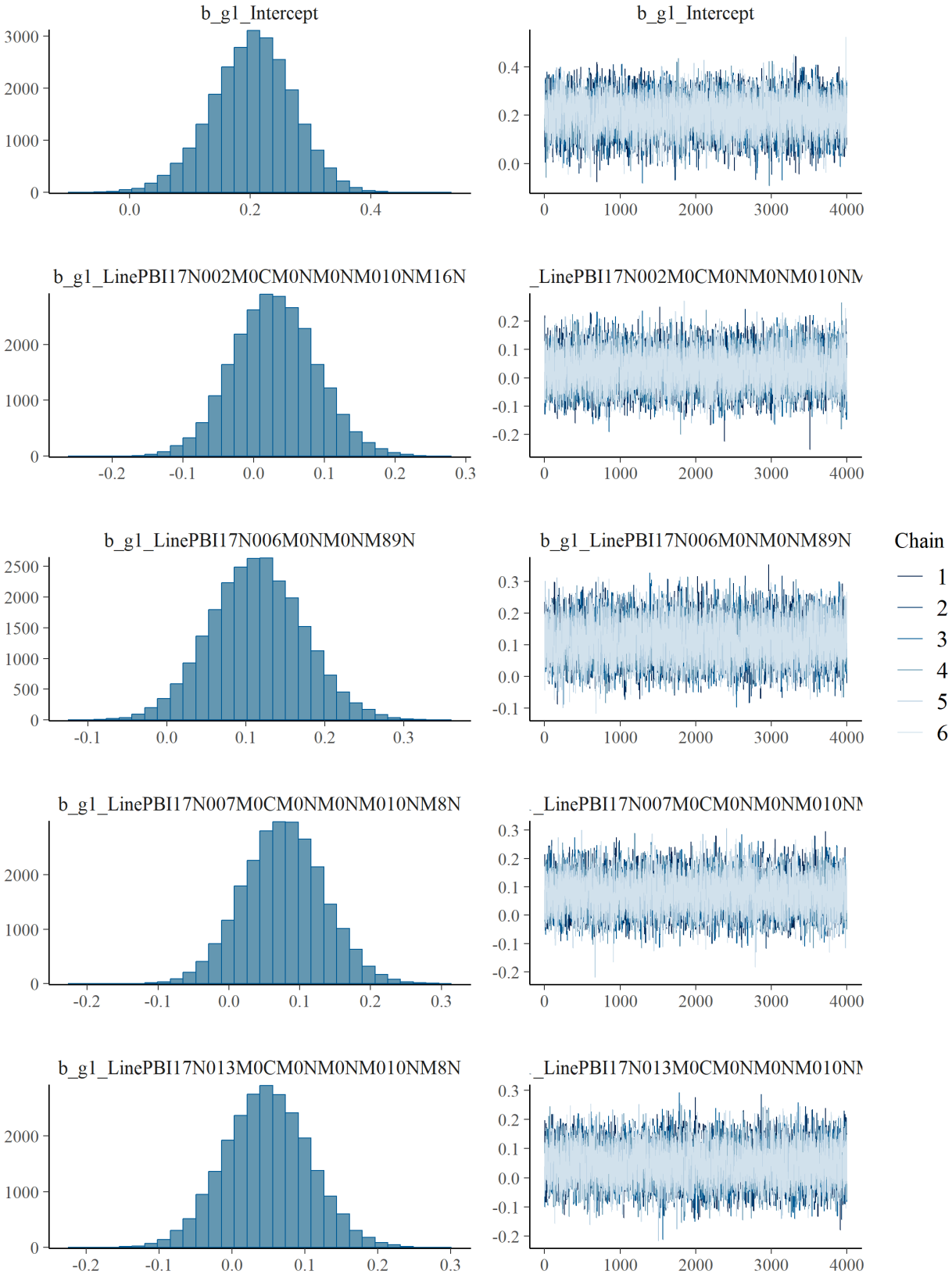


Figure 10: plot of chunk unnamed-chunk-11

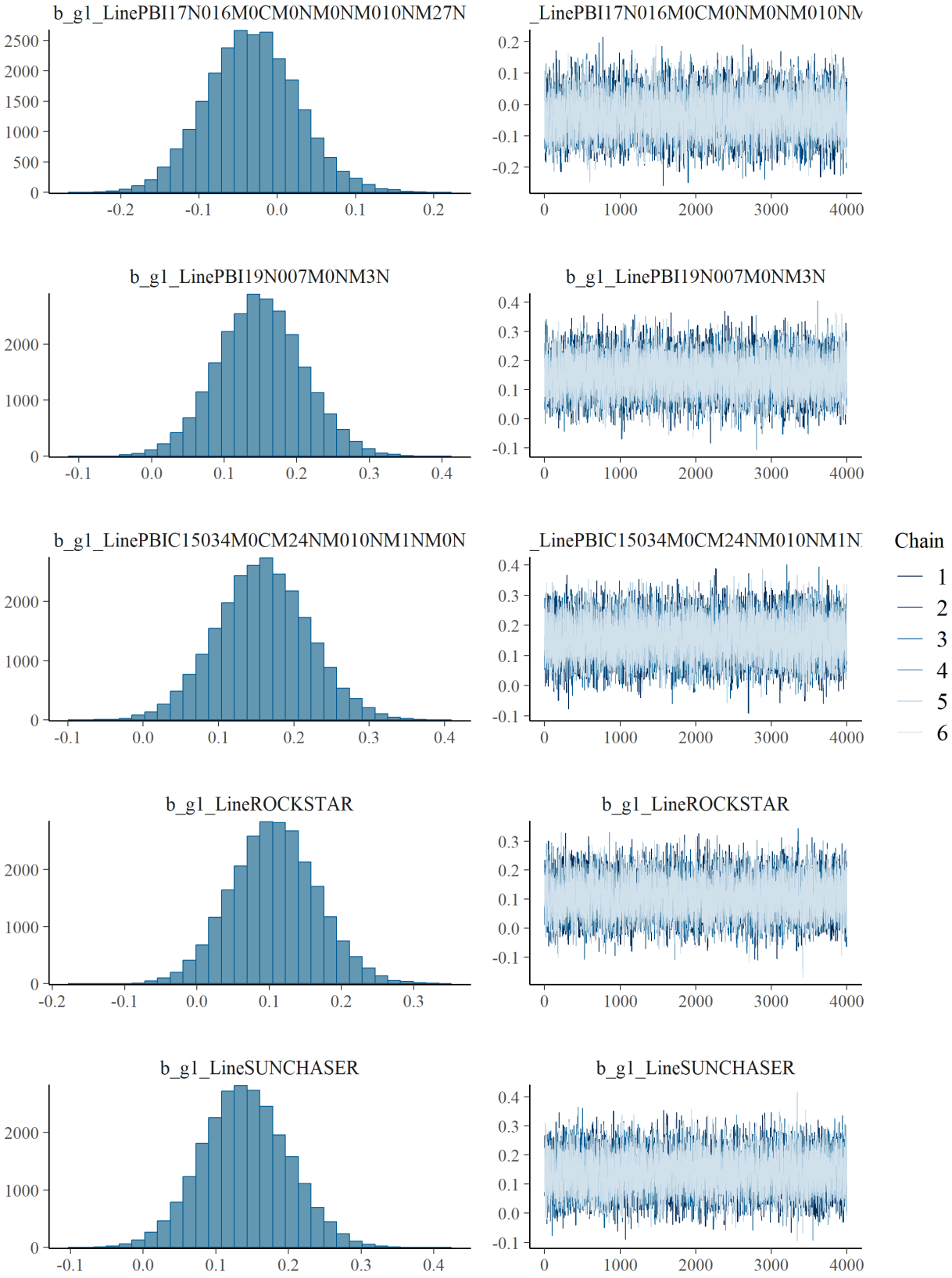


Figure 11: plot of chunk unnamed-chunk-11

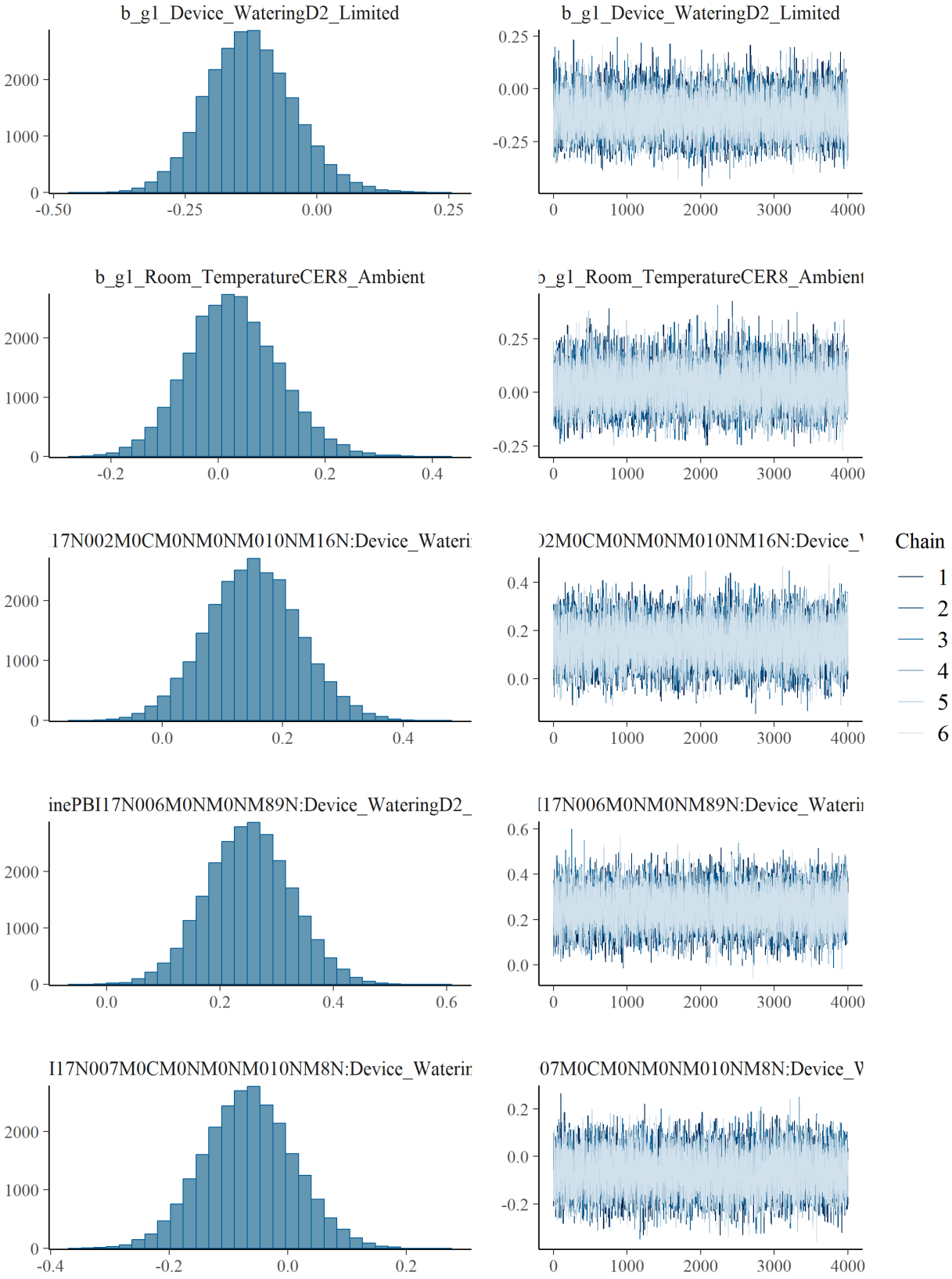


Figure 12: plot of chunk unnamed-chunk-11

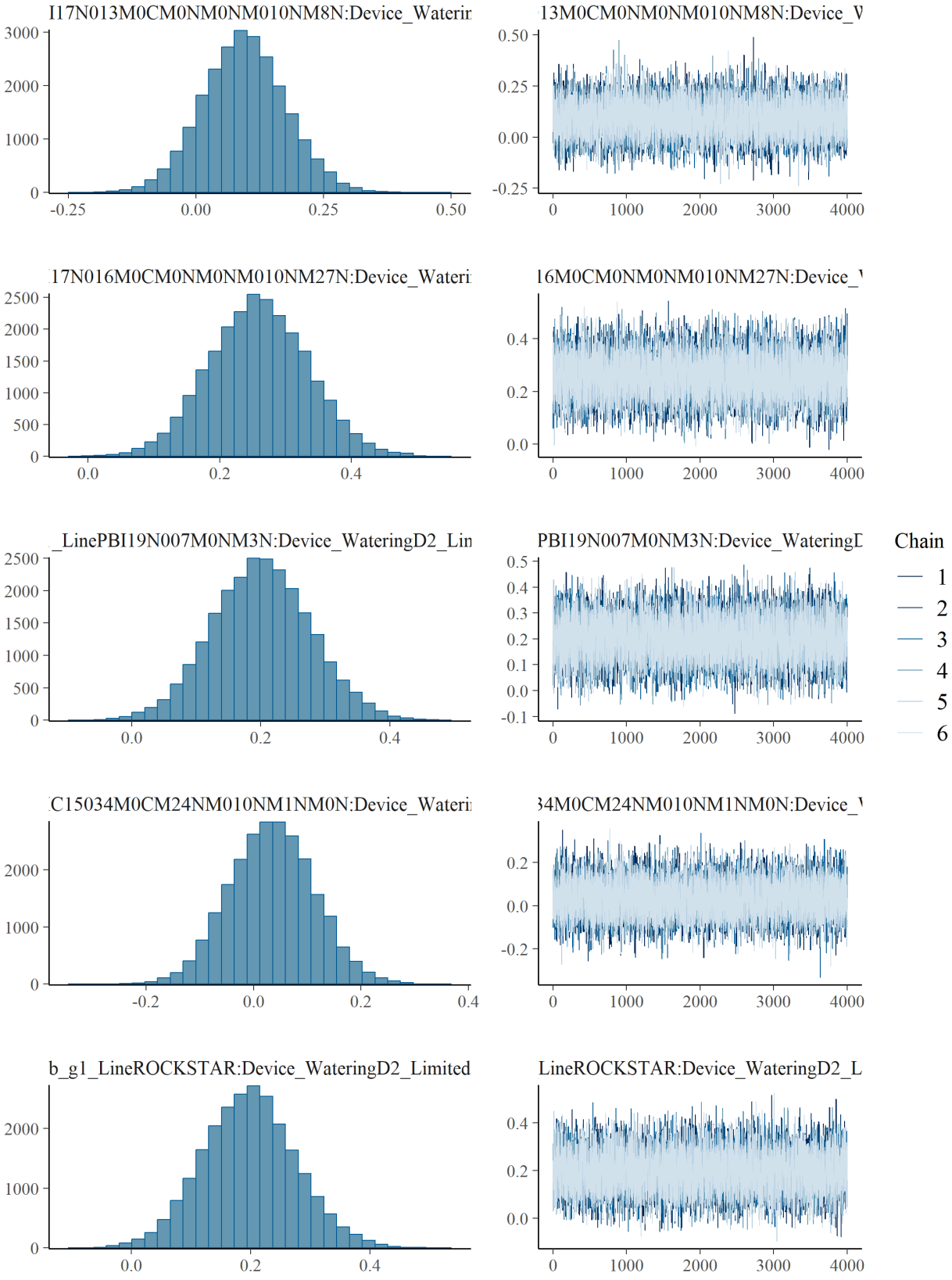


Figure 13: plot of chunk unnamed-chunk-11

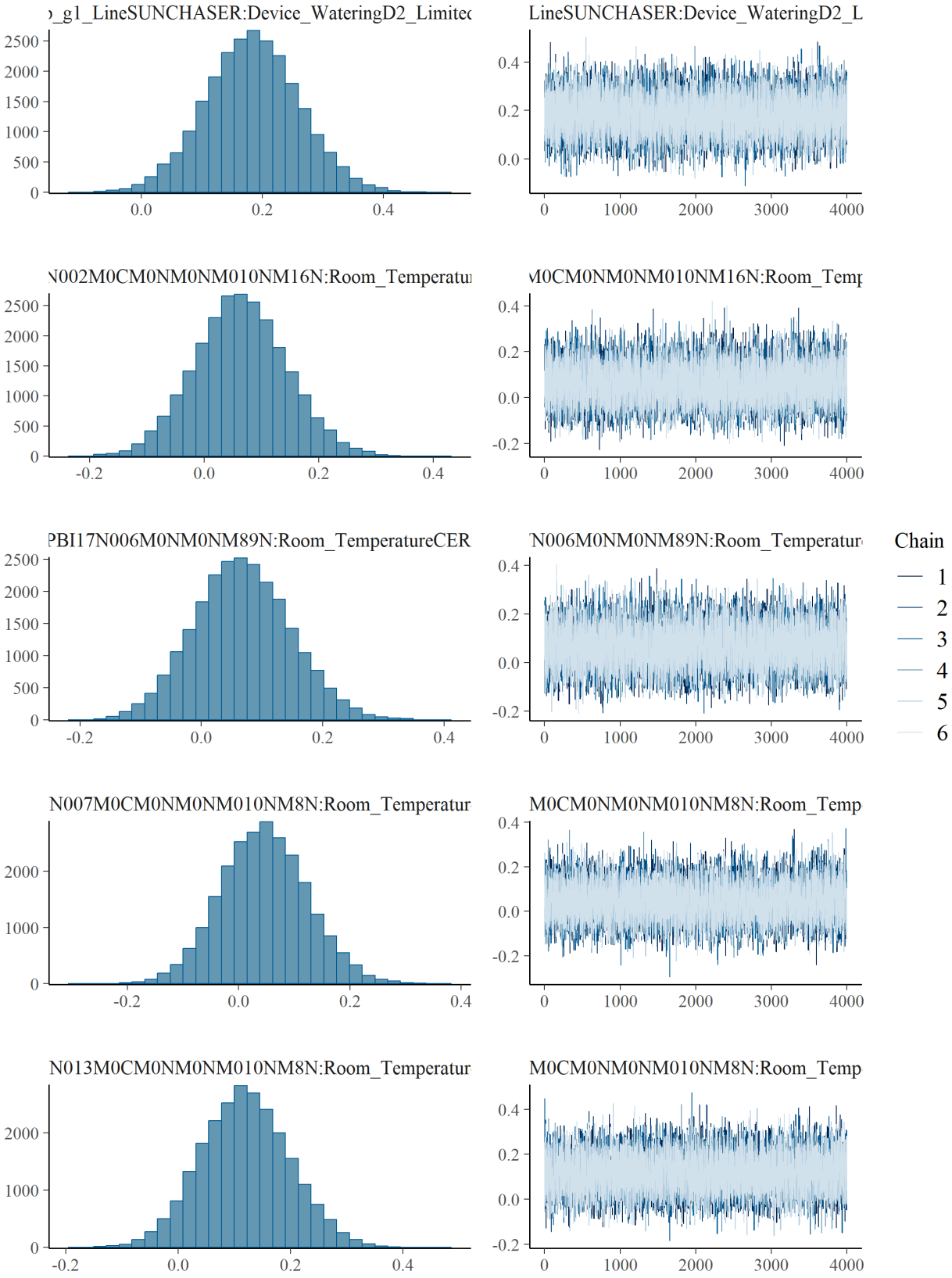


Figure 14: plot of chunk unnamed-chunk-11

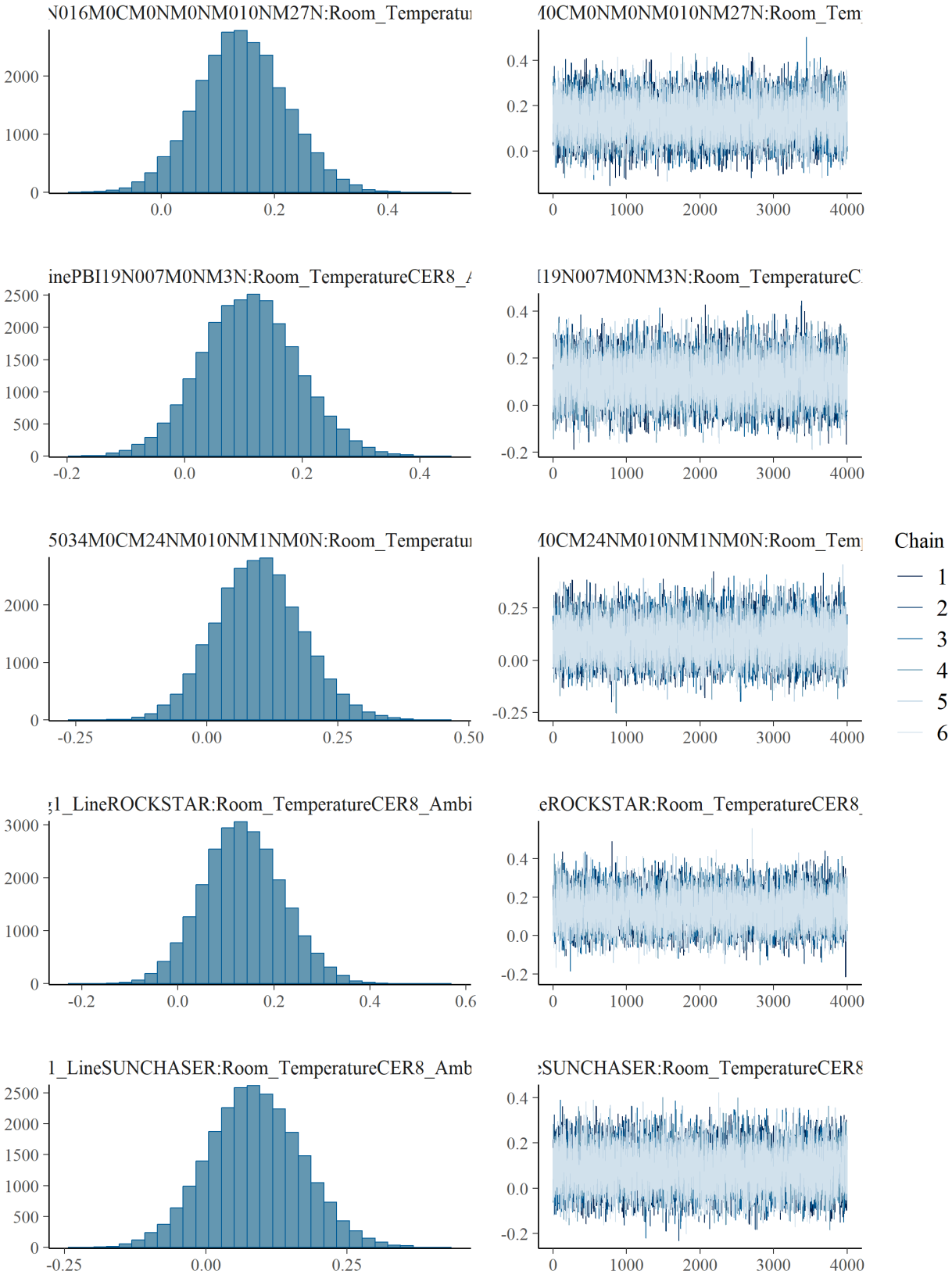


Figure 15: plot of chunk unnamed-chunk-11

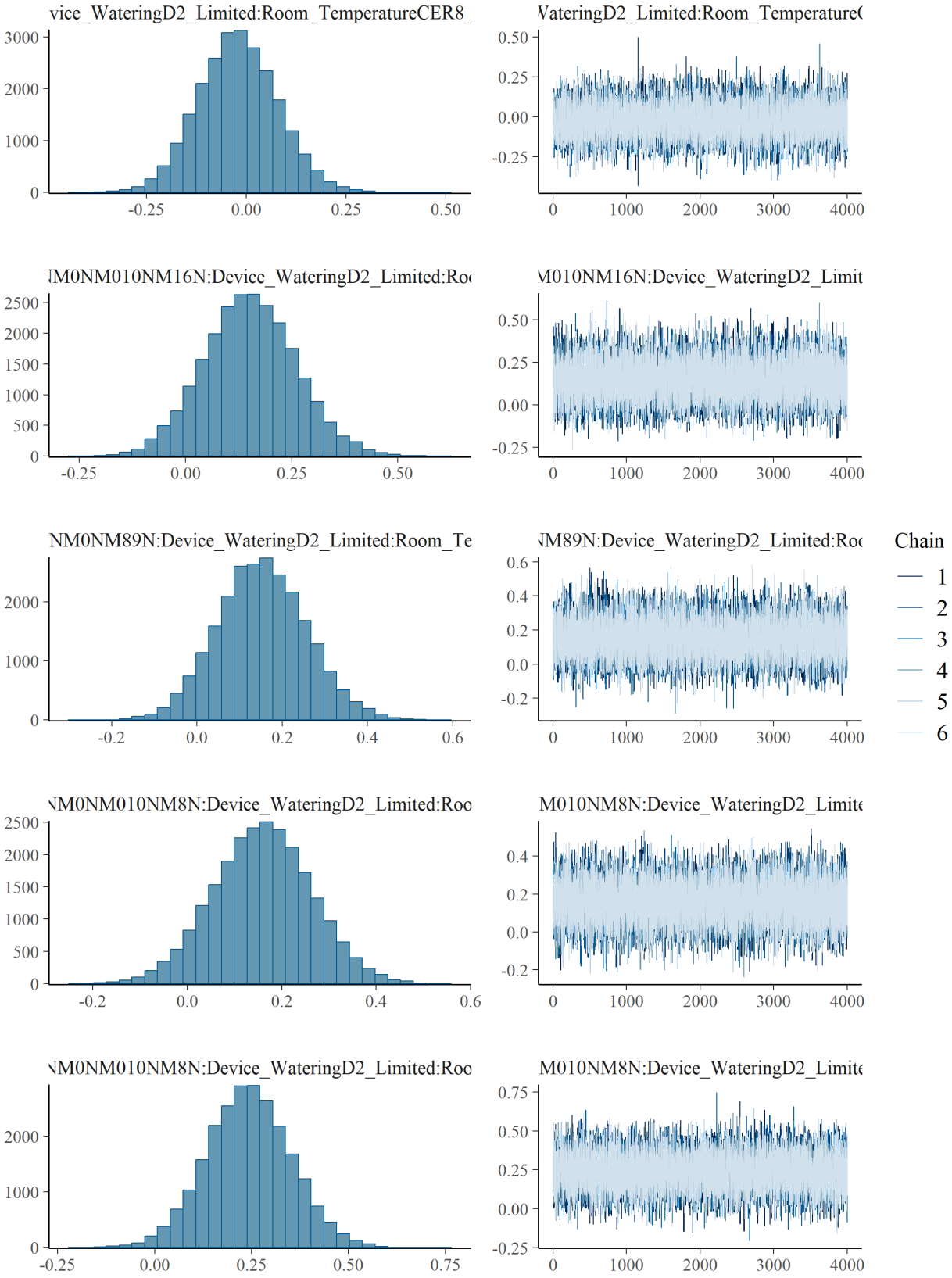


Figure 16: plot of chunk unnamed-chunk-11

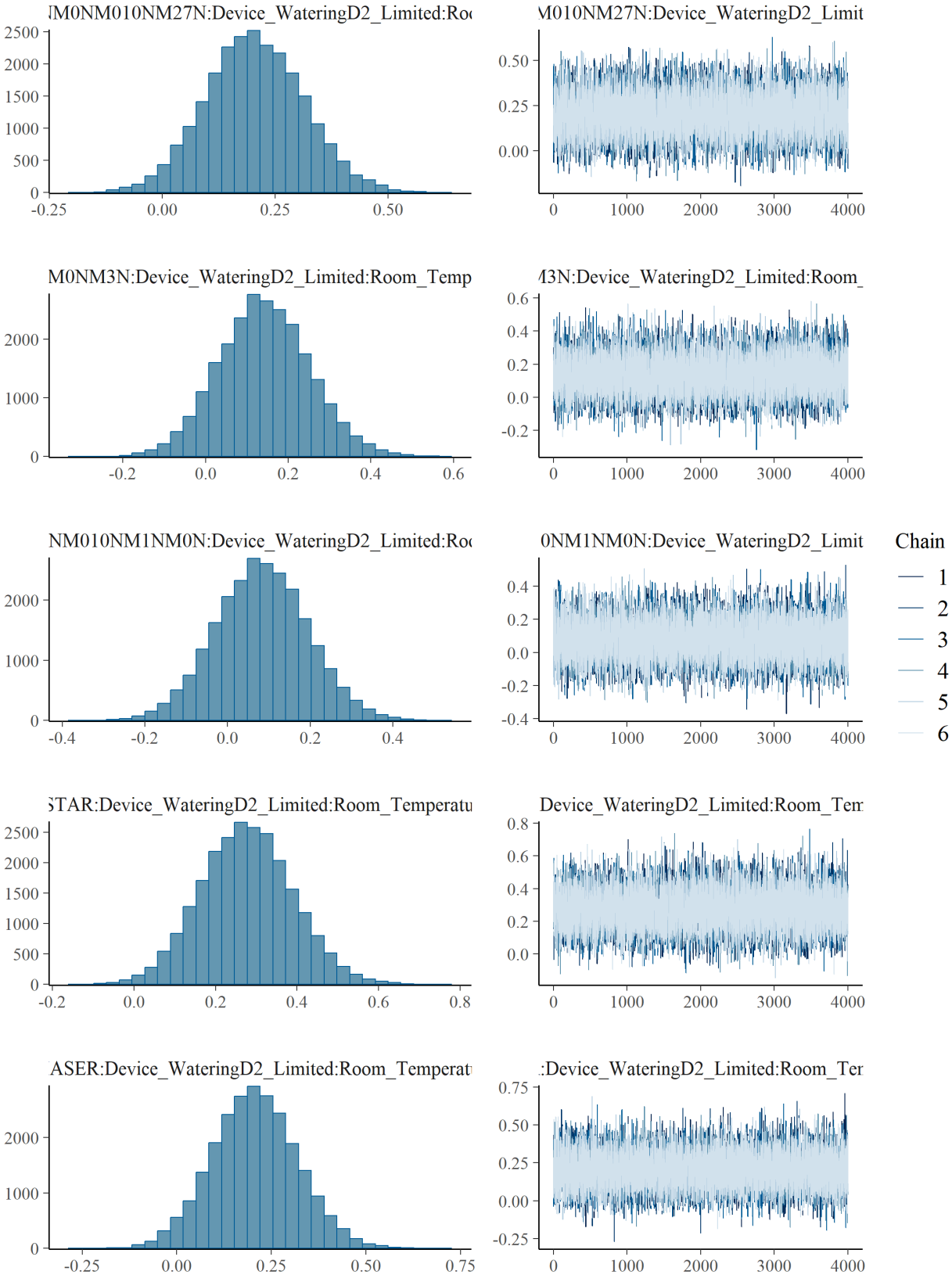


Figure 17: plot of chunk unnamed-chunk-11

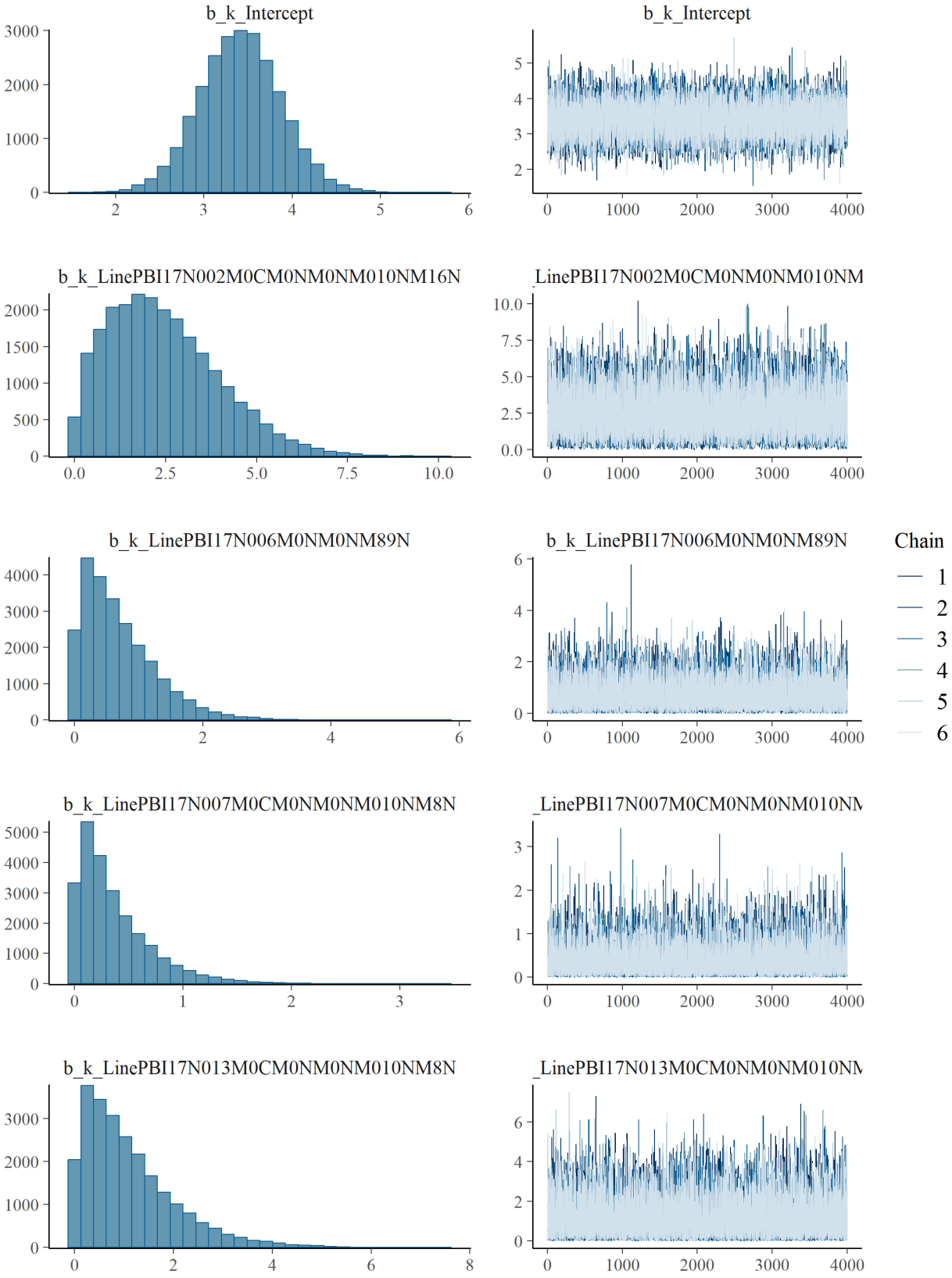


Figure 18: plot of chunk unnamed-chunk-11

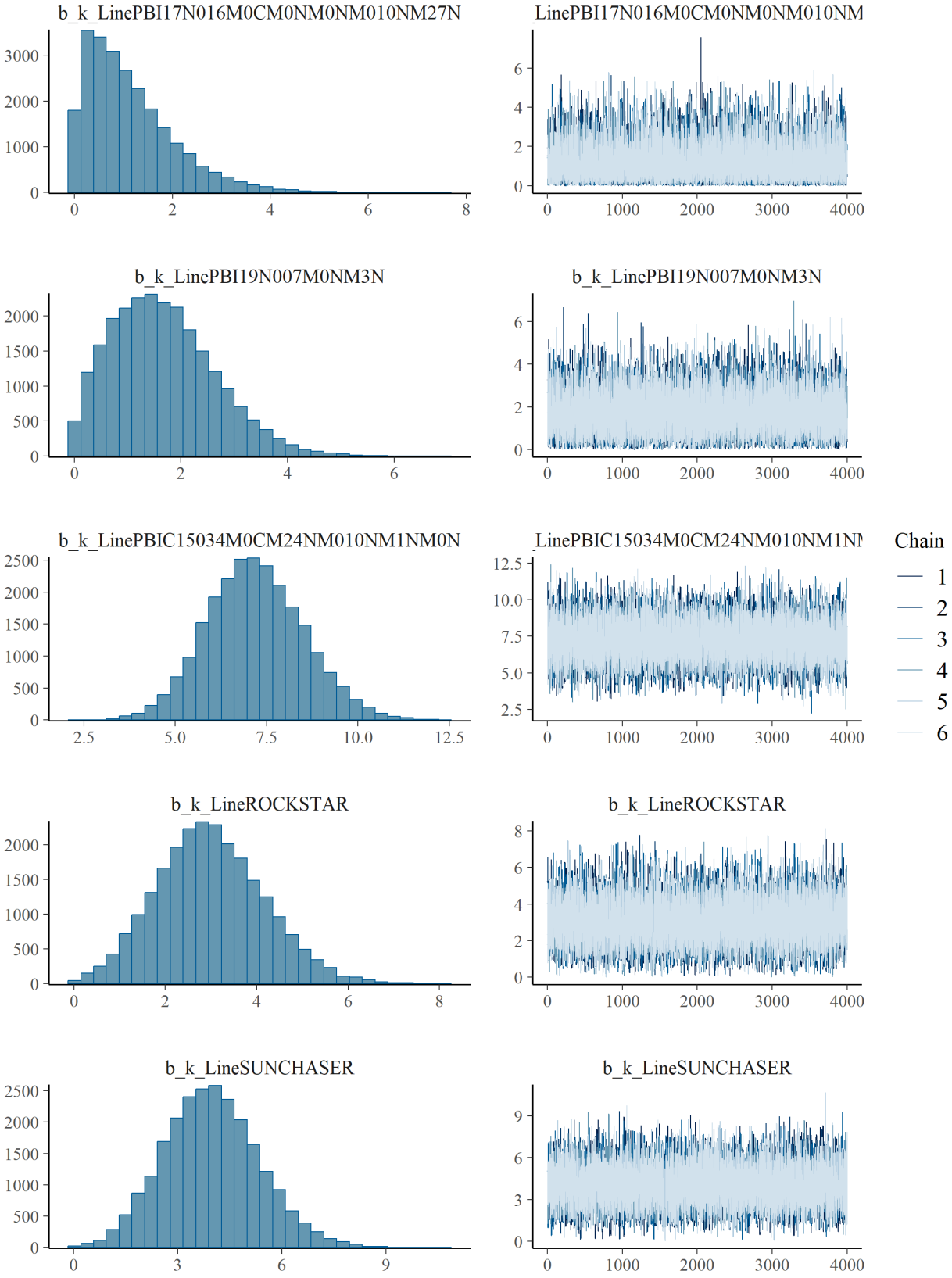


Figure 19: plot of chunk unnamed-chunk-11

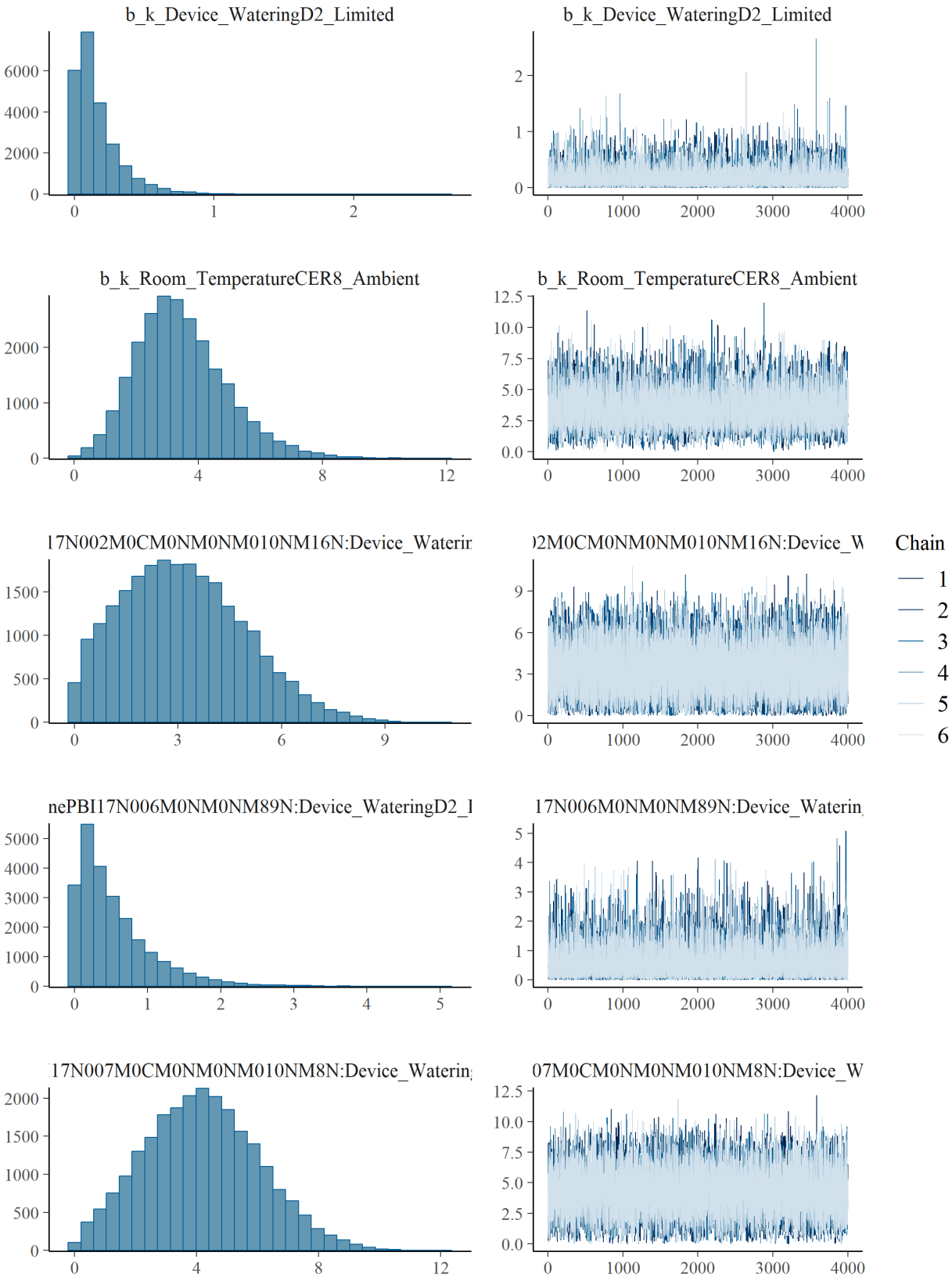


Figure 20: plot of chunk unnamed-chunk-11

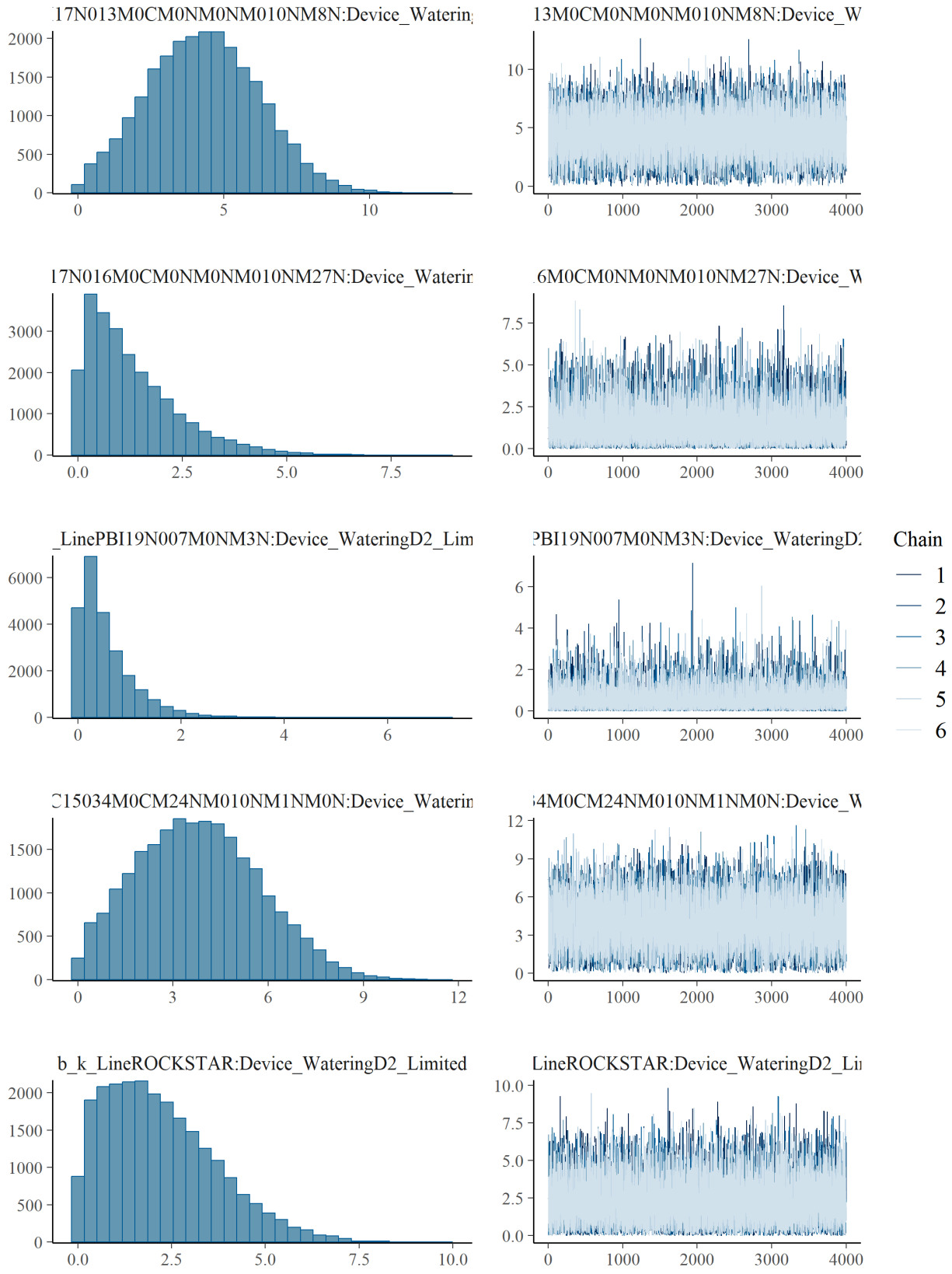


Figure 21: plot of chunk unnamed-chunk-11

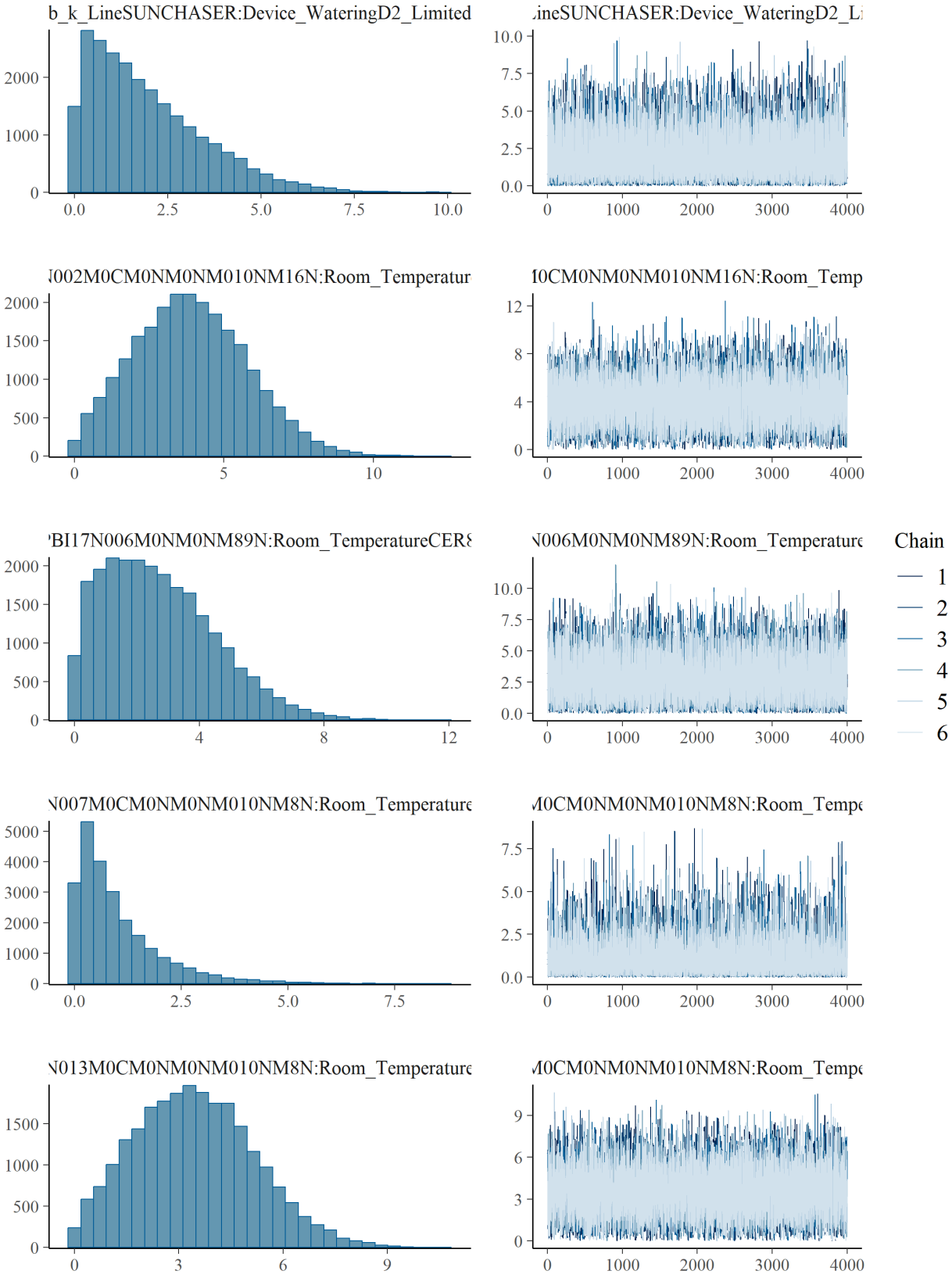


Figure 22: plot of chunk unnamed-chunk-11

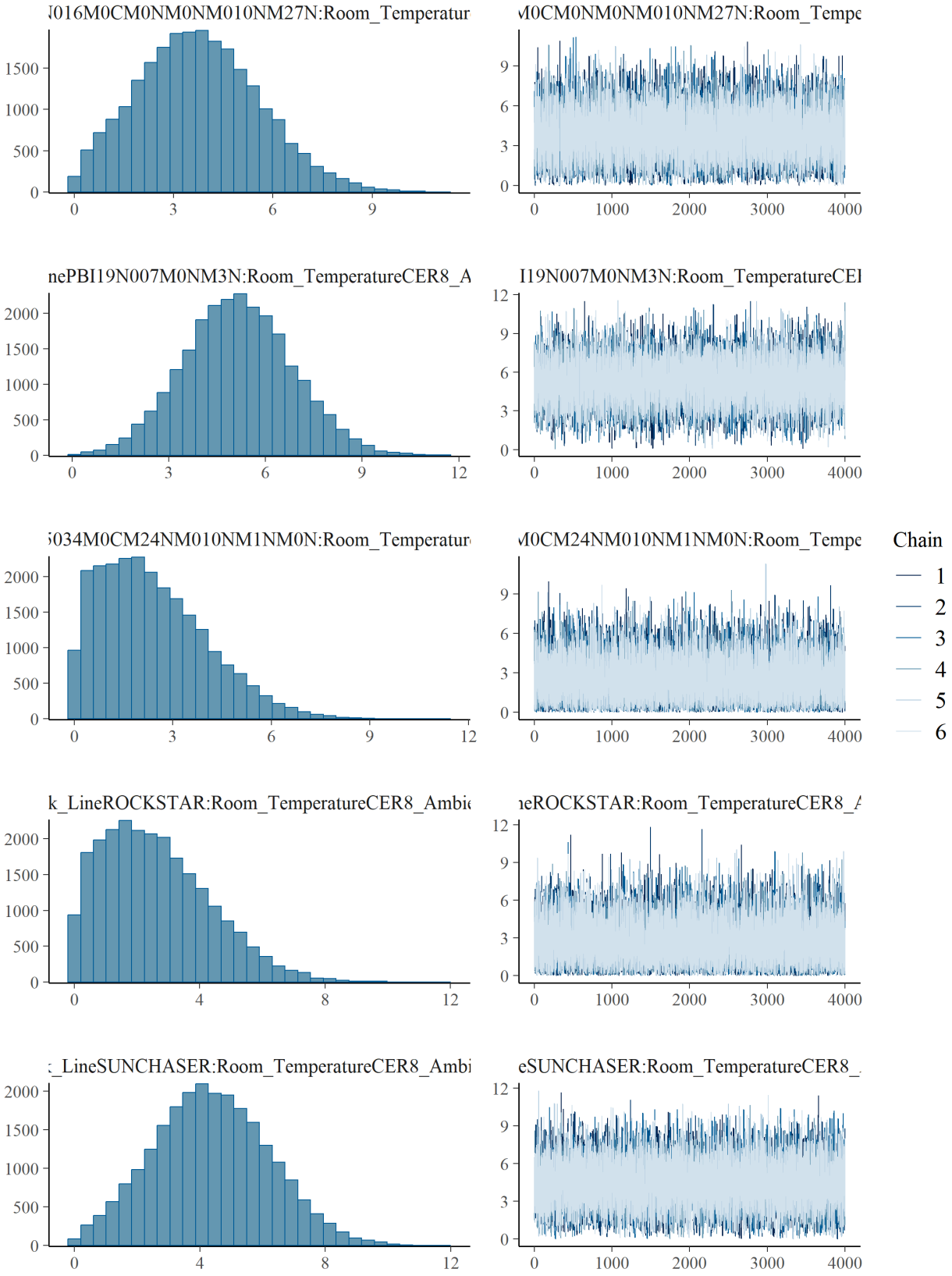


Figure 23: plot of chunk unnamed-chunk-11

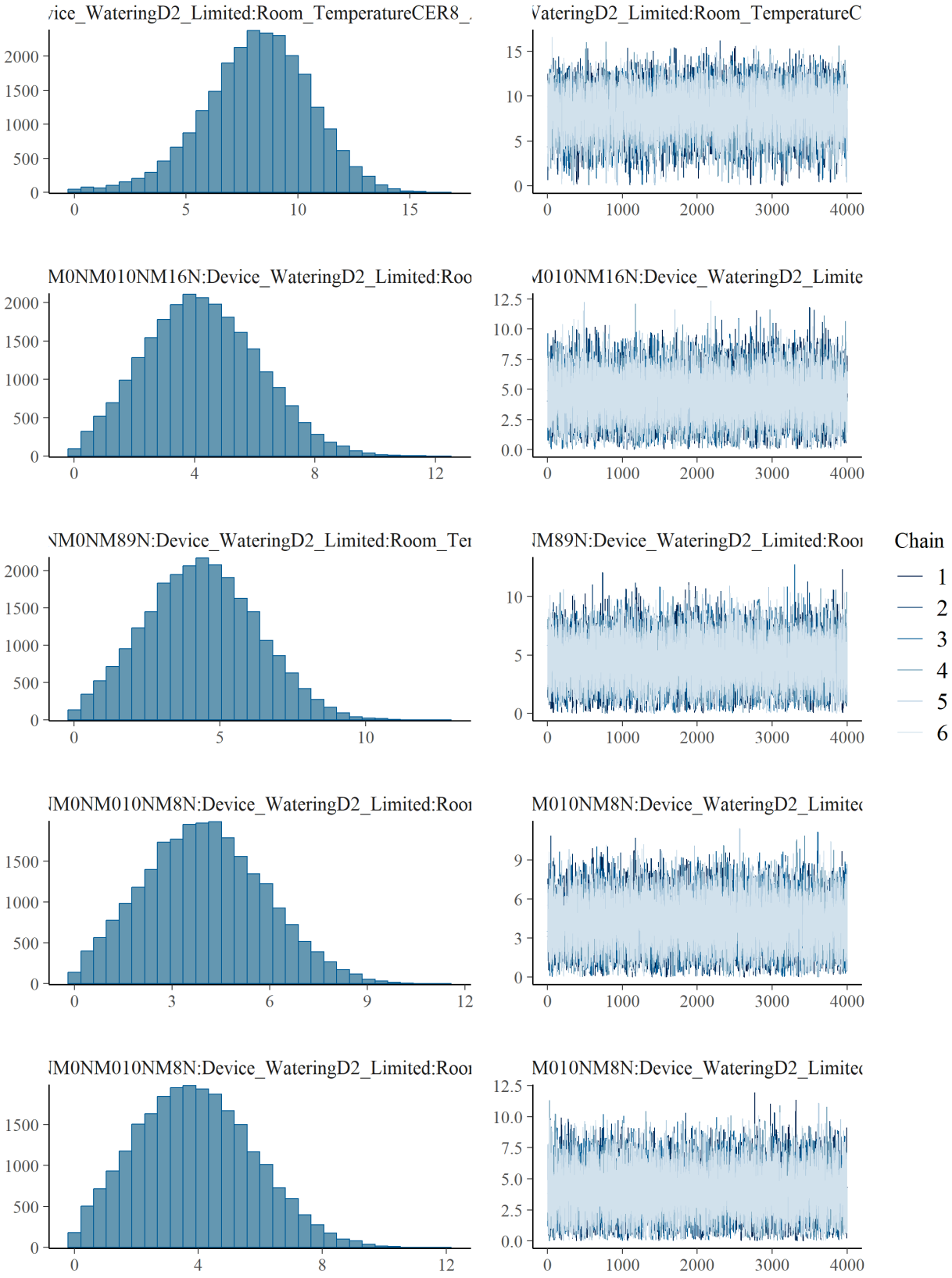


Figure 24: plot of chunk unnamed-chunk-11

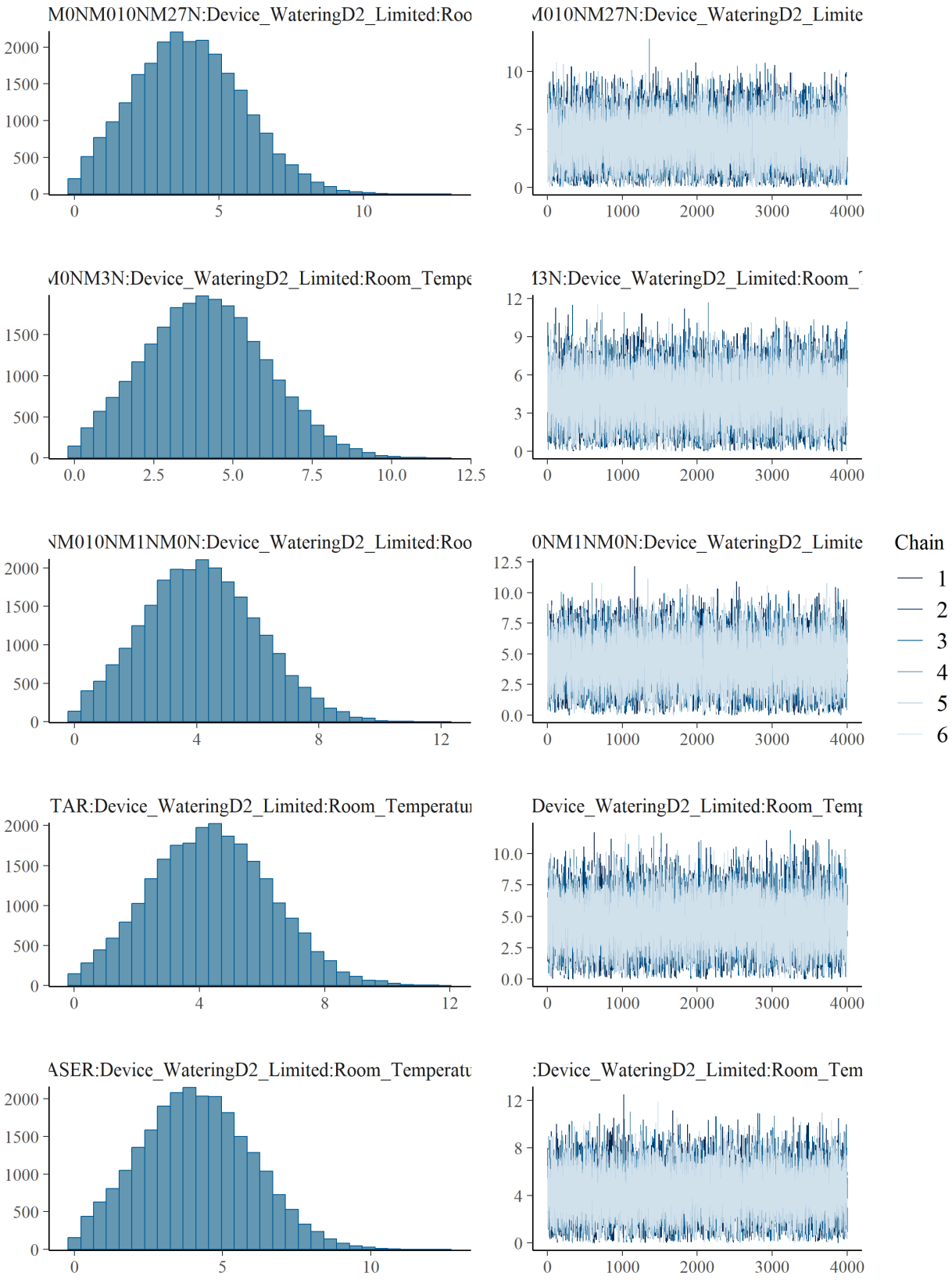


Figure 25: plot of chunk unnamed-chunk-11

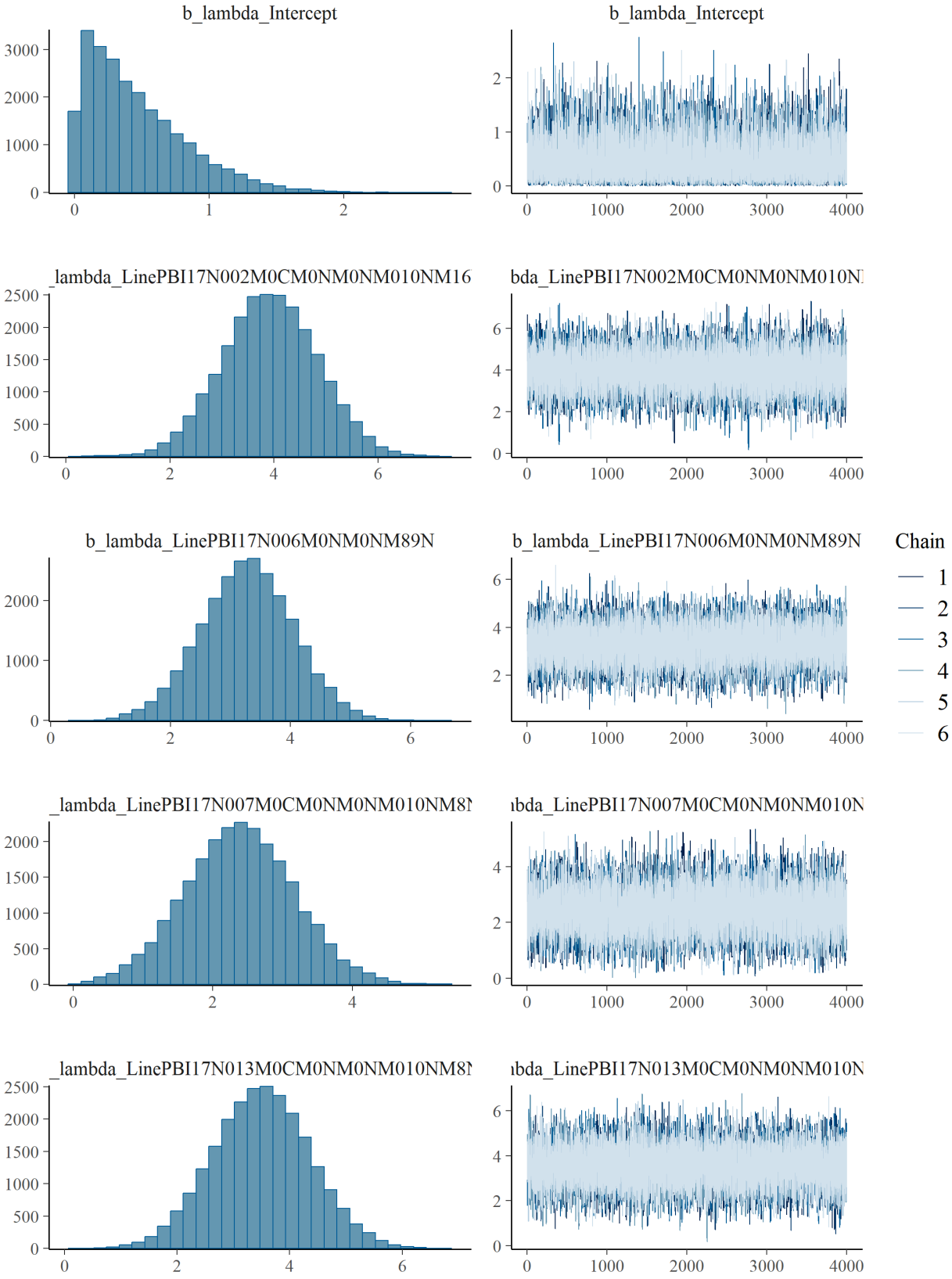


Figure 26: plot of chunk unnamed-chunk-11

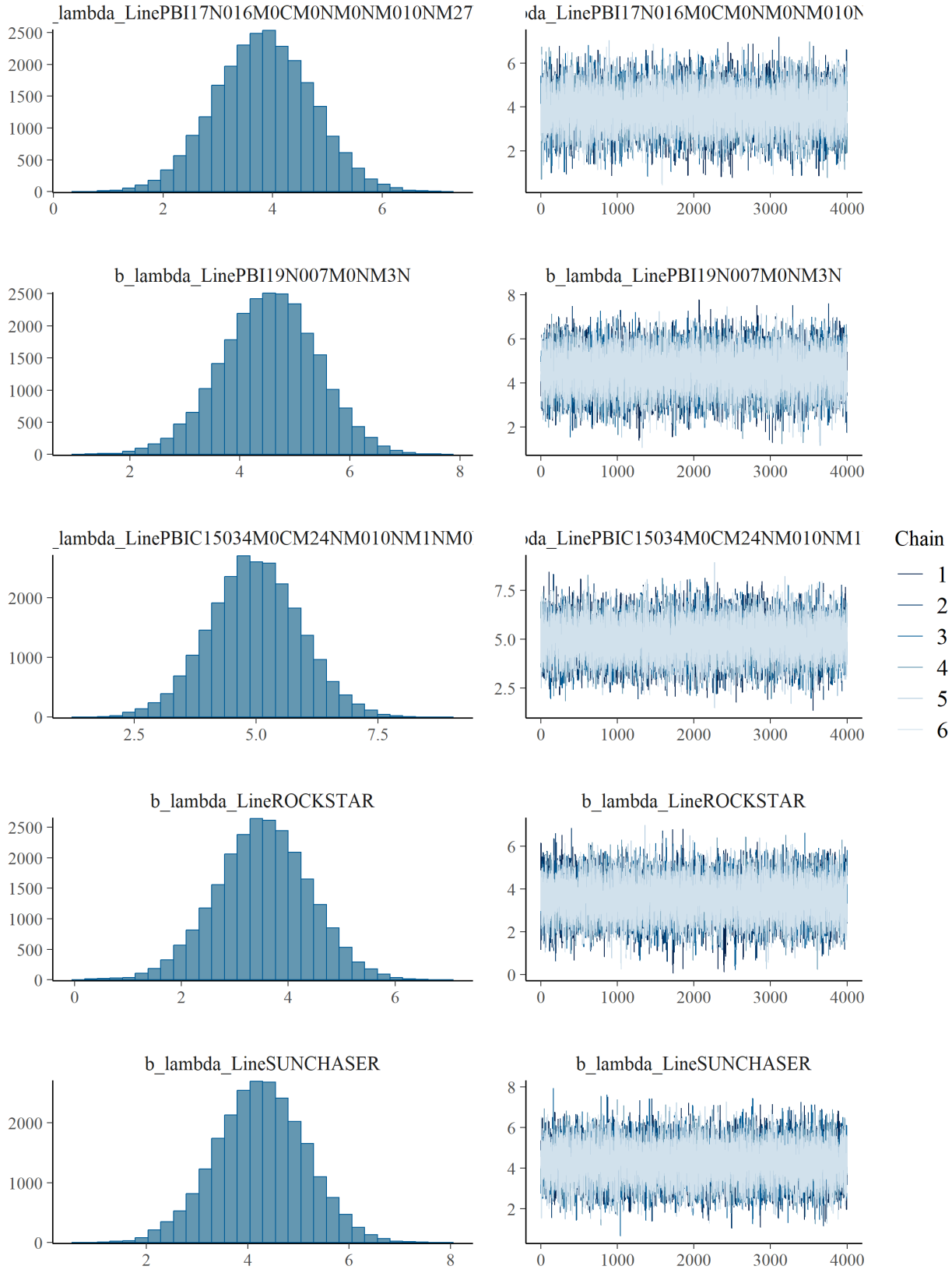


Figure 27: plot of chunk unnamed-chunk-11

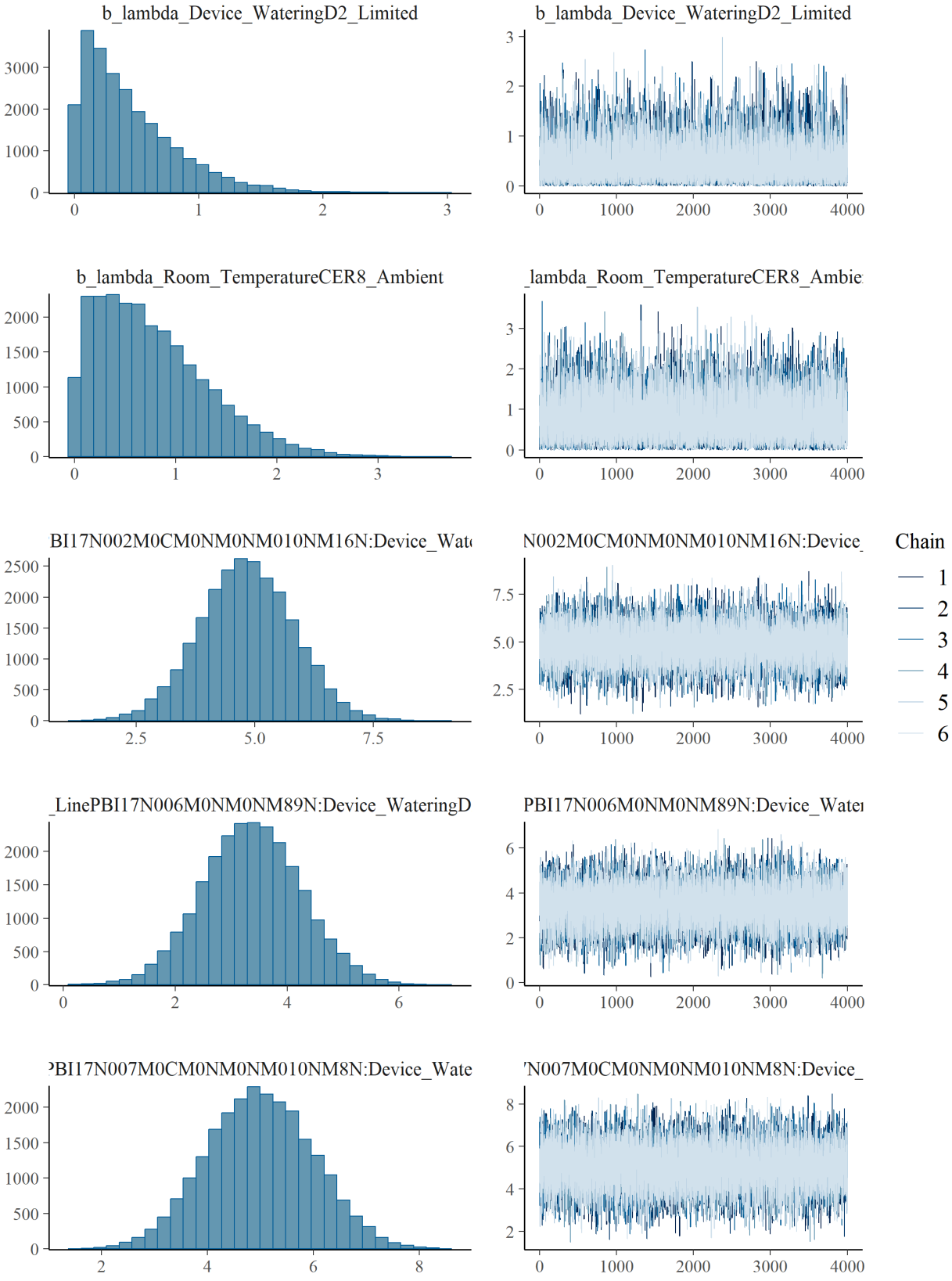


Figure 28: plot of chunk unnamed-chunk-11

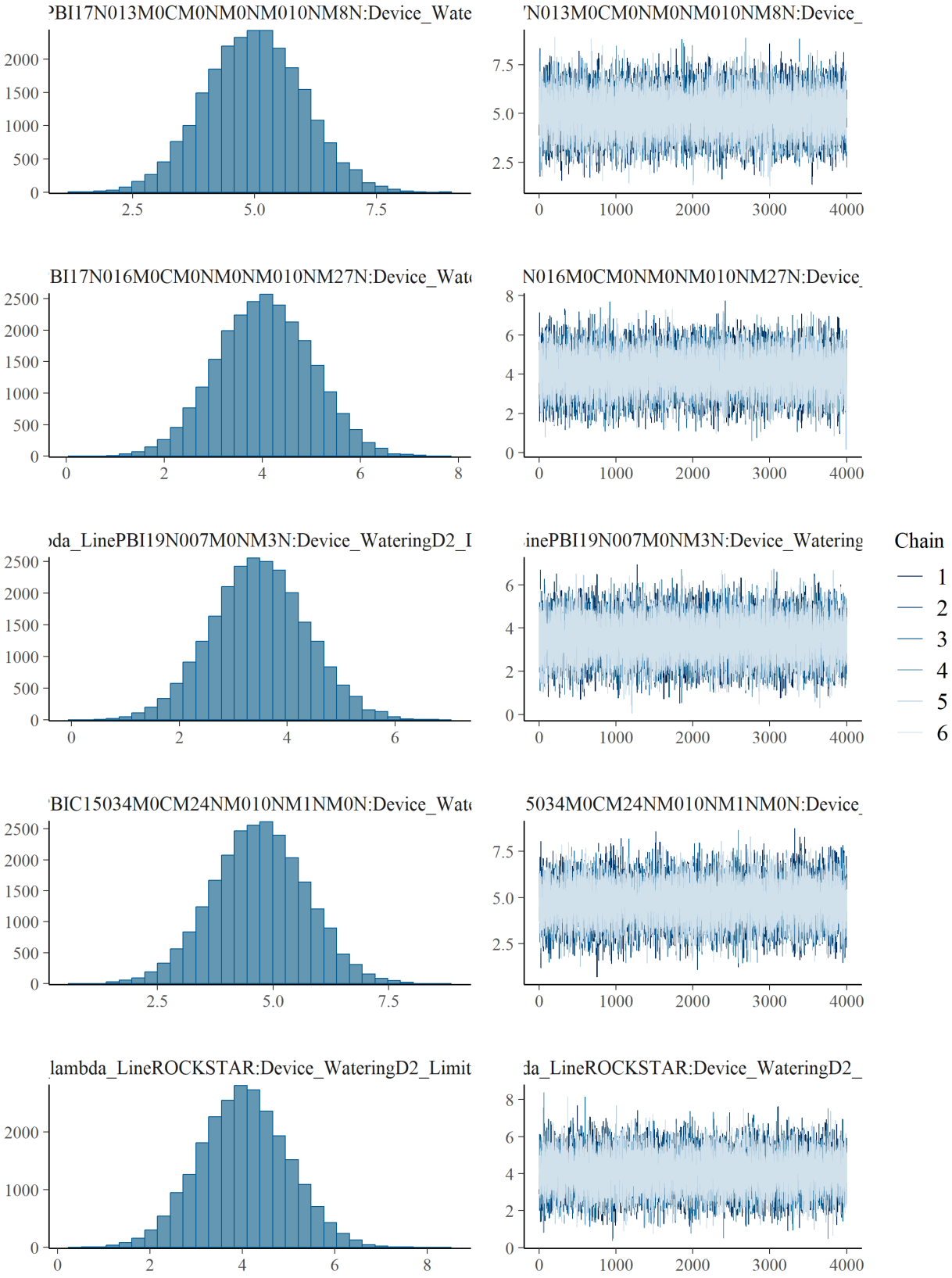


Figure 29: plot of chunk unnamed-chunk-11

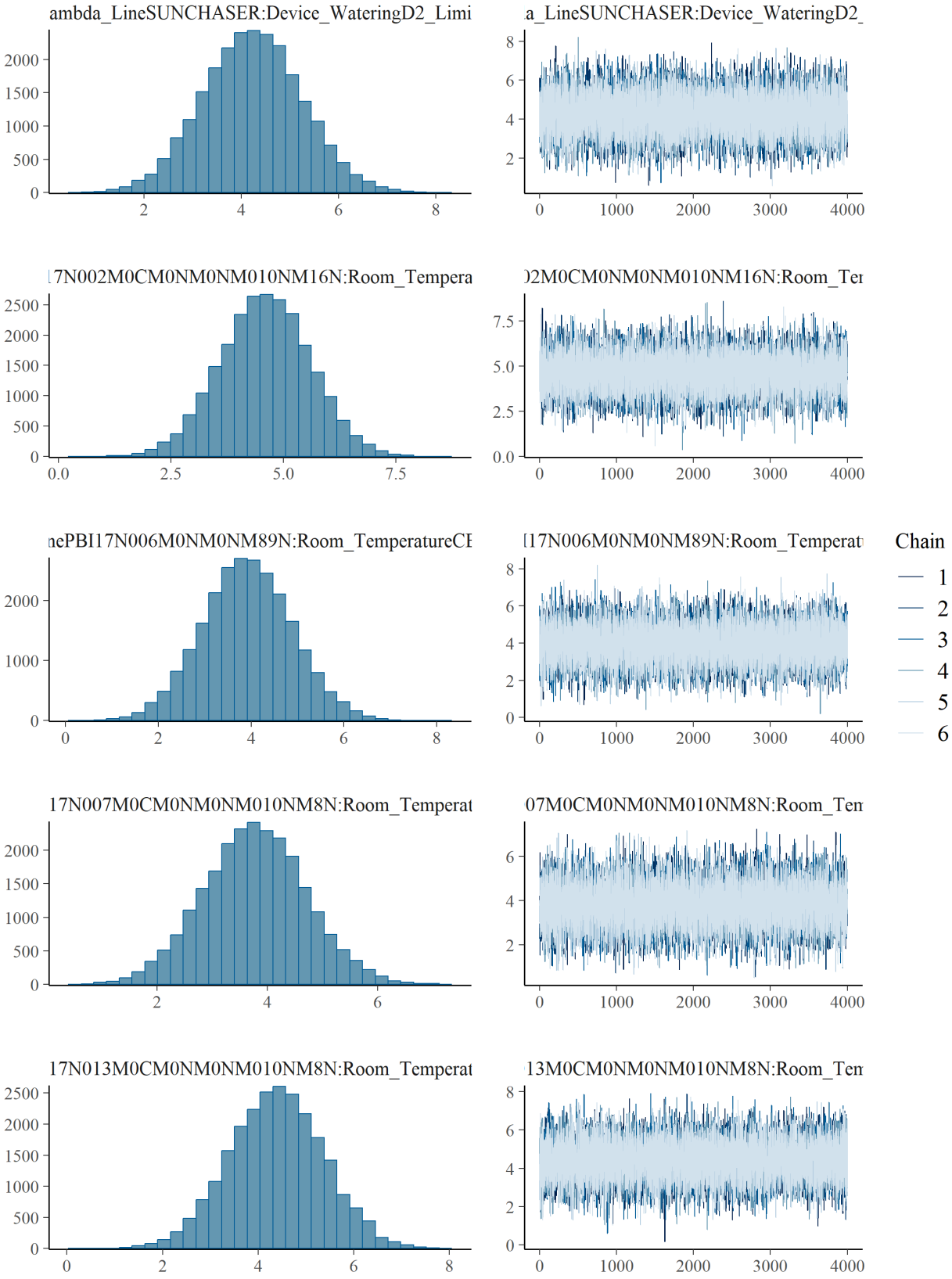


Figure 30: plot of chunk unnamed-chunk-11

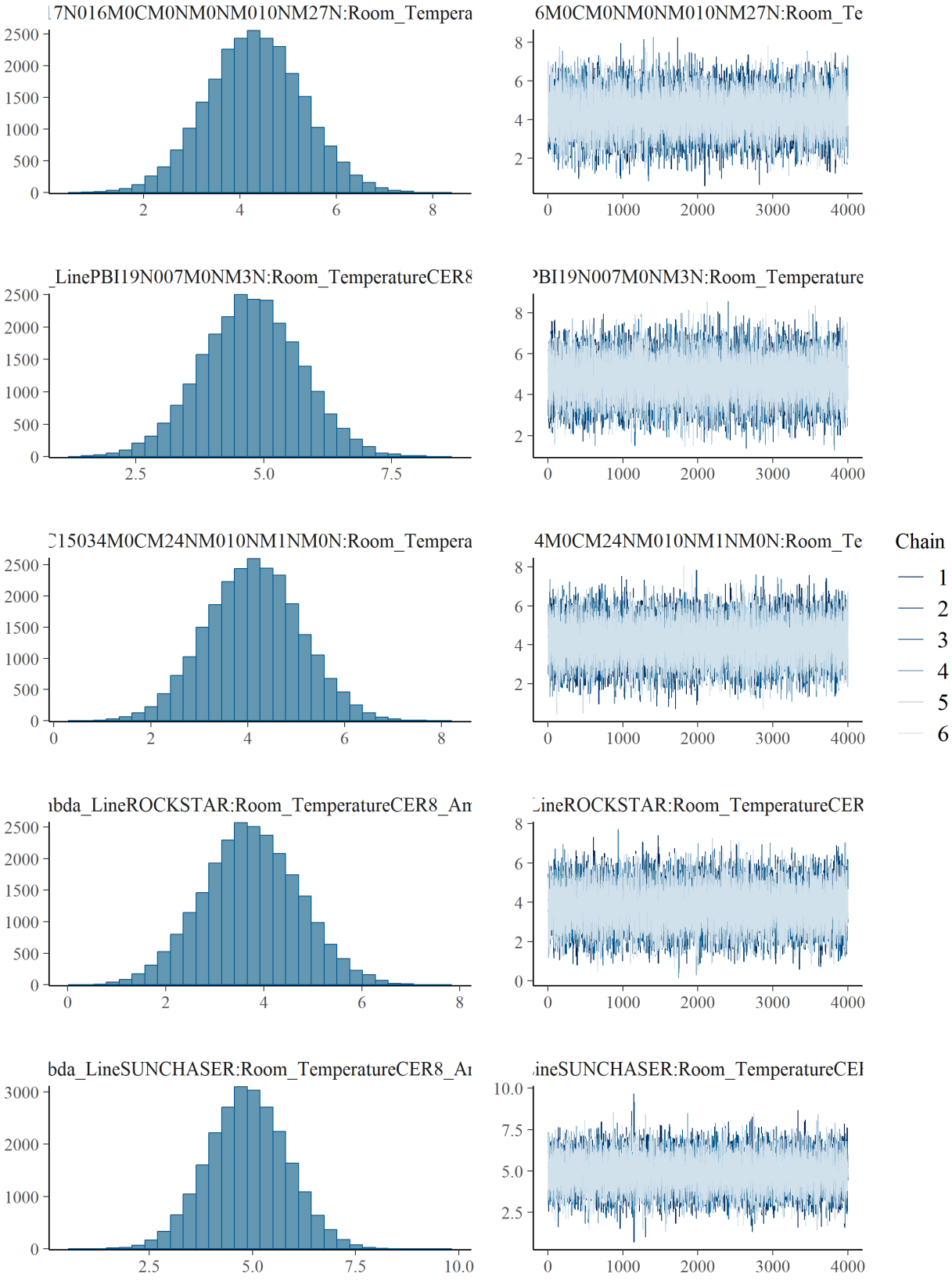


Figure 31: plot of chunk unnamed-chunk-11

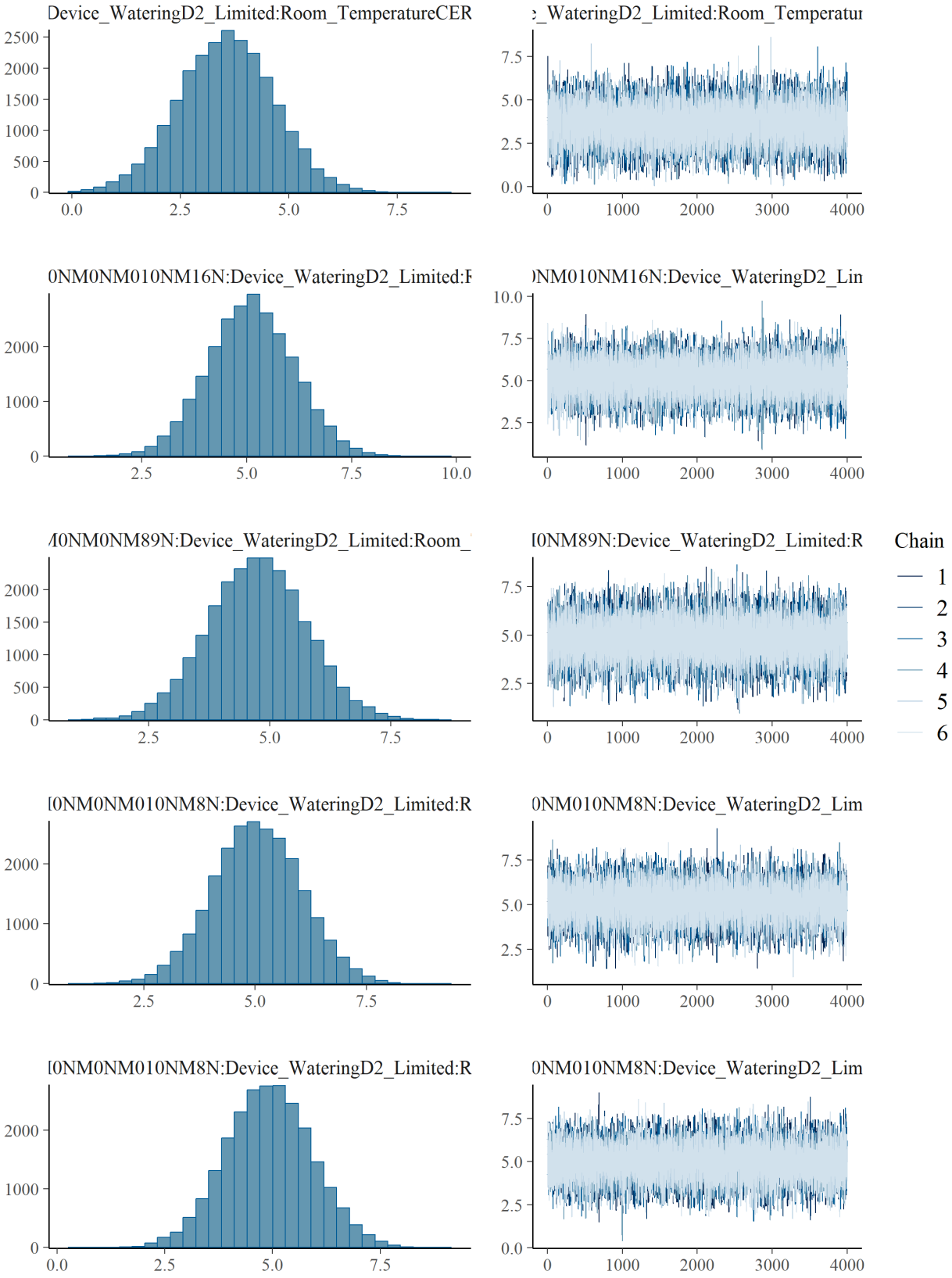


Figure 32: plot of chunk unnamed-chunk-11

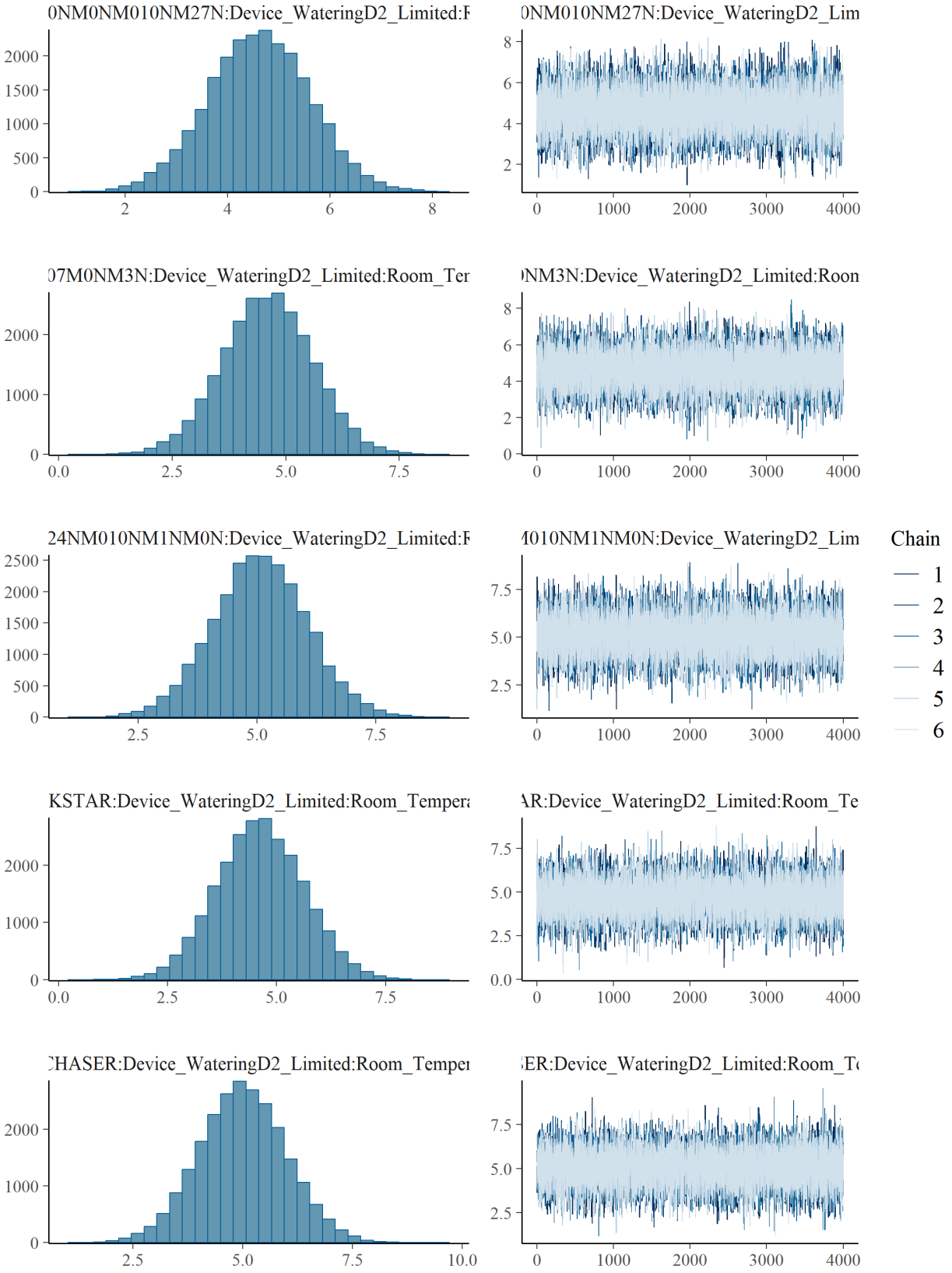


Figure 33: plot of chunk unnamed-chunk-11

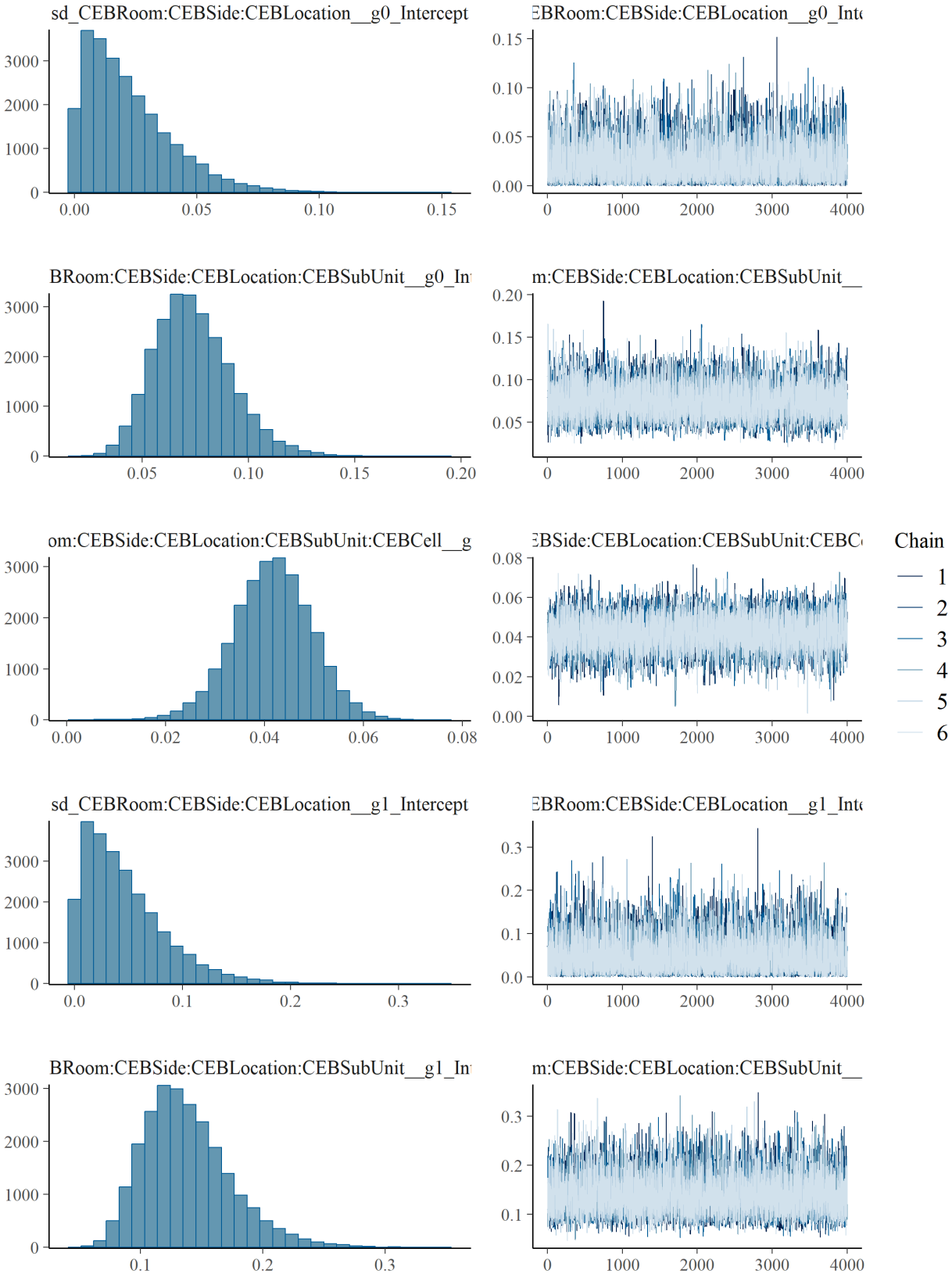


Figure 34: plot of chunk unnamed-chunk-11

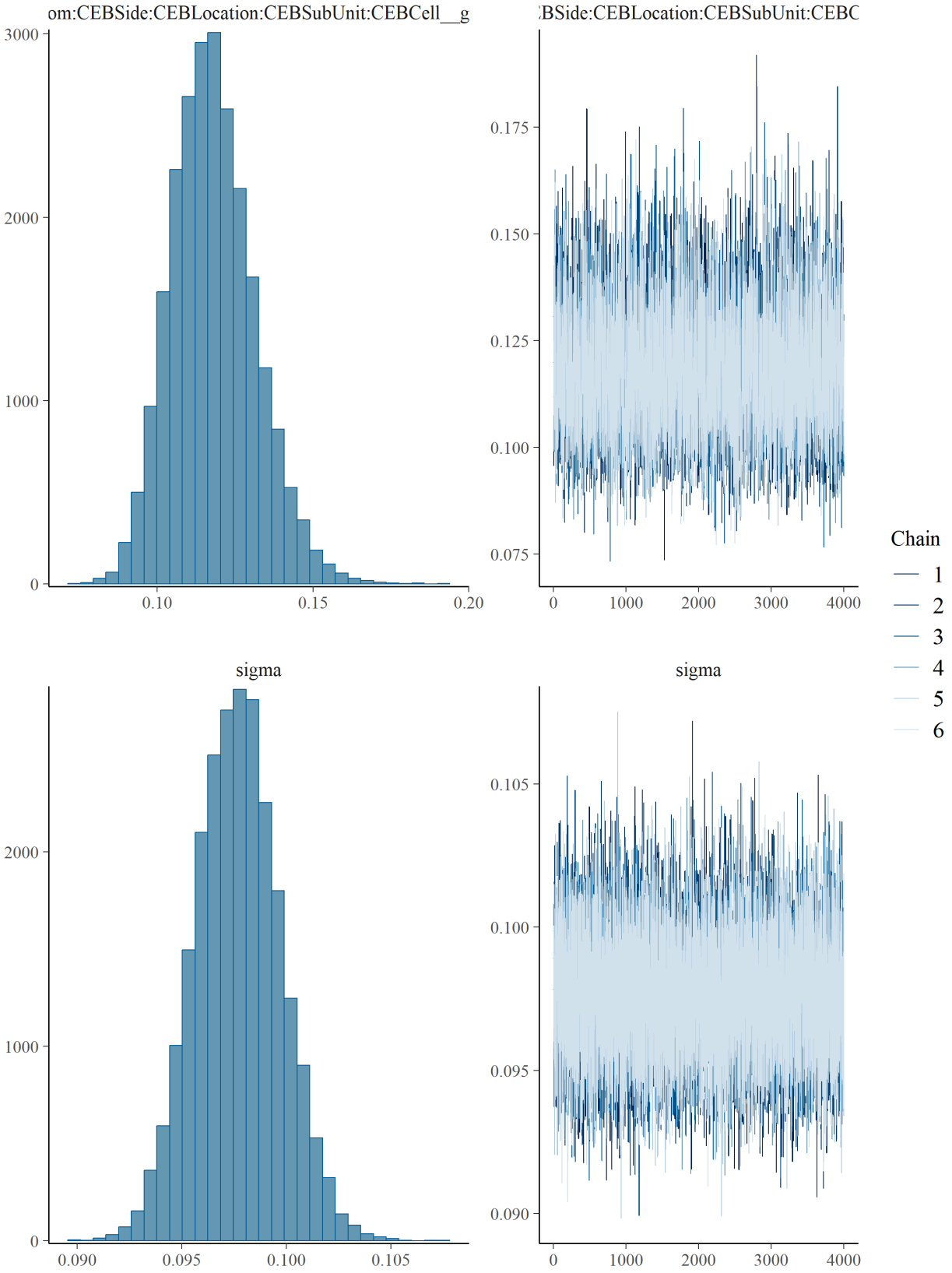


Figure 35: plot of chunk unnamed-chunk-11

```
E-BFMI indicated no pathological behavior.
```

```
loo(fit_on_max)
```

```
Computed from 24000 by 1682 log-likelihood matrix.

         Estimate   SE
elpd_loo   1406.0 39.7
p_loo       210.1  9.1
looic     -2812.0 79.4
------
MCSE of elpd_loo is 0.2.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.5, 1.6]).

All Pareto k estimates are good (k < 0.7).
See help('pareto-k-diagnostic') for details.
```

```
pp_check(fit_on_max)  # compares density of the observed data with data generated from posterior predic
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

```
pp_check(fit_on_max, type = "error_hist", ndraws = 12)
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
pp_check(fit_on_max, type = "scatter_avg", ndraws = 1000)
```

```
pp_check(fit_on_max, type = "stat_2d")
```

```
Using all posterior draws for ppc type 'stat_2d' by default.

Note: in most cases the default test statistic 'mean' is too weak to detect anything of interest.
```

```
pp_check(fit_on_max, type = "loo_ribbon")
```

```
Using all posterior draws for ppc type 'loo_ribbon' by default.
```

```
plot(gsw~time.elapsed, li600.on.dat, pch=16)
points(fitted(fit_on_max)[,1]~time.elapsed, li600.on.dat, pch=16, col="red")
```

```
##make the above plot prettier
li600.on.plot <- li600.on.dat
li600.on.plot$fitted <- fitted(fit_on_max)[,1]
li600.on.plot$fitted.fixed <- fitted(fit_on_max, re_formula = NA)[,1]

ggplot(li600.on.plot, aes(x= time.adj, y = gsw, group = Replicate)) +
  geom_point() +
  facet_grid(Room_Temperature + Device_Watering ~ Line) +
  geom_point(aes(y=fitted), colour = "red") +
  geom_line(aes(y=fitted), colour = "red")
```

```
ggplot(li600.on.plot, aes(x= time.adj, y = gsw, group = Replicate)) +
  geom_point() +
  facet_grid(Room_Temperature + Device_Watering ~ Line) +
  geom_point(aes(y=fitted), colour = "red") +
  geom_line(aes(y=fitted), colour = "red") +
  geom_point(aes(y=fitted.fixed), colour = "blue") +
  geom_line(aes(y=fitted.fixed), colour = "blue")
```

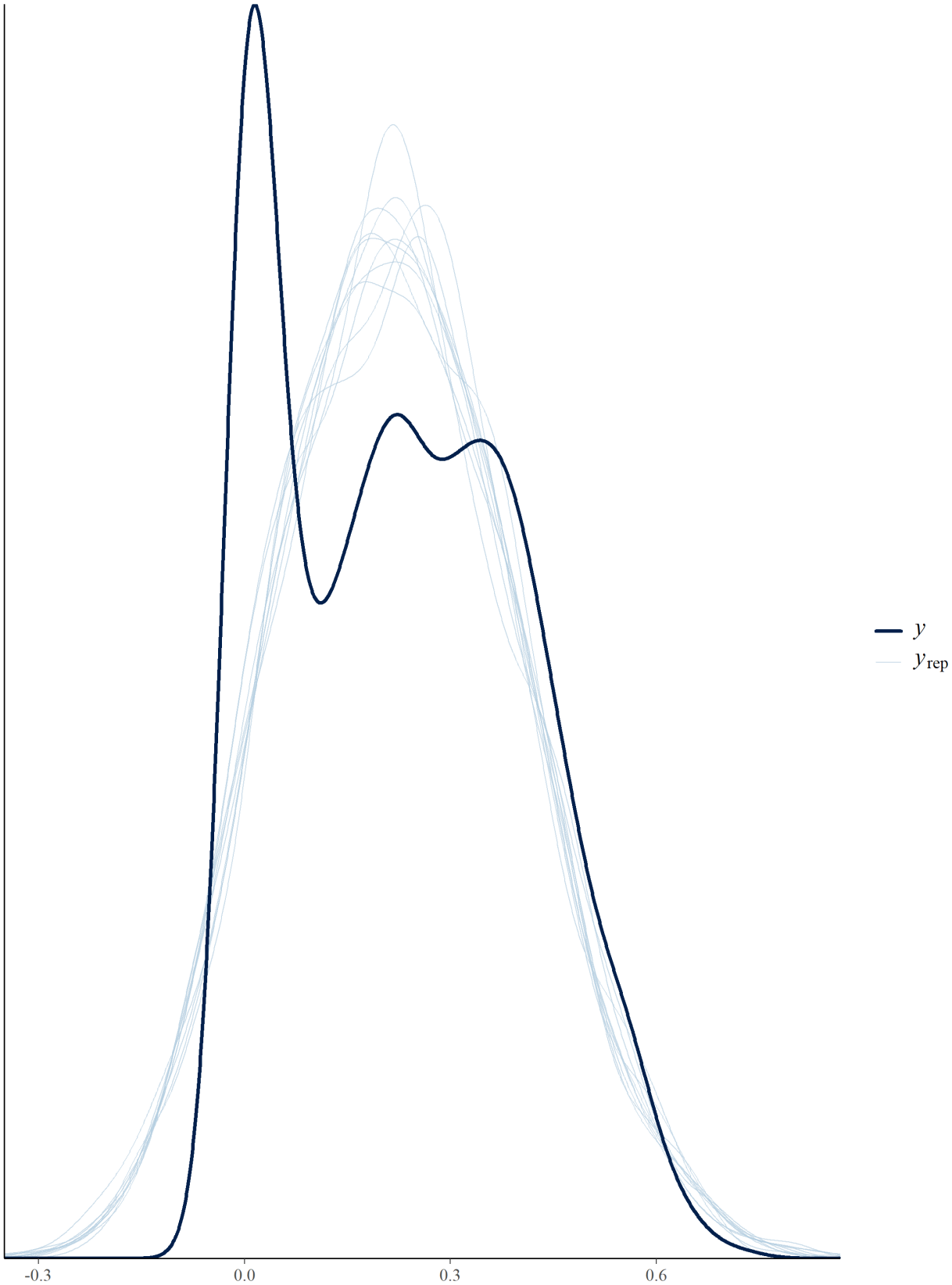


Figure 36: plot of chunk unnamed-chunk-11

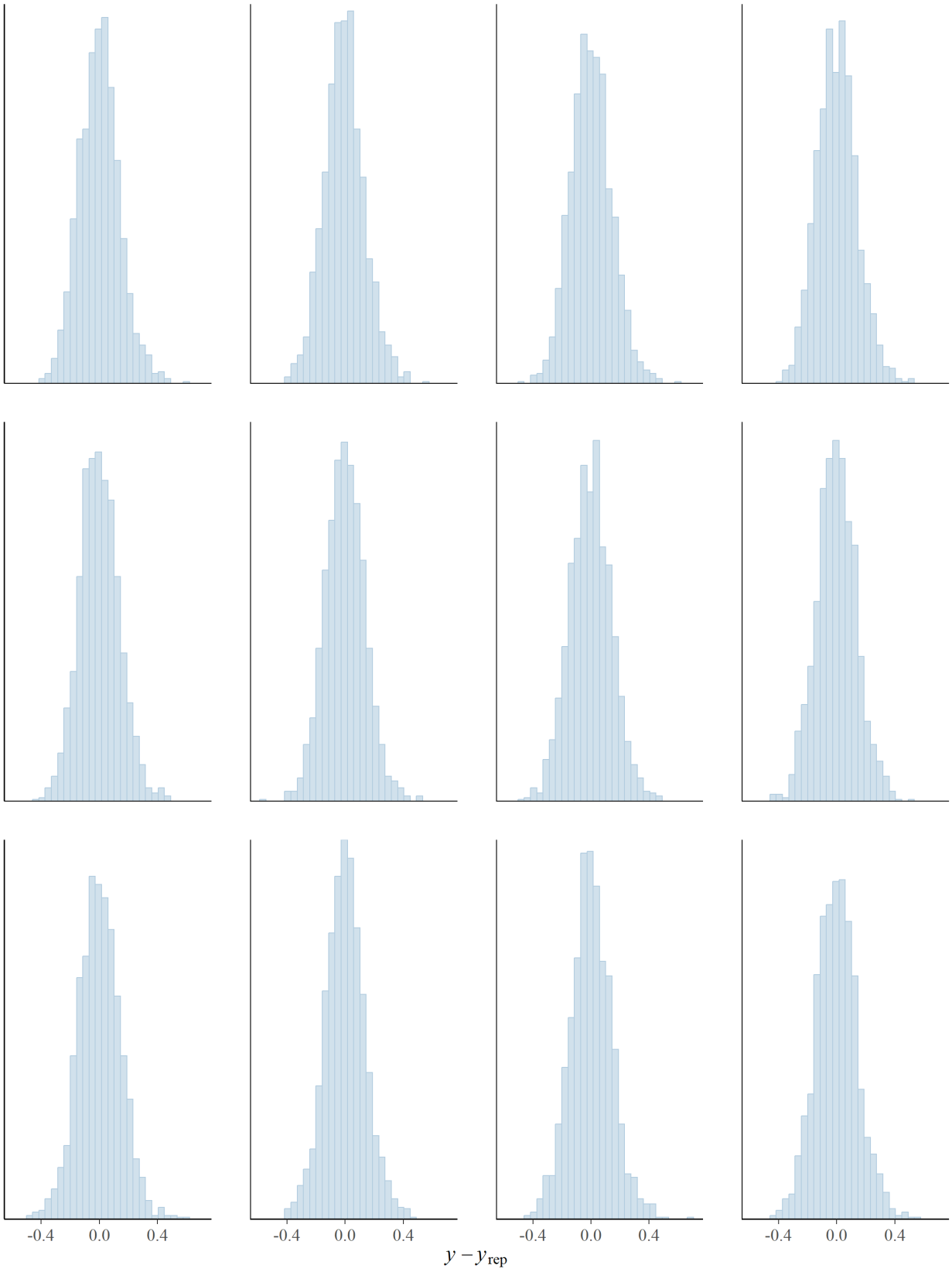


Figure 37: plot of chunk unnamed-chunk-11

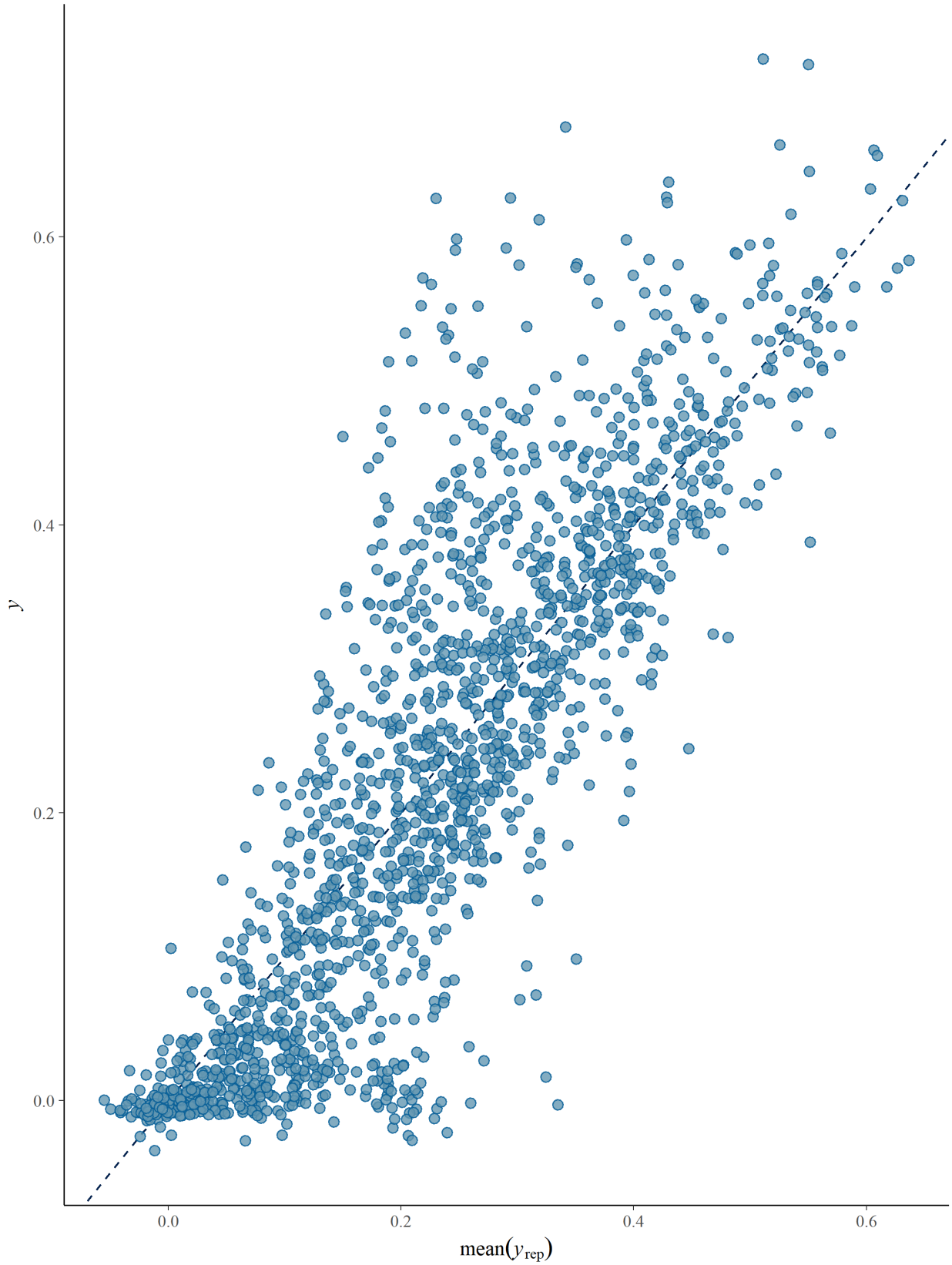


Figure 38: plot of chunk unnamed-chunk-11

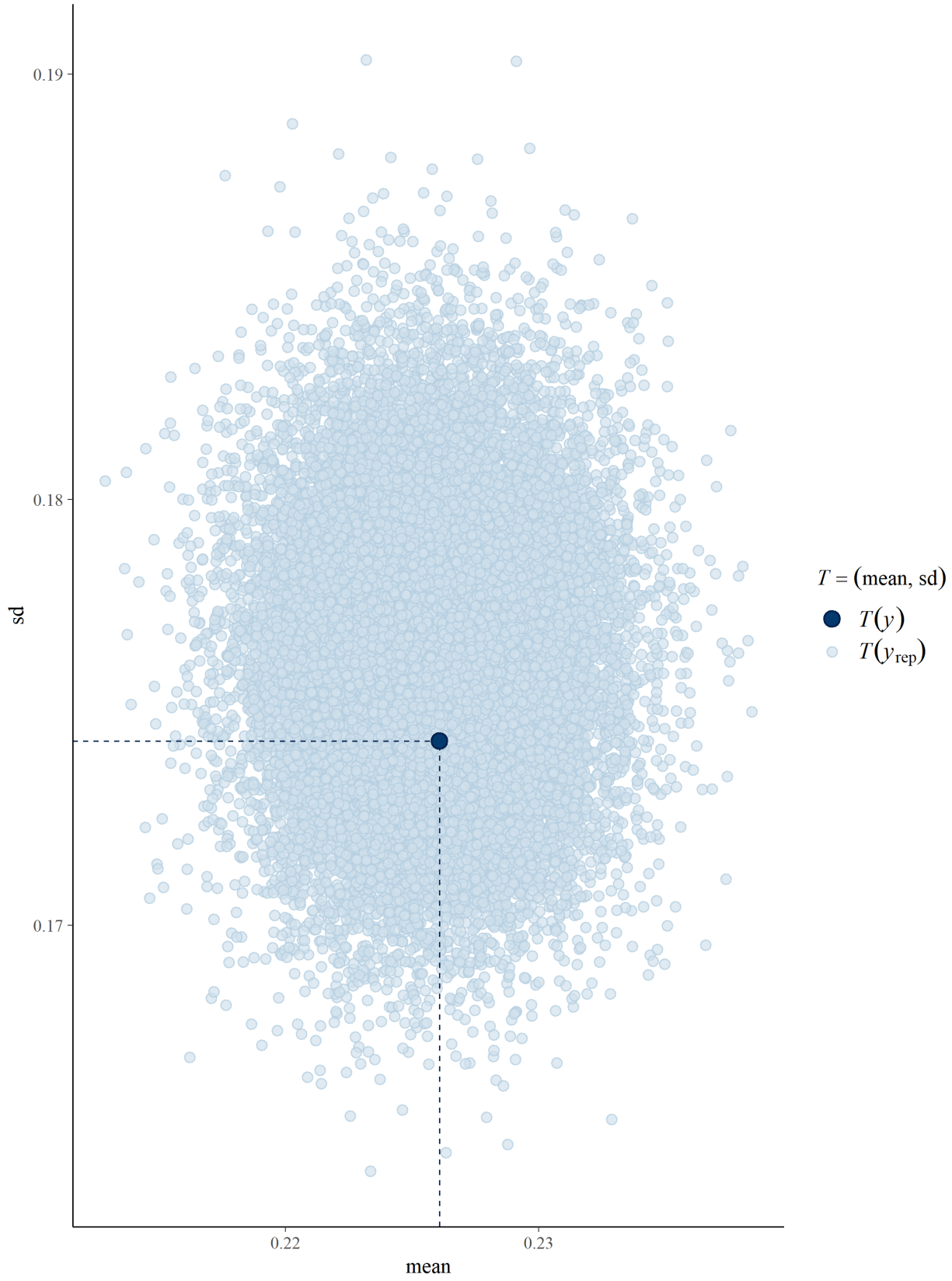


Figure 39: plot of chunk unnamed-chunk-11

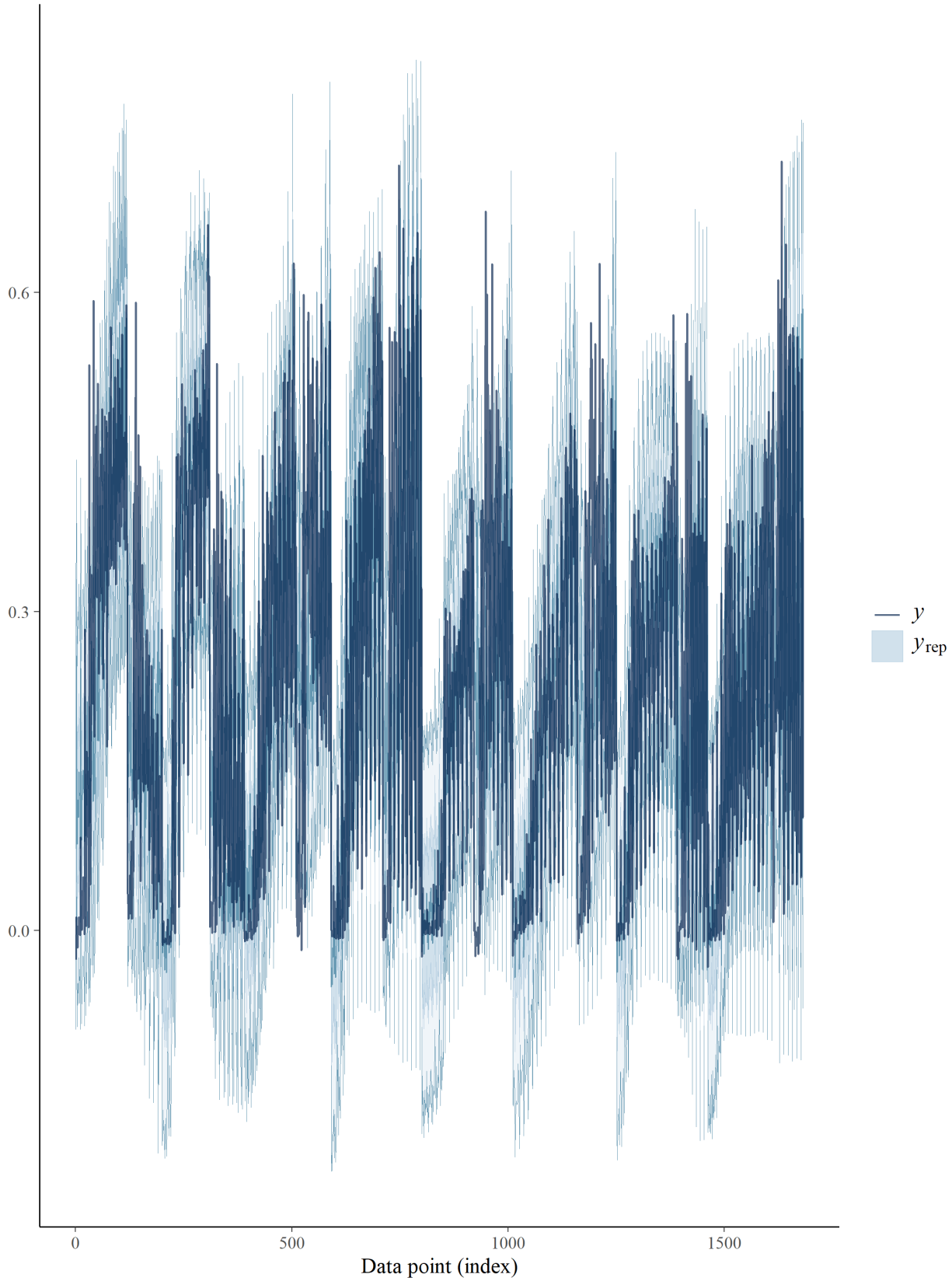


Figure 40: plot of chunk unnamed-chunk-11

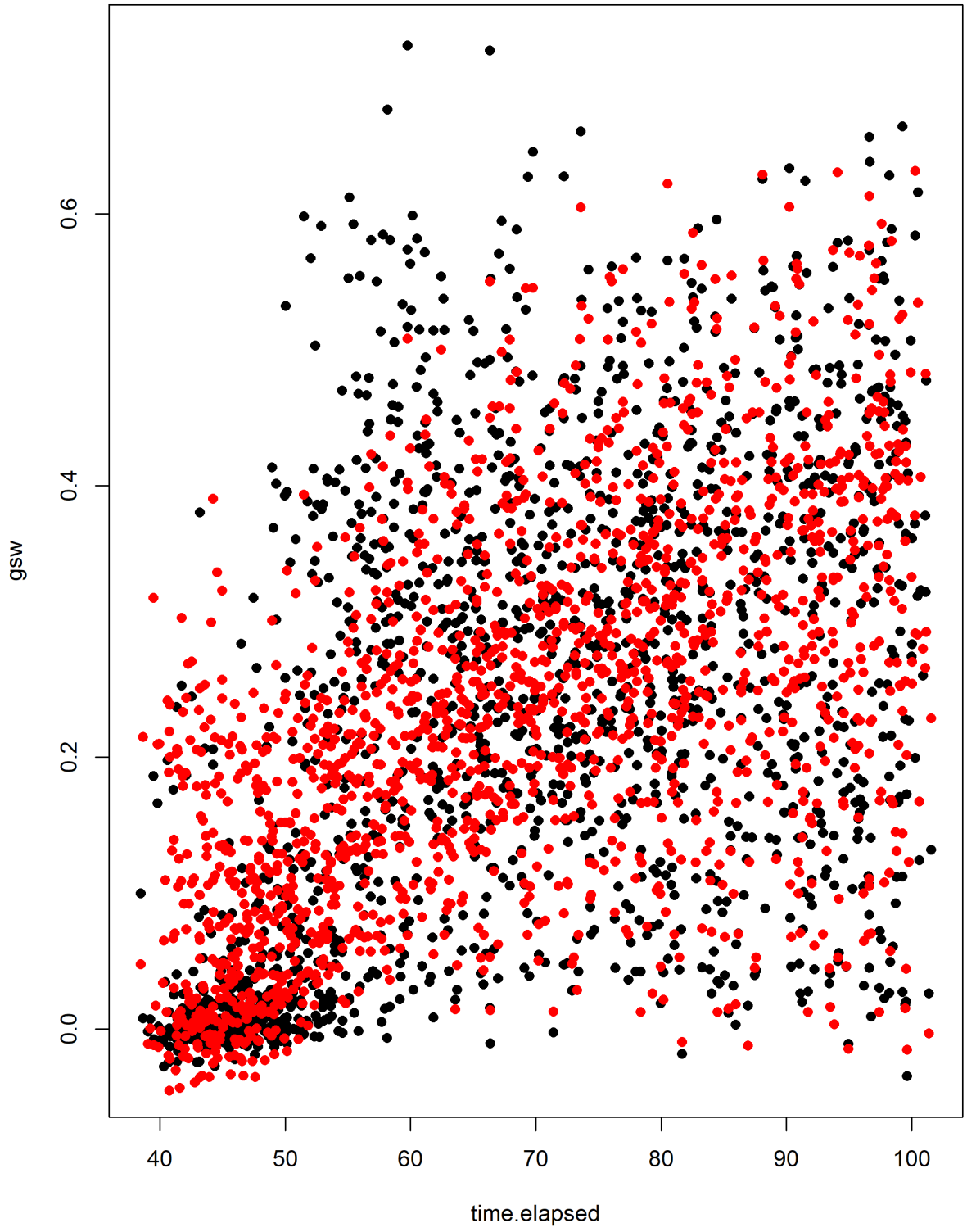


Figure 41: plot of chunk unnamed-chunk-11

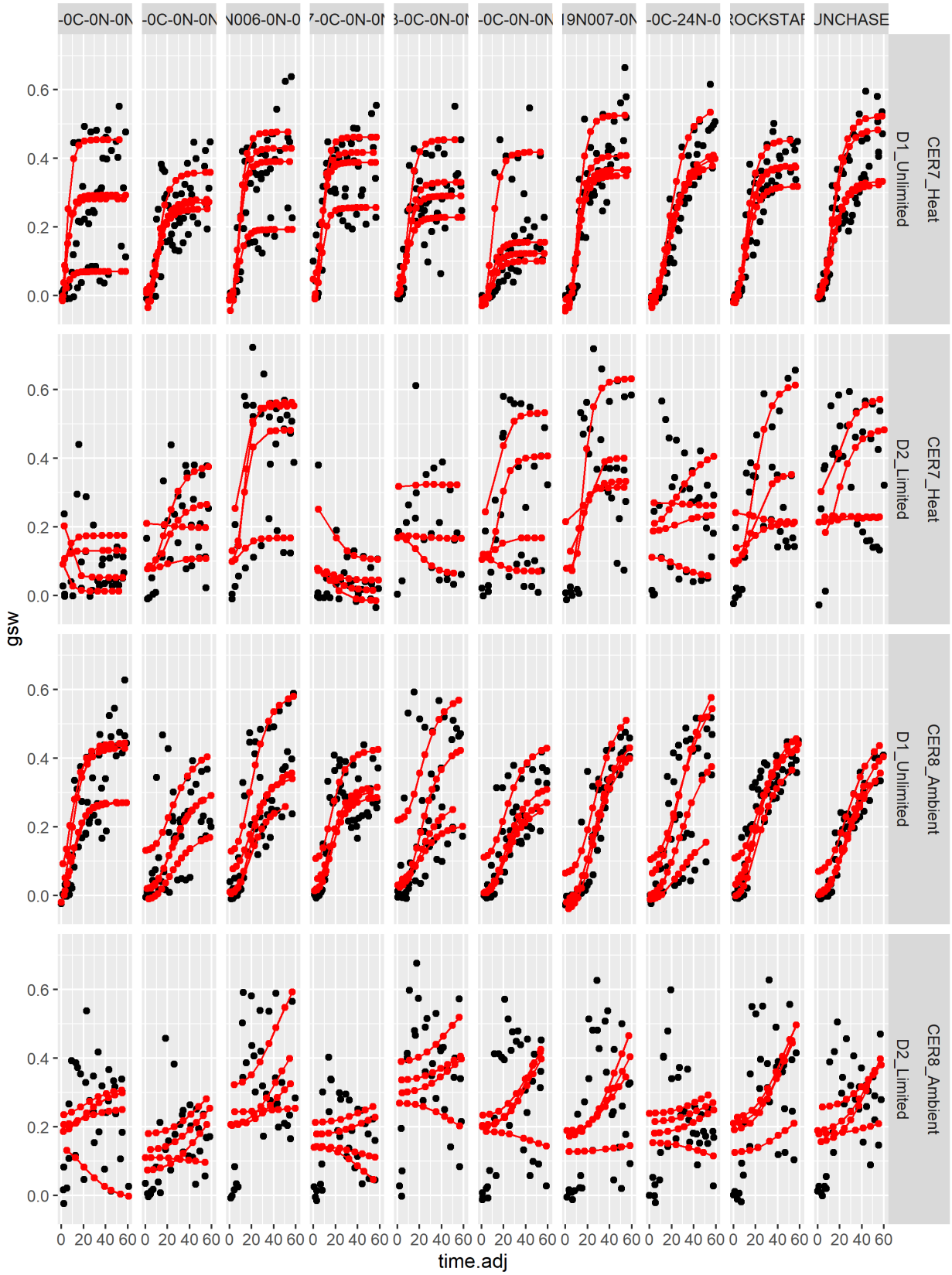


Figure 42: plot of chunk unnamed-chunk-11

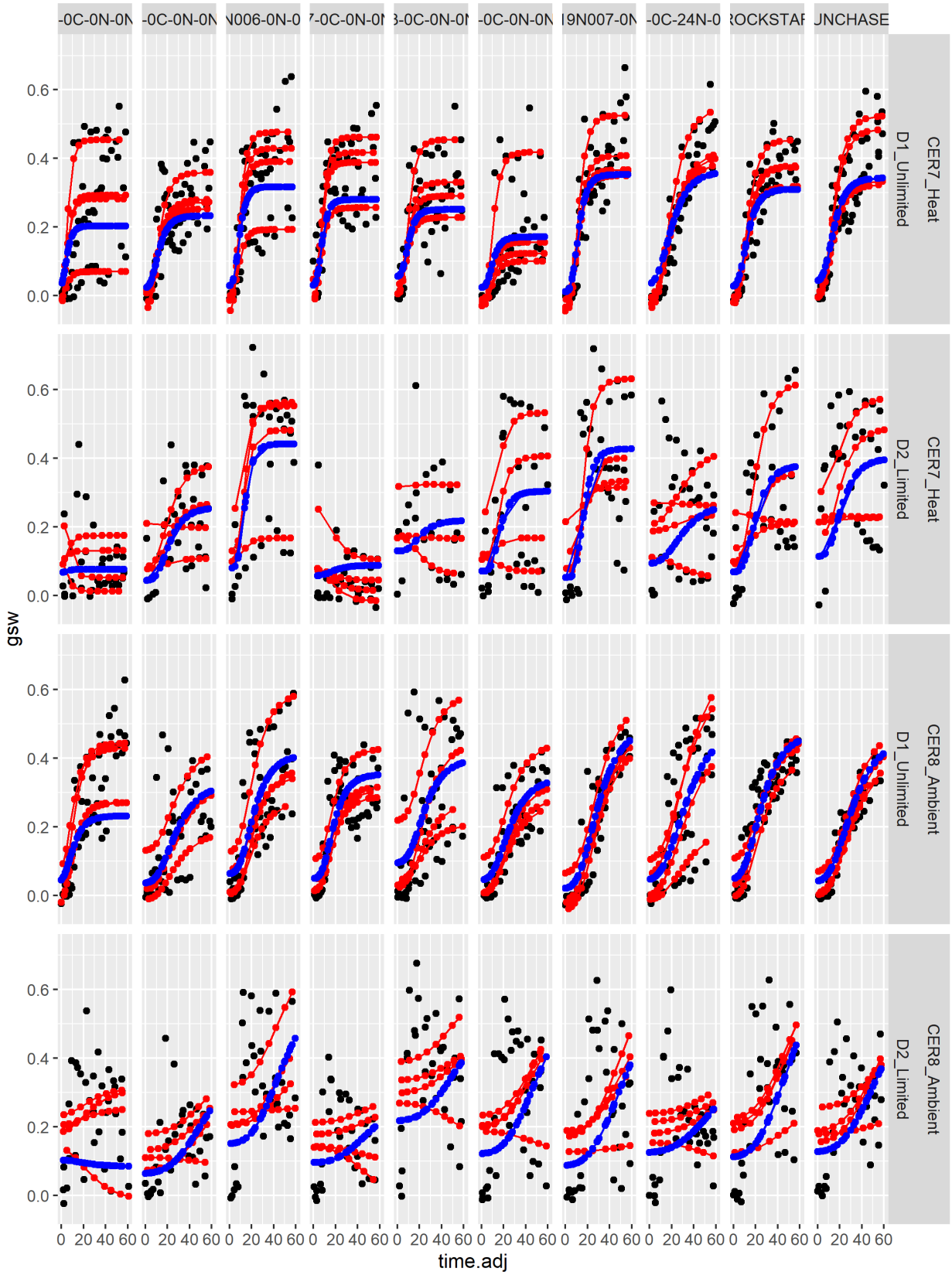


Figure 43: plot of chunk unnamed-chunk-11

```
##Testing contribution of effects in the model
# nlform.reduced <- bf(gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - time.adj)/k)+1)),
#               g0 ~ 1 + Line*Device_Watering*Room_Temperature +
#                 (1 | CEBRoom:CEBSide:CEBLocation +
#                    CEBRoom:CEBSide:CEBLocation:CEBSubUnit +
#                    CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell),
#               g1 ~ 1 + Line + Device_Watering + Room_Temperature +
#                 Line:Device_Watering +# Line:Room_Temperature +
#                 Device_Watering:Room_Temperature +
#                   (1 | CEBRoom:CEBSide:CEBLocation +
#                    CEBRoom:CEBSide:CEBLocation:CEBSubUnit +
#                    CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell),
#               k ~ 1 + Line,# + Device_Watering,# +  Room_Temperature
#                  #Line:Room_Temperature,#Line:Device_Watering +
#               lambda ~ 1,# + Line,#  + Room_Temperature,# + Device_Watering
#                #Line:Room_Temperature, # Line:Device_Watering
#               nl = TRUE)
#
# # Fit the model
# fit_on_reduced <- brm(
#   formula = nlform.reduced,
#   data = li600.on.dat,
#   prior = priors,
#   chains = 6,
#   cores = 6,
#   iter = 10000,
#   warmup = 6000,
#   control = list(
#     adapt_delta = 0.99,
#     max_treedepth = 12,
#     metric = "diag_e"
#   )
# )
#
# print(summary(fit_on_reduced))
# loo(fit_on_reduced)
# loo(fit_on_current)
# rstan::check_hmc_diagnostics(fit_on_reduced$fit)
# plot(gsw~time.adj, li600.on.dat, pch=16)
# points(fitted(fit_on_reduced)[,1]~time.adj, li600.on.dat, pch=16, col="red")
# length(fixef(fit_on_reduced))
#
# fit_on_current <- add_criterion(fit_on_current, "loo")
# fit_on_reduced <- add_criterion(fit_on_reduced, "loo")
#
# loo_compare(fit_on_current, fit_on_reduced)
##THREE WAY INTERACTION
##Lambda: Drop Room_Temperature:Device_Watering:Line interaction
##k: Drop Room_Temperature:Device_Watering:Line interaction
##g0: Retain Room_Temperature:Device_Watering:Line interaction
##g1: Drop Room_Temperature:Device_Watering:Line interaction

##TWO WAY INTERACTION
```

```
##Lambda: Drop Room_Temperature:Device_Watering
##k: Drop Room_Temperature:Device_Watering
##g1: Retain Room_Temperature:Device_Watering
##lambda: Drop Line:Device_Watering
##lambda: Drop Line:Room_Temperature
##k: Drop Line:Device_Watering
##k: Drop Line:Room_Temperature
##g1: Drop Line:Room_Temperature
##g1: Retain Line:Device_Watering

##MAIN EFFECTS
##lambda: Drop Device_Watering
##lambda: Drop Room_Temperature
##lambda: Drop Line
##k: Drop Device_Watering
##k: Drop Room_Temperature
##k: Retain Line

# fit_on_current <- fit_on_reduced
# print(summary(fit_on_current))
# pp_check(fit_on_current)
```

```
##Final model:
##g0: Room_Temperature:Device_Watering:Line interaction
##g1: Room_Temperature:Device_Watering + Line:Device_Watering
##lambda: 1
##k: Line

nlform_final <- bf(gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - time.adj)/k)+1)),
                     g0 ~ 1 + Line*Device_Watering*Room_Temperature +
                       (1 | CEBRoom:CEBSide:CEBLocation +
                          CEBRoom:CEBSide:CEBLocation:CEBSubUnit +
                          CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell),
                     g1 ~ 1 + Line + Device_Watering + Room_Temperature +
                       Line:Device_Watering + Device_Watering:Room_Temperature +
                       (1 | CEBRoom:CEBSide:CEBLocation +
                          CEBRoom:CEBSide:CEBLocation:CEBSubUnit +
                          CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell),
                     k ~ 1 + Line,
                     lambda ~ 1,
                     nl = TRUE)

# Fit the model
fit_on_final <- brm(
  formula = nlform_final,
  data = li600.on.dat,
  prior = priors,
  chains = 6,
  cores = 6,
  iter = 10000,
  warmup = 6000,
  control = list(
```

```
    adapt_delta = 0.99,
    max_treedepth = 12,
    metric = "diag_e"
  )
)
```

Fit final model

```
Compiling Stan program...
```

```
Start sampling
```

```
print(summary(fit_on_final))
```

```
 Family: gaussian
  Links: mu = identity
Formula: gsw ~ g0 + (g1 - g0) * exp(-exp(((lambda - time.adj)/k) + 1))
         g0 ~ 1 + Line * Device_Watering * Room_Temperature + (1 | CEBRoom:CEBSide:CEBLocation + CEBRoo
         g1 ~ 1 + Line + Device_Watering + Room_Temperature + Line:Device_Watering + Device_Watering:Ro
         k ~ 1 + Line
         lambda ~ 1
   Data: li600.on.dat (Number of observations: 1682)
  Draws: 6 chains, each with iter = 10000; warmup = 6000; thin = 1;
         total post-warmup draws = 24000

Multilevel Hyperparameters:
~CEBRoom:CEBSide:CEBLocation (Number of levels: 8)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.02      0.02     0.00     0.06 1.00     6963    11013
sd(g1_Intercept)     0.03      0.03     0.00     0.10 1.00     9505    13168

~CEBRoom:CEBSide:CEBLocation:CEBSubUnit (Number of levels: 16)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.05      0.01     0.02     0.08 1.00     7495    10063
sd(g1_Intercept)     0.09      0.03     0.06     0.15 1.00     6857    11296

~CEBRoom:CEBSide:CEBLocation:CEBSubUnit:CEBCell (Number of levels: 159)
                Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(g0_Intercept)     0.01      0.01     0.00     0.02 1.00     9583    11234
sd(g1_Intercept)     0.09      0.01     0.08     0.11 1.00     7107    13229

Regression Coefficients:
                                                                                Estimate Est.
g0_Intercept                                                                        0.01
g0_LinePBI17N002M0CM0NM0NM010NM16N                                                 -0.00
g0_LinePBI17N006M0NM0NM89N                                                         -0.00
g0_LinePBI17N007M0CM0NM0NM010NM8N                                                   0.02
g0_LinePBI17N013M0CM0NM0NM010NM8N                                                   0.02
g0_LinePBI17N016M0CM0NM0NM010NM27N                                                 -0.01
g0_LinePBI19N007M0NM3N                                                             -0.01
g0_LinePBIC15034M0CM24NM010NM1NM0N                                                 -0.01
g0_LineROCKSTAR                                                                    -0.01
g0_LineSUNCHASER                                                                    0.01
g0_Device_WateringD2_Limited                                                        0.03
g0_Room_TemperatureCER8_Ambient                                                    -0.01
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited                       -0.02
```

g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	-0.01
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	-0.01
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	0.01
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	-0.00
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited	-0.01
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	0.04
g0_LineROCKSTAR:Device_WateringD2_Limited	-0.01
g0_LineSUNCHASER:Device_WateringD2_Limited	0.03
g0_LinePBI17N002M0CM0NM0NM010NM16N:Room_TemperatureCER8_Ambient	0.00
g0_LinePBI17N006M0NM0NM89N:Room_TemperatureCER8_Ambient	0.00
g0_LinePBI17N007M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	-0.01
g0_LinePBI17N013M0CM0NM0NM010NM8N:Room_TemperatureCER8_Ambient	0.01
g0_LinePBI17N016M0CM0NM0NM010NM27N:Room_TemperatureCER8_Ambient	-0.00
g0_LinePBI19N007M0NM3N:Room_TemperatureCER8_Ambient	-0.01
g0_LinePBIC15034M0CM24NM010NM1NM0N:Room_TemperatureCER8_Ambient	0.01
g0_LineROCKSTAR:Room_TemperatureCER8_Ambient	0.00
g0_LineSUNCHASER:Room_TemperatureCER8_Ambient	-0.02
g0_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.00
g0_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.01
g0_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.00
g0_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.00
g0_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.01
g0_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.00
g0_LinePBI19N007M0NM3N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.00
g0_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.01
g0_LineROCKSTAR:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.00
g0_LineSUNCHASER:Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	-0.01
g1_Intercept	0.23
g1_LinePBI17N002M0CM0NM0NM010NM16N	0.01
g1_LinePBI17N006M0NM0NM89N	0.09
g1_LinePBI17N007M0CM0NM0NM010NM8N	0.08
g1_LinePBI17N013M0CM0NM0NM010NM8N	0.05
g1_LinePBI17N016M0CM0NM0NM010NM27N	-0.02
g1_LinePBI19N007M0NM3N	0.13
g1_LinePBIC15034M0CM24NM010NM1NM0N	0.10
g1_LineROCKSTAR	0.11
g1_LineSUNCHASER	0.09
g1_Device_WateringD2_Limited	-0.14
g1_Room_TemperatureCER8_Ambient	0.01
g1_LinePBI17N002M0CM0NM0NM010NM16N:Device_WateringD2_Limited	0.12
g1_LinePBI17N006M0NM0NM89N:Device_WateringD2_Limited	0.23
g1_LinePBI17N007M0CM0NM0NM010NM8N:Device_WateringD2_Limited	-0.03
g1_LinePBI17N013M0CM0NM0NM010NM8N:Device_WateringD2_Limited	0.19
g1_LinePBI17N016M0CM0NM0NM010NM27N:Device_WateringD2_Limited	0.25
g1_LinePBI19N007M0NM3N:Device_WateringD2_Limited	0.16
g1_LinePBIC15034M0CM24NM010NM1NM0N:Device_WateringD2_Limited	0.08
g1_LineROCKSTAR:Device_WateringD2_Limited	0.16
g1_LineSUNCHASER:Device_WateringD2_Limited	0.18
g1_Device_WateringD2_Limited:Room_TemperatureCER8_Ambient	0.04
k_Intercept	2.03
k_LinePBI17N002M0CM0NM0NM010NM16N	2.12
k_LinePBI17N006M0NM0NM89N	1.13
k_LinePBI17N007M0CM0NM0NM010NM8N	0.50
k_LinePBI17N013M0CM0NM0NM010NM8N	0.15

```
k_LinePBI17N016M0CM0NM0NM010NM27N                                    2.15
k_LinePBI19N007M0NM3N                                               4.68
k_LinePBIC15034M0CM24NM010NM1NM0N                                   5.22
k_LineROCKSTAR                                                      3.11
k_LineSUNCHASER                                                     3.98
lambda_Intercept                                                    5.47

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     0.08      0.00     0.08     0.09 1.00    31612    18569

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
pp_check(fit_on_final)
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```

```
rstan::check_hmc_diagnostics(fit_on_final$fit)
```

```
Divergences:

0 of 24000 iterations ended with a divergence.

Tree depth:

0 of 24000 iterations saturated the maximum tree depth of 12.

Energy:

E-BFMI indicated no pathological behavior.
```

```
plot(fit_on_final, ask = FALSE)
```

```
##make the above plot prettier
li600.on.plot <- li600.on.dat
li600.on.plot$fitted <- fitted(fit_on_final)[,1]
li600.on.plot$fitted.fixed <- fitted(fit_on_final, re_formula = NA)[,1]

ggplot(li600.on.plot, aes(x= time.adj, y = gsw, group = Replicate)) +
  geom_point() +
  facet_grid(Room_Temperature + Device_Watering ~ Line) +
  geom_point(aes(y=fitted), colour = "red") +
  geom_line(aes(y=fitted), colour = "red")
```

```
ggplot(li600.on.plot, aes(x= time.adj, y = gsw, group = Replicate)) +
  geom_point() +
  facet_grid(Room_Temperature + Device_Watering ~ Line) +
  geom_point(aes(y=fitted), colour = "red") +
  geom_line(aes(y=fitted), colour = "red") +
  geom_point(aes(y=fitted.fixed), colour = "blue") +
  geom_line(aes(y=fitted.fixed), colour = "blue")
```

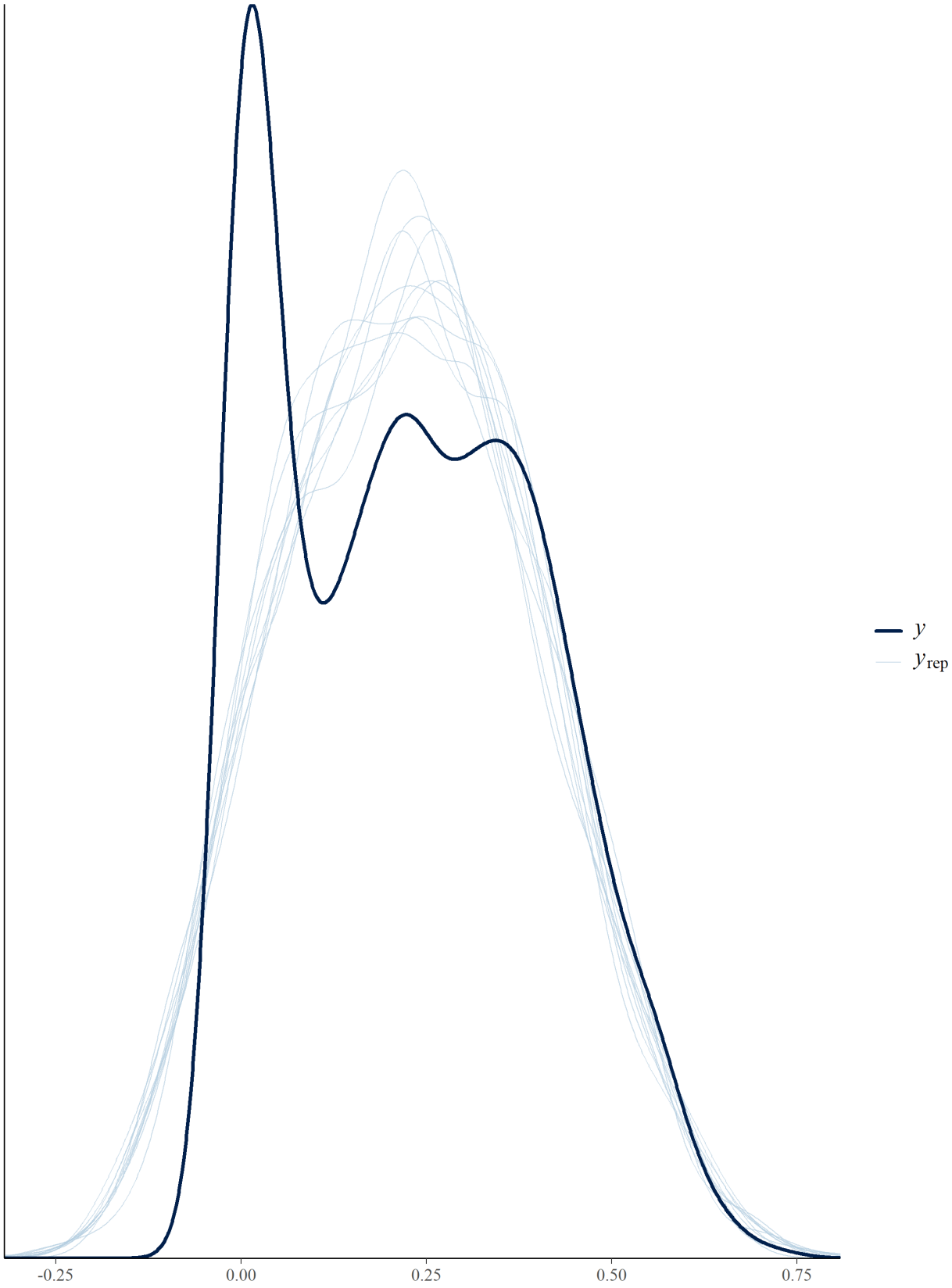


Figure 44: plot of chunk unnamed-chunk-12

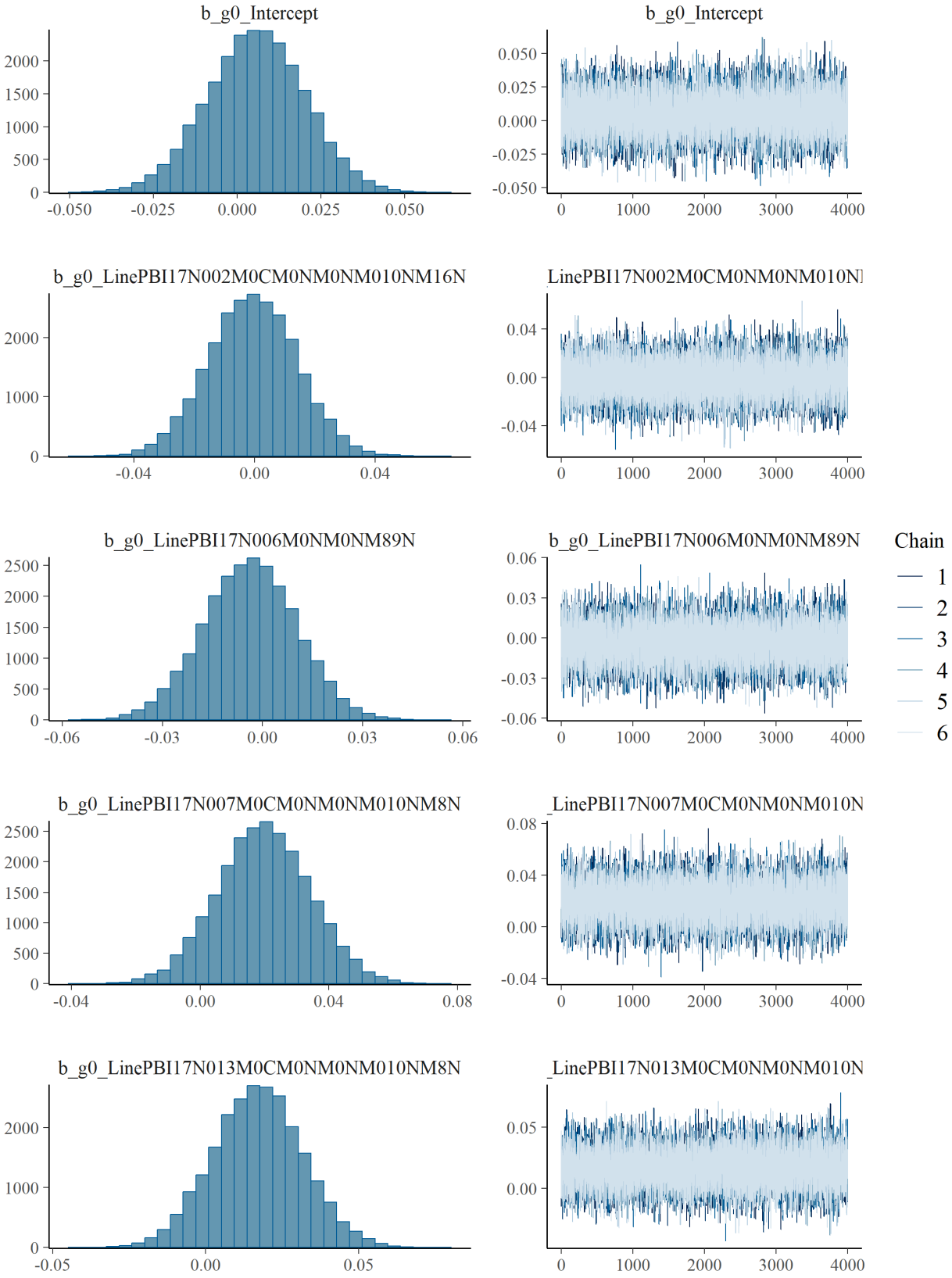


Figure 45: plot of chunk unnamed-chunk-12

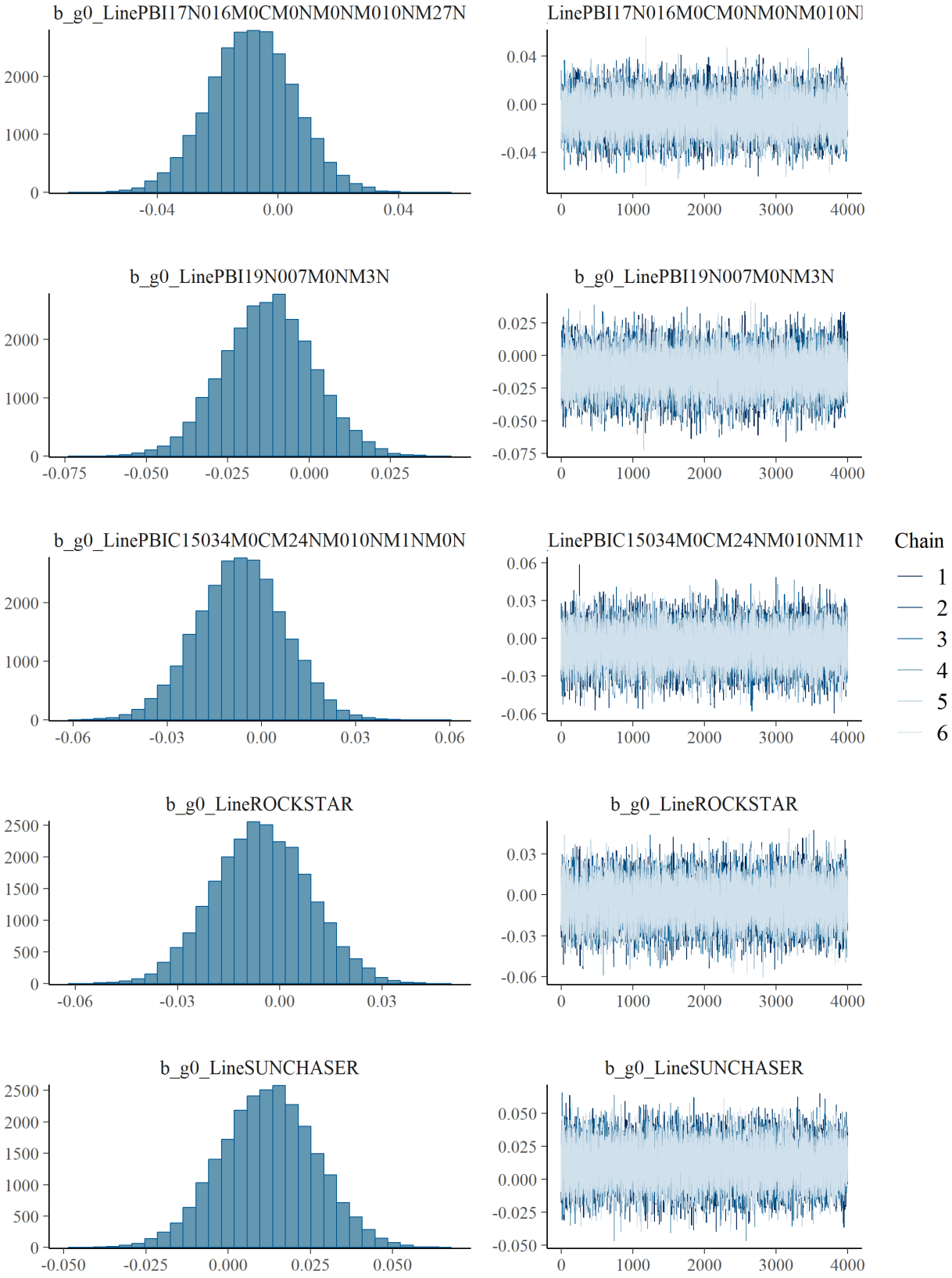


Figure 46: plot of chunk unnamed-chunk-12

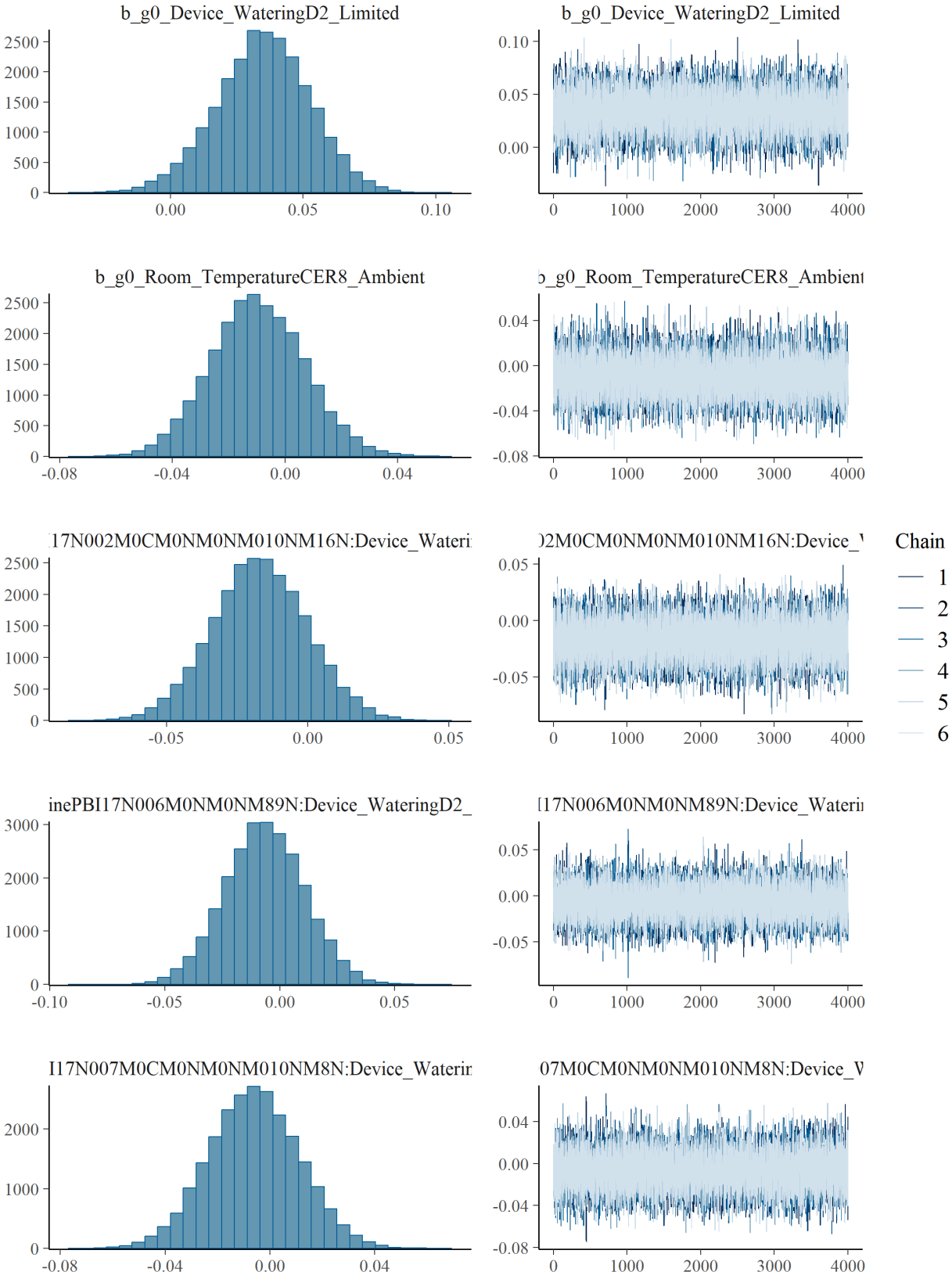


Figure 47: plot of chunk unnamed-chunk-12

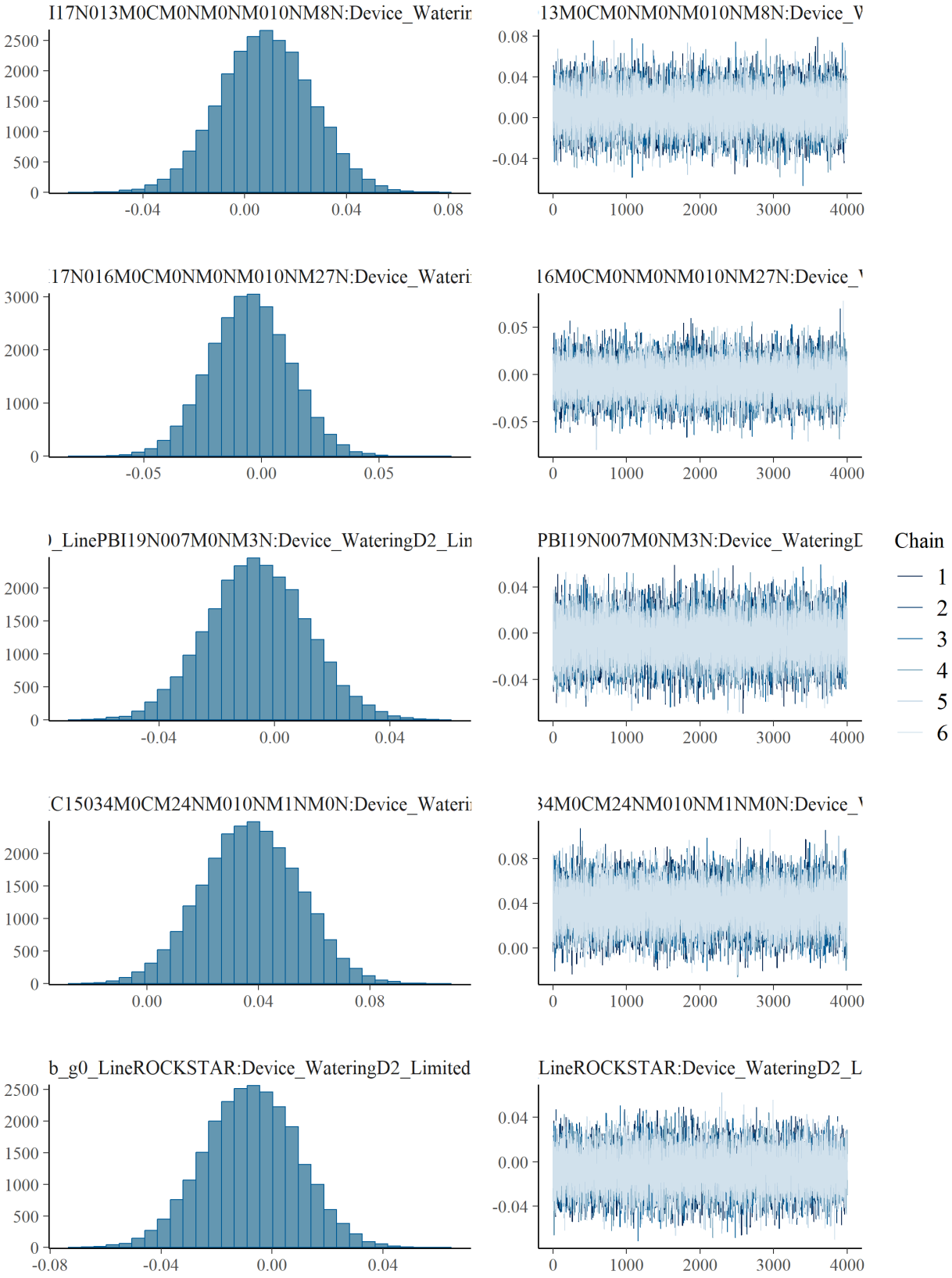


Figure 48: plot of chunk unnamed-chunk-12

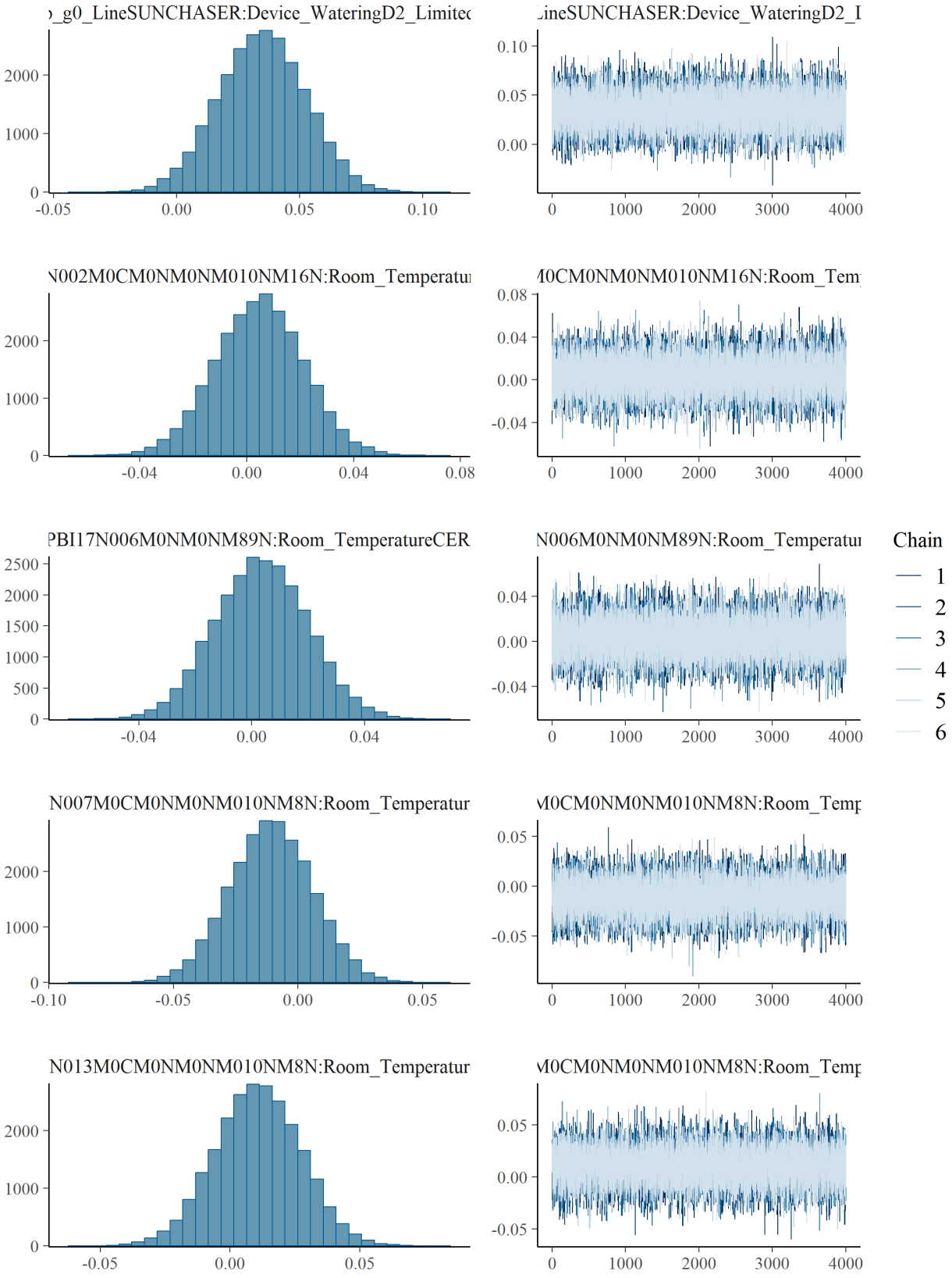


Figure 49: plot of chunk unnamed-chunk-12

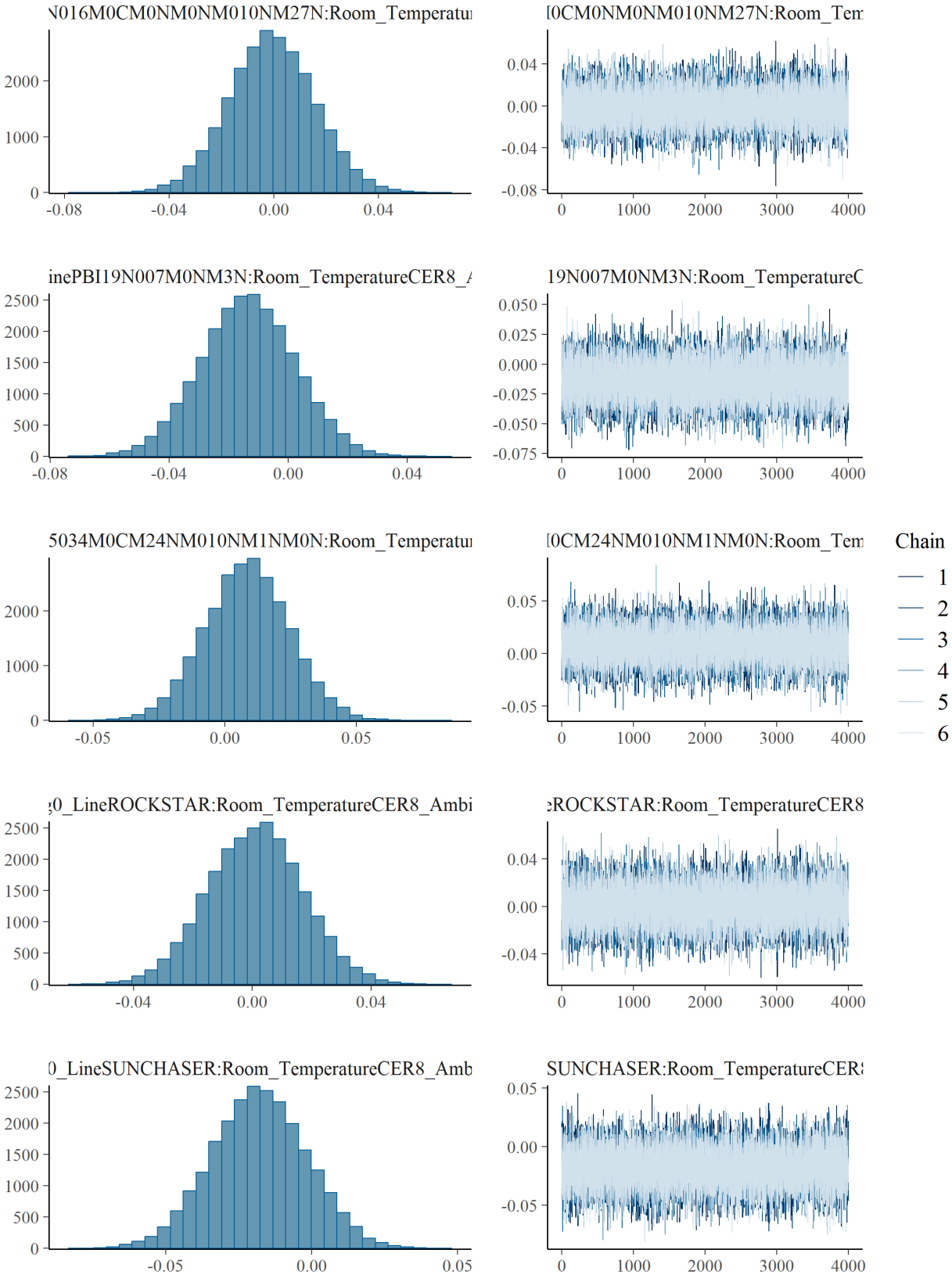


Figure 50: plot of chunk unnamed-chunk-12

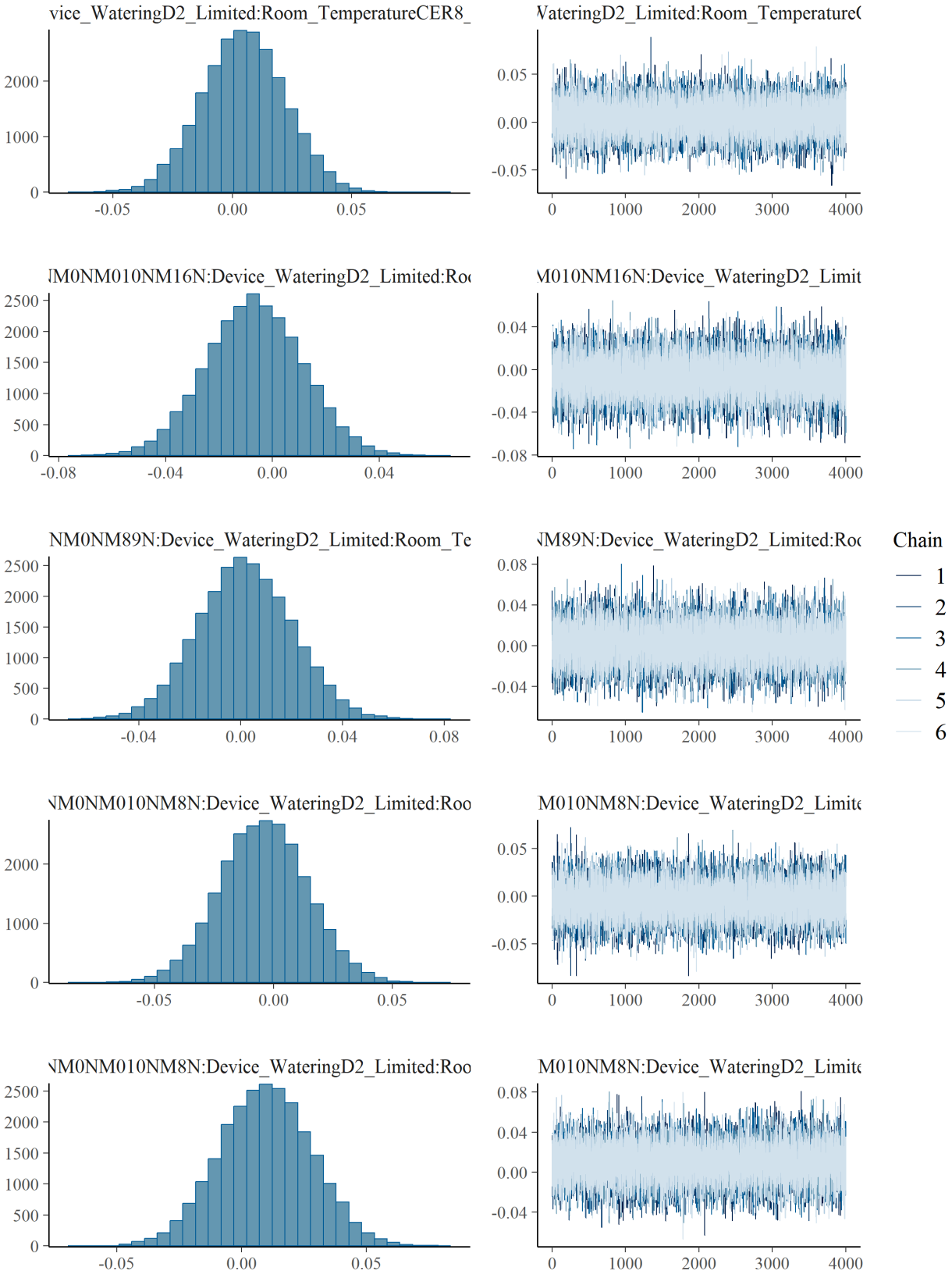


Figure 51: plot of chunk unnamed-chunk-12

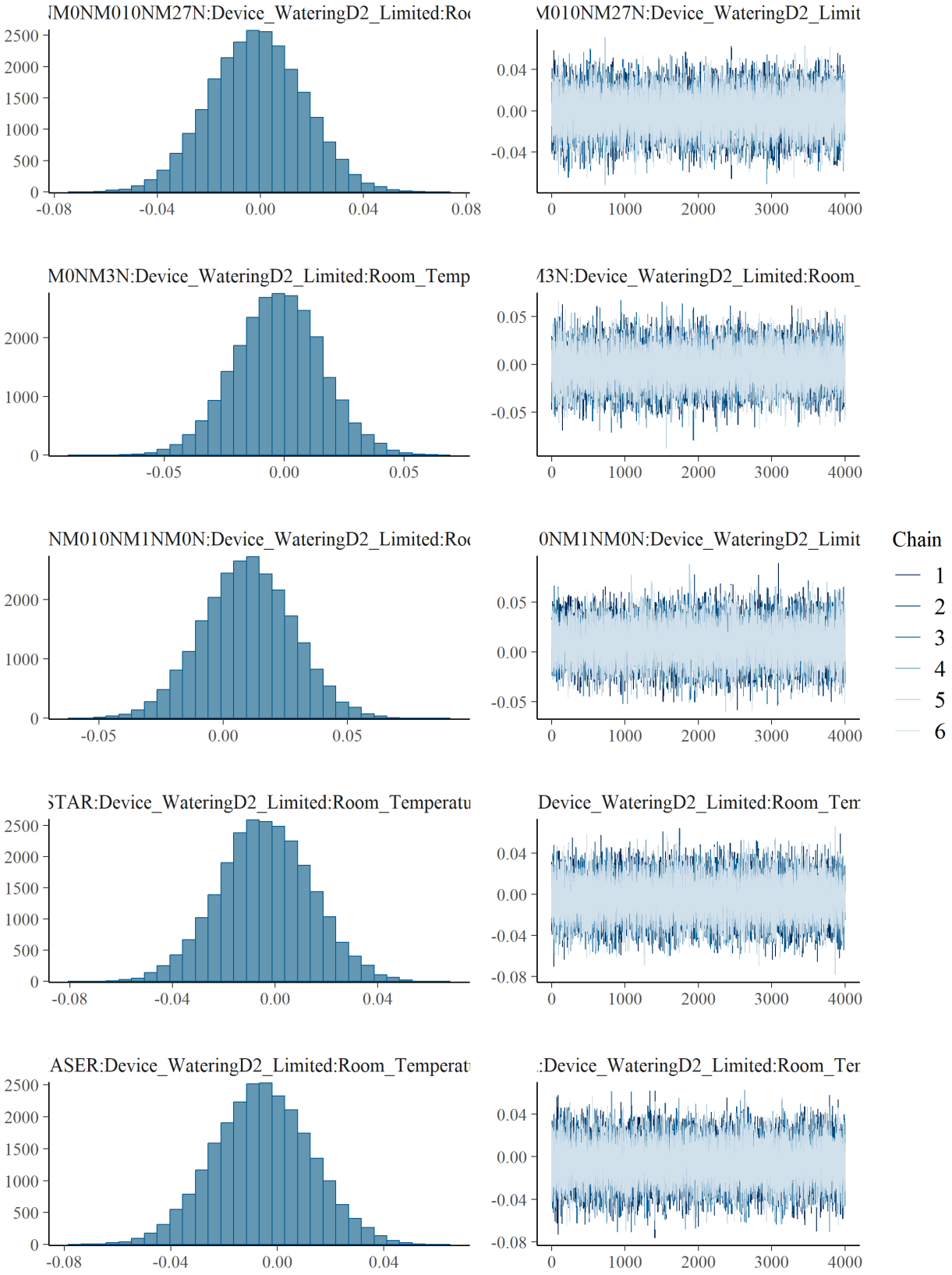


Figure 52: plot of chunk unnamed-chunk-12

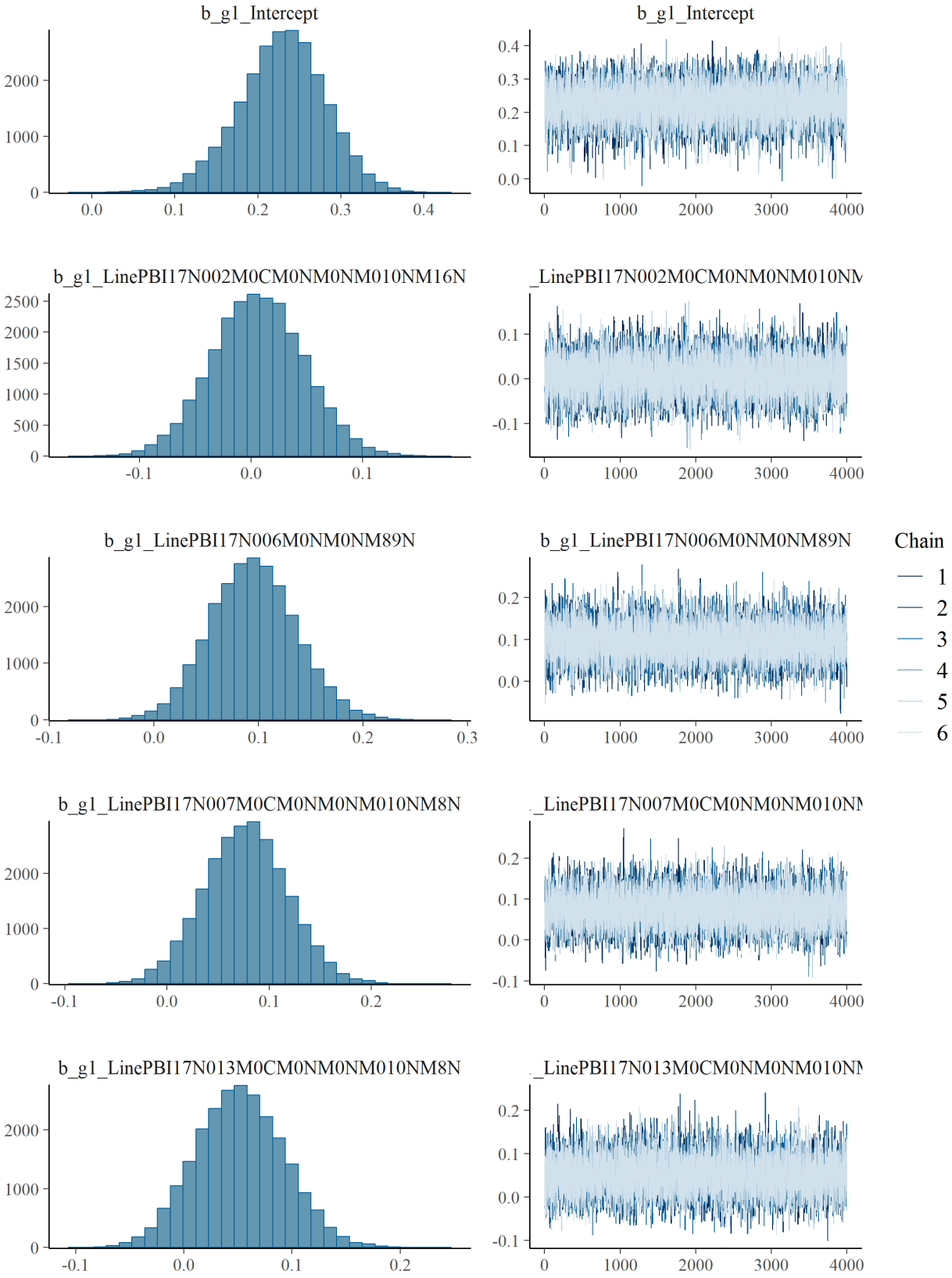


Figure 53: plot of chunk unnamed-chunk-12

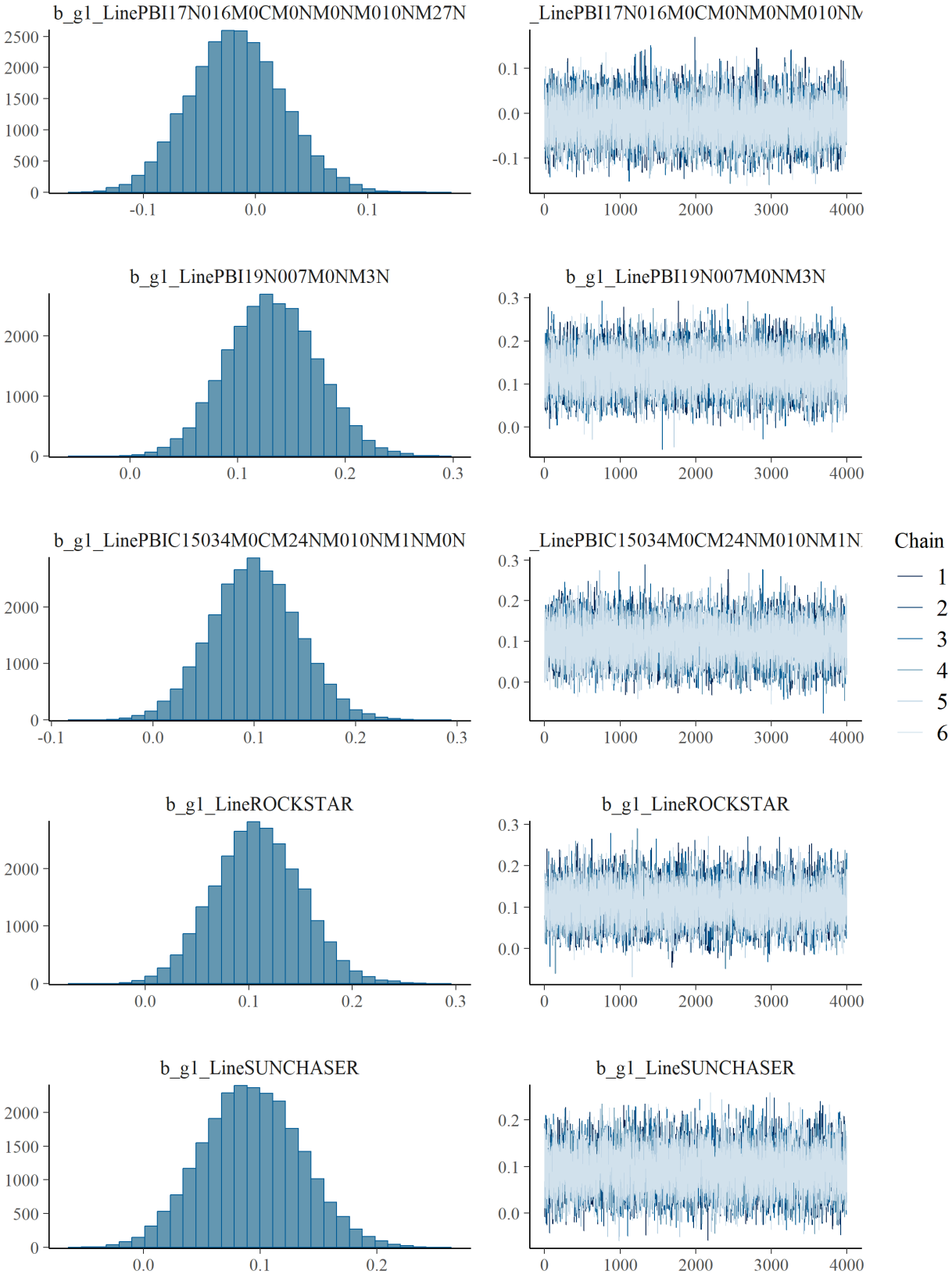


Figure 54: plot of chunk unnamed-chunk-12

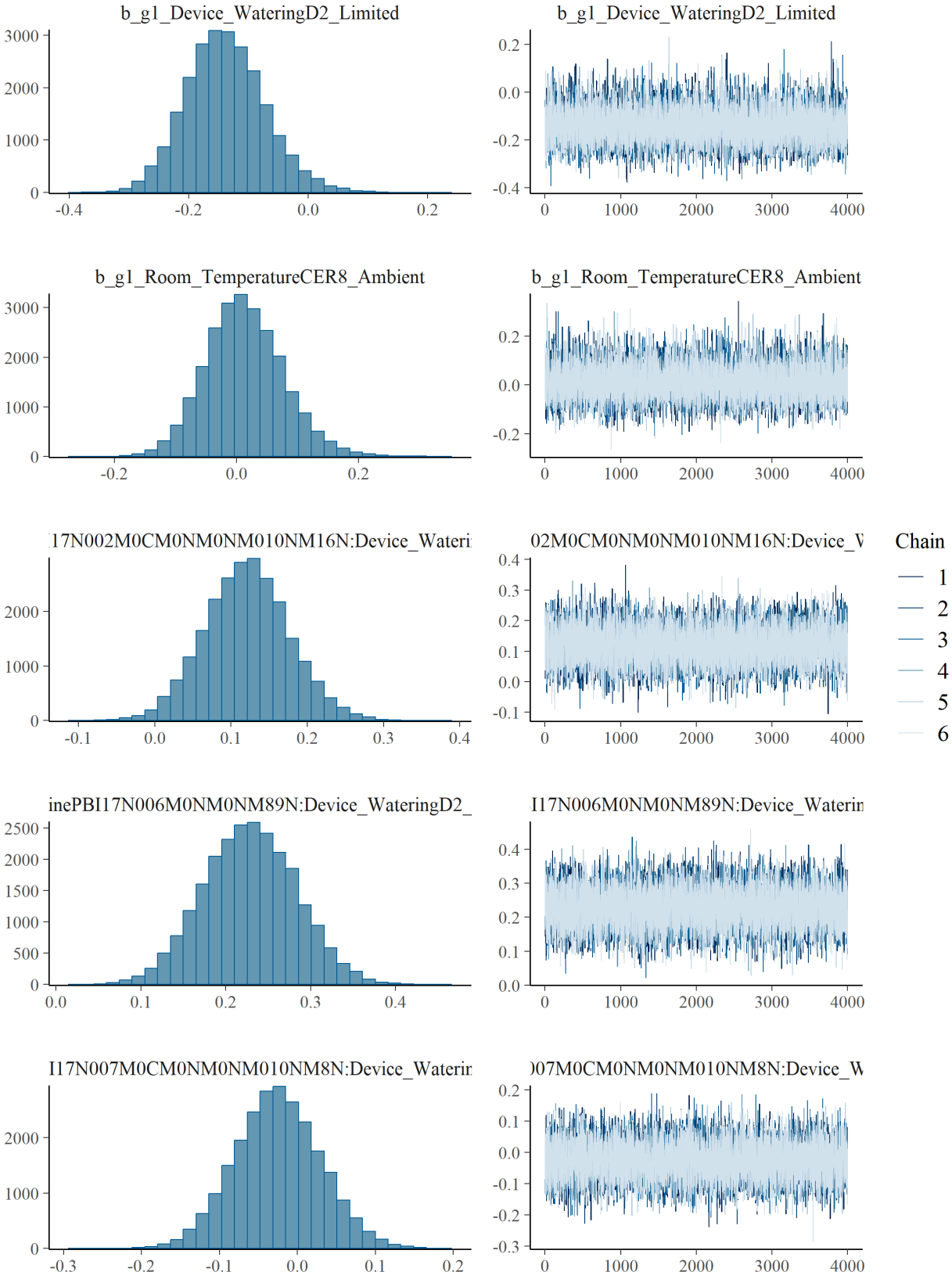


Figure 55: plot of chunk unnamed-chunk-12

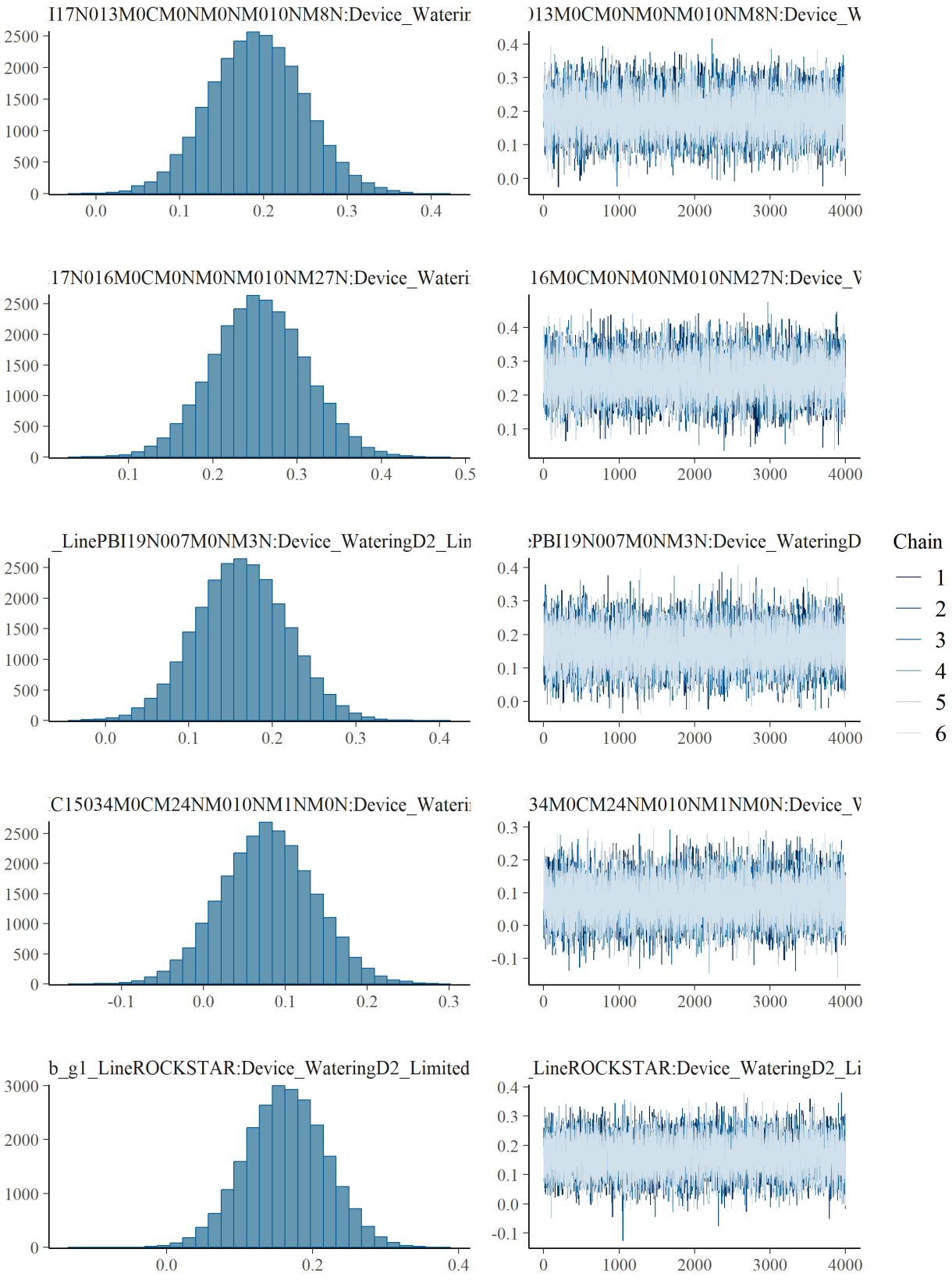


Figure 56: plot of chunk unnamed-chunk-12

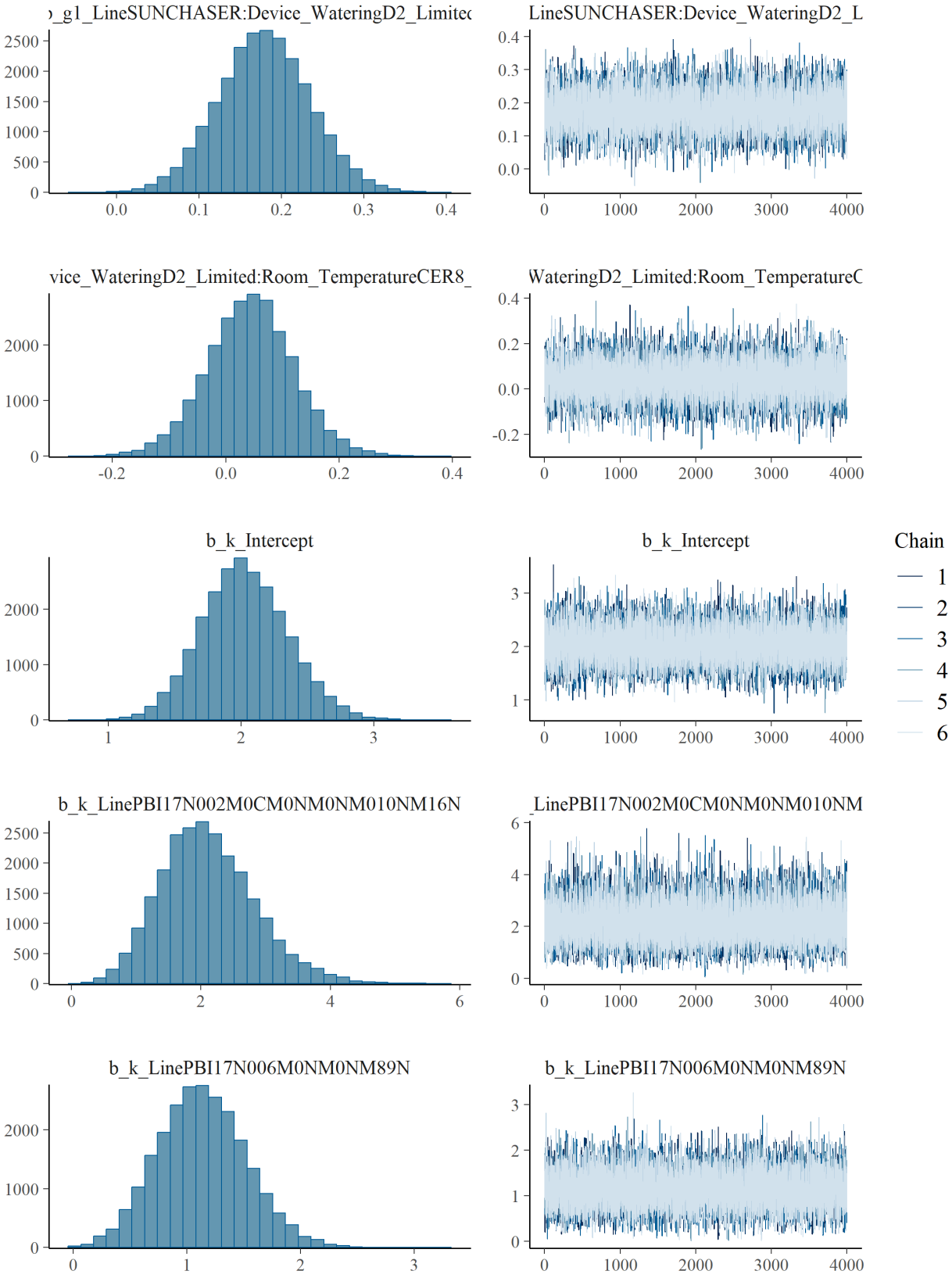


Figure 57: plot of chunk unnamed-chunk-12

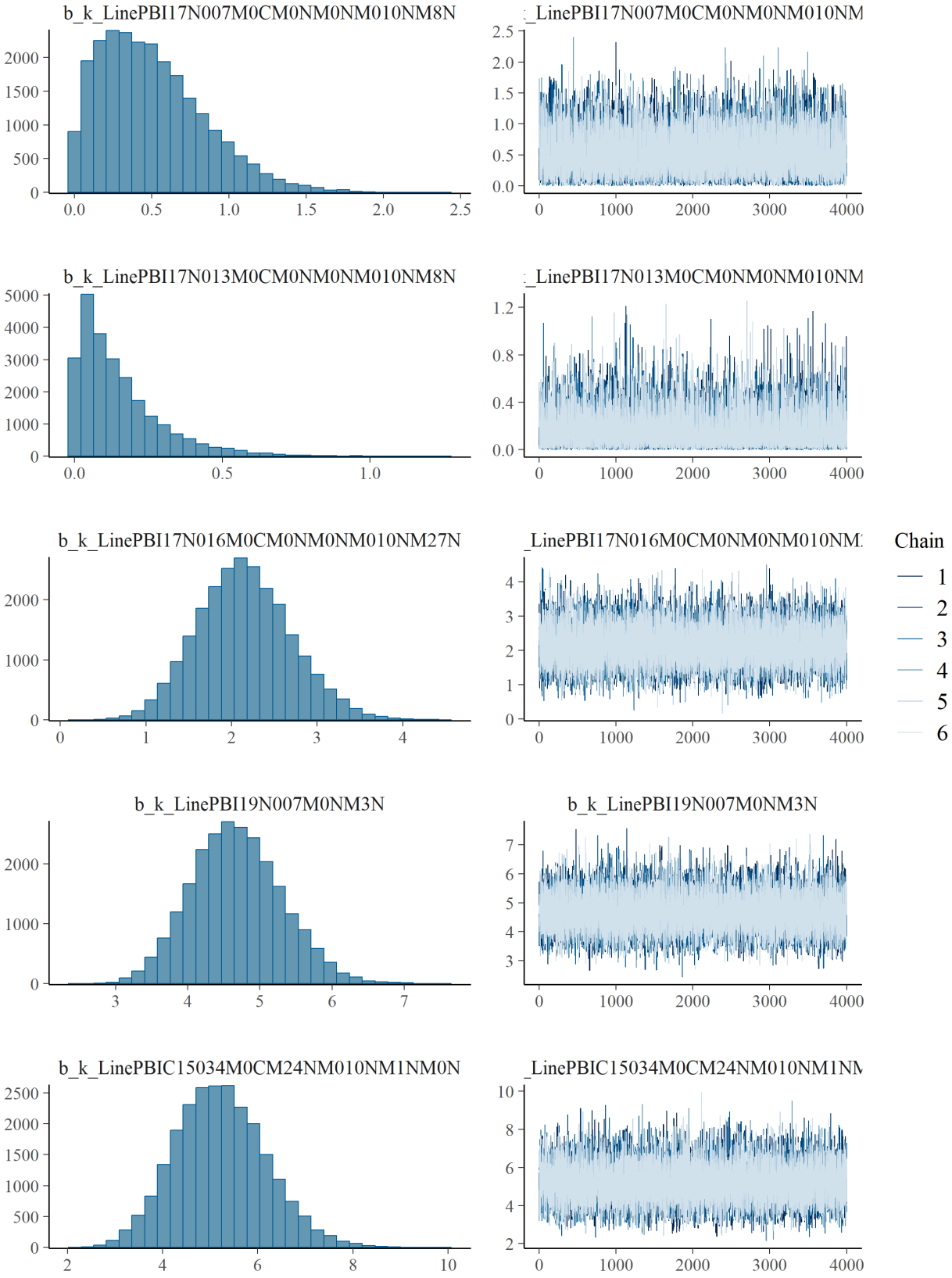


Figure 58: plot of chunk unnamed-chunk-12

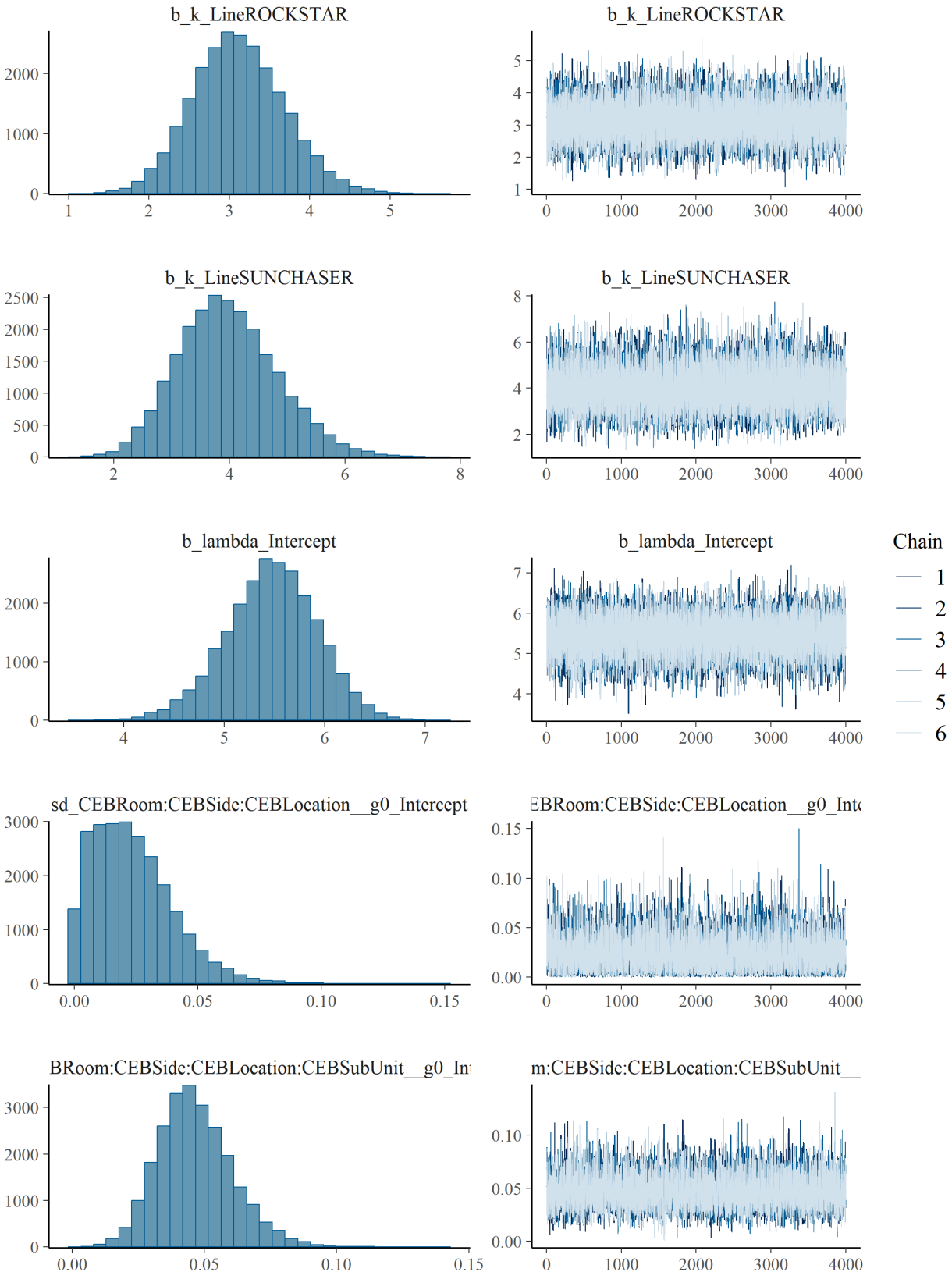


Figure 59: plot of chunk unnamed-chunk-12

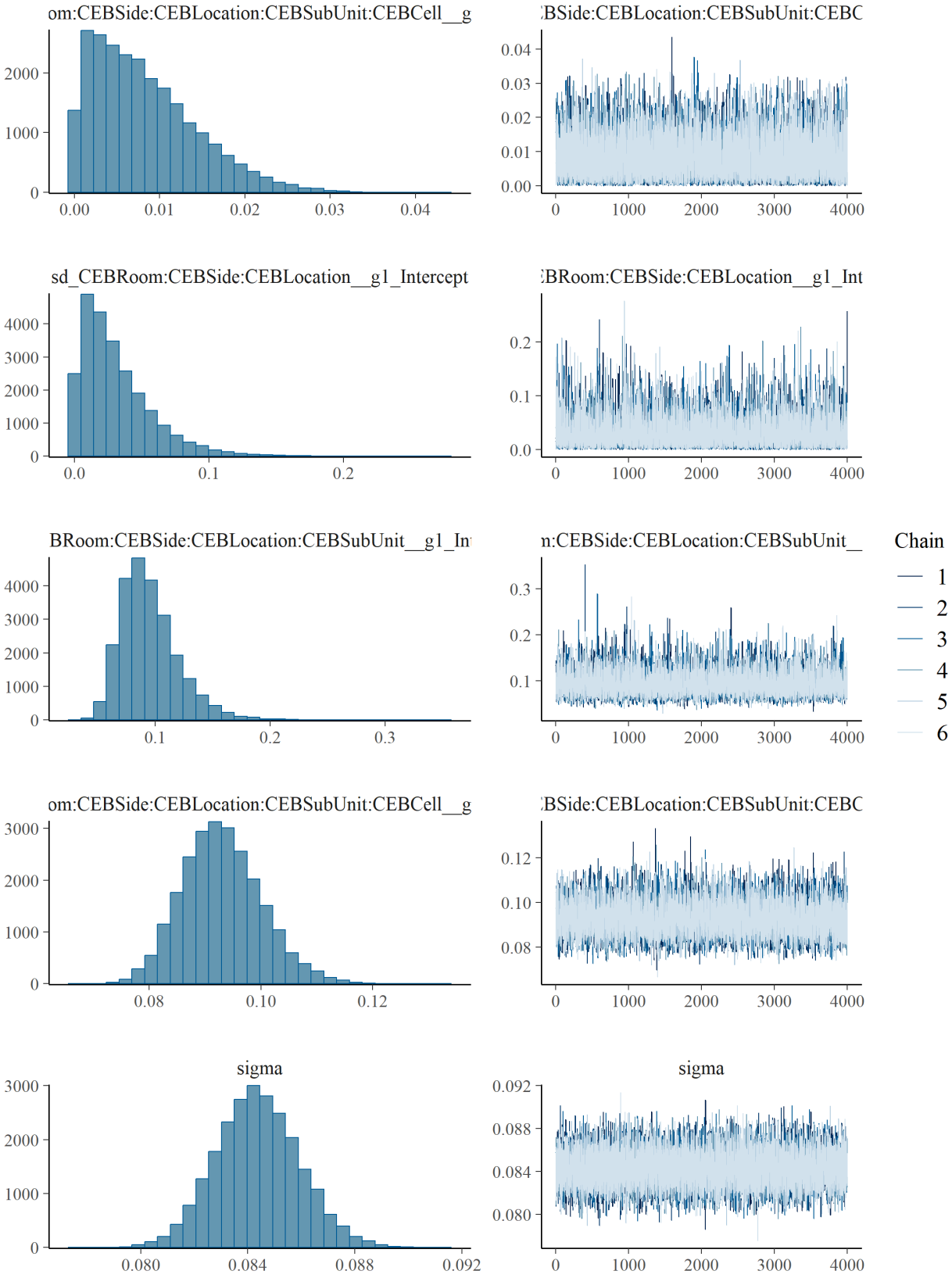


Figure 60: plot of chunk unnamed-chunk-12

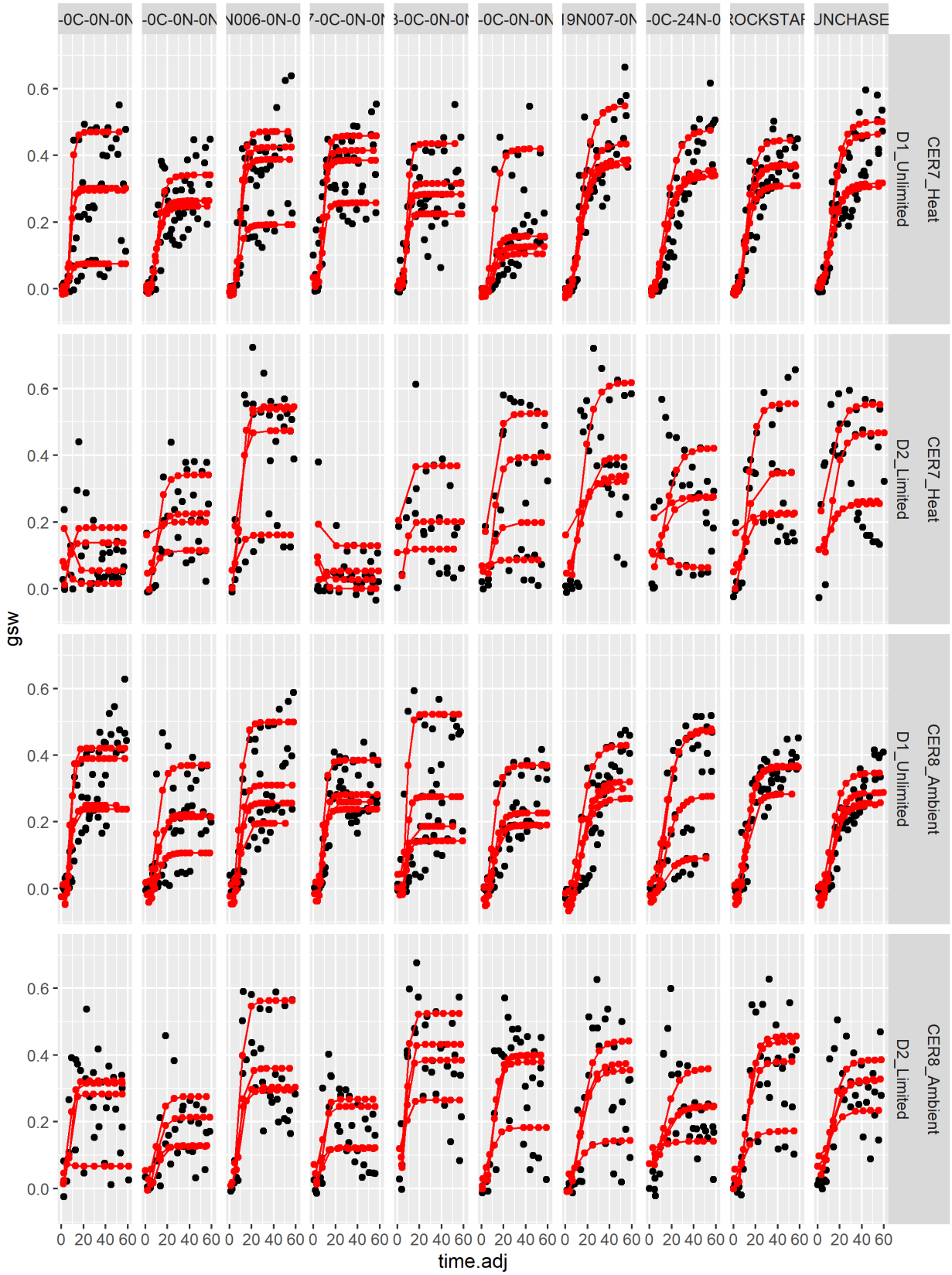


Figure 61: plot of chunk unnamed-chunk-12

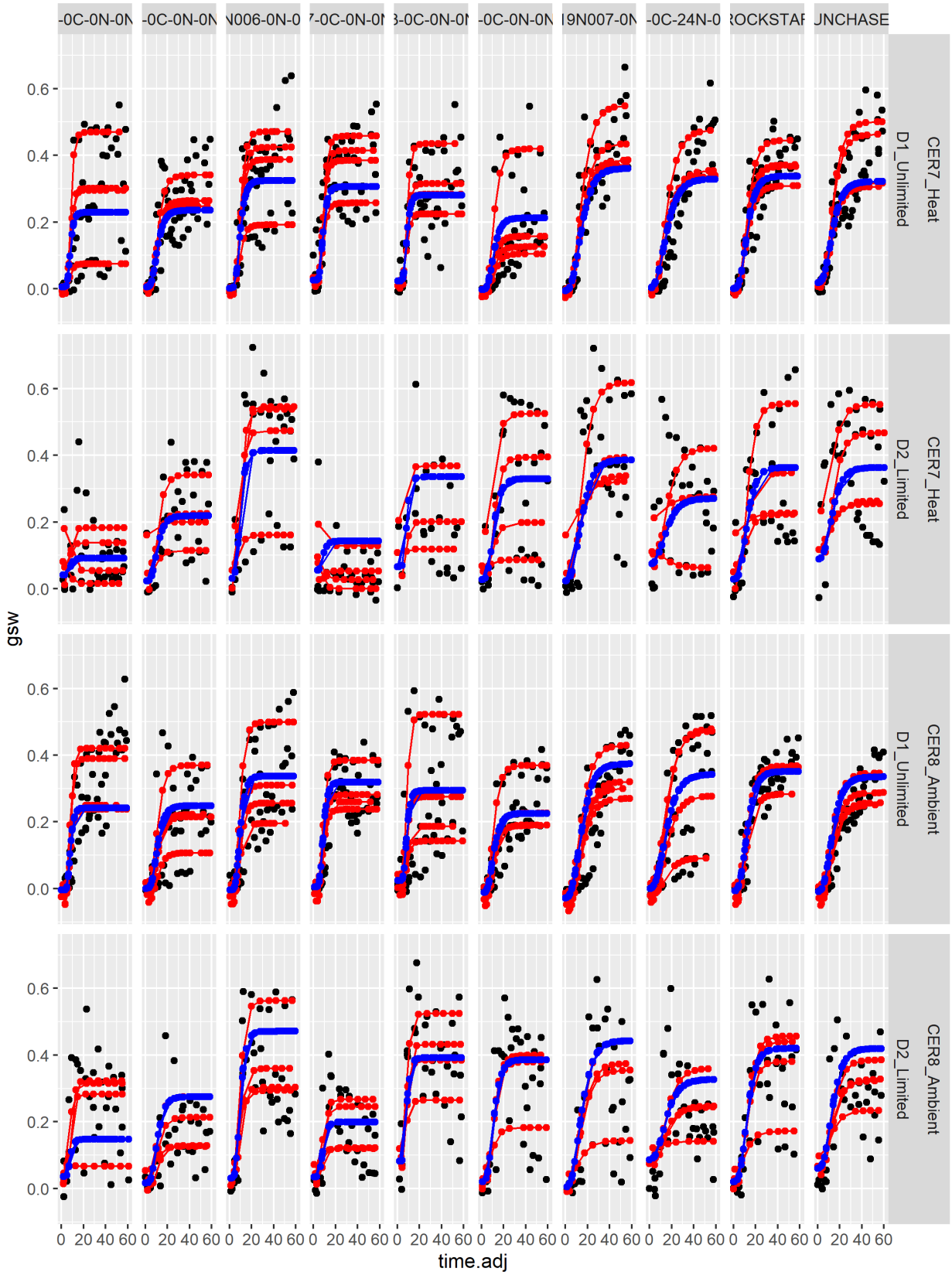


Figure 62: plot of chunk unnamed-chunk-12

Save data

```
save.image(saveFile)
```